# Compressing Global Dynamics of Deep Nonlinear Feature Learning

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**Abstract**

We give a modular proof of three trajectory-compression constructions for Gaussian-initialized deep nonlinear networks with learned hidden features. Legendre compression retains $n^{5/4+o(1)}$ moving coordinates and fixed dense mixers. Harmonic compression retains $O((\log n)^{3d+2})$ real coordinates for the whole input sphere. Taylor compression retains $O((\log n)^3)$ real coordinates on a fixed panel declared at initialization. All three have error negligible relative to actual independent dense-run variability, uniformly over their query domain and the complete training trajectory, including the fitted limit. The two selected constructions also attain absolute error $Y/n$. The common ingredients are finite residual activity, analytic response approximation, and geometry-preserving reduction; the selected dynamics use an additional readout correction to cancel unstable feedback. A separate short-time deep-linear benchmark establishes a matching literal-storage bound for a frozen-top neural tangent hierarchy. This manuscript also gives polynomial sufficient widths for Legendre and Taylor, and exact-real initialization bounds for both selected constructions: near-quadratic work from supplied dense matrices or near-linear work for a fresh Gaussian reference sampled implicitly. It makes no empirical or finite-precision claim.

## 1 Setup and the headline theorem

There are $m \geq 2$ training pairs $(x_a, y_a)$ with $x_a \in \mathbb{R}^d$, $\|x_a\|_2 = \sqrt{d}$, width $n$, and a fixed number $L \geq 2$ of hidden layers. Write $v = x/\sqrt{d}$. The first matrix is $W^{(1)} \in \mathbb{R}^{n\times d}$, the hidden matrices are $W^{(\ell)} \in \mathbb{R}^{n\times n}$, and the readout is $w \in \mathbb{R}^n$. The activation may depend on the layer; the superscript is suppressed when the same equation applies to every layer. When a response is indexed by the normalized query $v$, its network input is $x = \sqrt{d}\, v$. The canonical network is

$$z^{(1)}(x) = W^{(1)}v, \qquad z^{(\ell)}(x) = W^{(\ell)}h^{(\ell-1)}(x), \qquad h^{(\ell)}(x) = \phi_\ell(z^{(\ell)}(x)),$$
$$f_n(x) = \frac{w^\top h^{(L)}(x)}{n}, \qquad r_a = f_n(x_a) - y_a, \qquad \mathcal{L} = \frac{1}{m}\sum_{a=1}^{m} r_a^2. \tag{1}$$

The residual-free backward responses are

$$\delta_a^{(L)} = w \odot \phi_L'(z_a^{(L)}), \qquad \delta_a^{(\ell)} = \phi_\ell'(z_a^{(\ell)}) \odot W^{(\ell+1)\top}\delta_a^{(\ell+1)}. \tag{2}$$

At a passive input use the same forward and backward recursions, without supplying a label or adding a training term. Dots denote physical training time. Gradient flow with block mobilities $(n, 1, \ldots, 1, n)$ is exactly

$$\dot{W}^{(1)} = -\frac{2}{m}\sum_a r_a \delta_a^{(1)} v_a^\top,$$
$$\dot{W}^{(\ell)} = -\frac{2}{nm}\sum_a r_a \delta_a^{(\ell)} h_a^{(\ell-1)\top} \quad (2 \leq \ell \leq L), \qquad \dot{w} = -\frac{2}{m}\sum_a r_a h_a^{(L)}. \tag{3}$$

First-layer entries are independent $\mathcal{N}(0,1)$; hidden entries are independent $\mathcal{N}(0,1/n)$; all blocks are independent; and $w(0)=0$. Nothing freezes the hidden features.

**Activations, data, and constants.** Each $\phi_\ell$ is real on the real axis, holomorphic on $|\operatorname{Im} z|<a$ for a common $a>0$, and has bounded first derivative on that strip. Its values need not be bounded. For explicit estimates put

$$\beta=\max\left\{10,1+\max_{\ell}|\phi_\ell(0)|,16/a,\max_{\ell,\,j\in\{1,2\}}\sup_{|\operatorname{Im} z|\le a/2}|\phi_\ell^{(j)}(z)|\right\}.$$

The second derivative is bounded on the smaller strip by Cauchy's formula. Define the uncentered population feature Gram and the label scale by

$$\begin{gathered}Q_{ab}^{(0)}=v_a^\top v_b,\qquad Q_{ab}^{(\ell)}=\mathbb{E}[\phi_\ell(Z_a)\phi_\ell(Z_b)],\quad Z\sim\mathcal{N}(0,Q^{(\ell-1)}),\\ \gamma=\lambda_{\min}(Q^{(L)})>0,\qquad Y=\|y\|_2/\sqrt{m}>0.\end{gathered}\tag{4}$$

Assume $m/\gamma\ge 1$ and the same sufficient small-label condition

$$0<Y\le\frac{\gamma}{m}\beta^{-30L}.\tag{5}$$

No input-rank, centering, orthogonality, or label-sign hypothesis is imposed. All data, $m,d,L,\gamma,Y$, and activations are fixed separately as $n$ grows. Throughout the stated bounds, $C>0$ depends only on activations and depth, never on $n,m,d,\gamma,Y$ or a passive-panel size. Its value may change between occurrences. All other constants are defined in the statement or proof using them. Initialization costs use a separate fixed-parameter big-$O$ convention, specified in Proposition 1.2; its constants are not the activation-and-depth-only storage constant $C$.

**Query domains and probability.** For a query set $\mathcal{X}$, use

$$\|f-g\|_{\mathcal{X}}=\sup_{t\in[0,\infty]}\sup_{x\in\mathcal{X}}|f(t,x)-g(t,x)|.\tag{6}$$

The endpoint $t=\infty$ means the fitted limit. For Legendre and Harmonic, $\mathcal{X}=\{x:\|x\|_2=\sqrt{d}\}$. For Taylor, fix $p\ge 0$ additional inputs on this same sphere before initialization and take $\mathcal{X}=\{x_1,\dots,x_{m+p}\}$. Their labels are never supplied. Let $\widetilde{f}_n$ be an independently initialized dense run on the same training problem.

An eventual assertion at confidence $1-\delta$ means: for every fixed admissible problem and $0<\delta<1$, there is a finite width threshold such that the assertion holds at each larger individual width. That threshold may depend on every fixed problem parameter, including the positive $Y$. The headline theorem quantifies this threshold polynomially for Legendre and Taylor; Harmonic retains the eventual qualification. Polynomial sufficient widths need not be practically small. Each probability bound concerns one fixed problem at one individual width, not a simultaneous event over all datasets or infinitely many independently initialized widths. No unrestricted growing-data limit is asserted. For a ratio below, missing fitted limits or a zero denominator count as failure.

The compression and nonlinear feature-learning statements below inherit this setup and these probability conventions; the quantitative width improvement does not extend to the separate nonlinear feature-learning certificates. The separate NTH benchmark in Section 11 states its own assumptions and probability convention. Every storage count charges a nonprimitive activation evaluator's description and scratch separately.

**What must be remembered.** The following is the complete persistent notation. A row with several entries counts each entry separately; layer and sample indices do not create new objects. The two uses of $\delta$ are counted separately.

| Symbols | Meaning | Count |
|---|---|---|
| $n, m, d, L, p$ | width, training count, dimension, depth, passive count | 5 |
| $x, v = x/\sqrt{d}, y, t$ | input, normalized input, labels, physical time | 4 |
| $W, w, z, h, r, \delta_a^{(\ell)}$ | weights, readout, preactivations, features, residuals, backward responses | 6 |
| $\phi, a, \beta, Q, \gamma, Y$ | activations, strip, envelope, population Gram, gap, label RMS | 6 |
| $\mathcal{L}, \mathcal{X}, \|\cdot\|_{\mathcal{X}}$ | loss, query domain, trajectory norm | 3 |
| $\delta, C$ | failure probability, activation/depth constant | 2 |
| $f_n, \widetilde{f}_n, f_{\mathrm{model}}, f_{\mathrm{NTK},n}$ | dense, independent dense, compressed and frozen-kernel predictors | 4 |
| | | 30 |

Subscripts naming a compression method specialize $f_{\mathrm{model}}$. Ordinary mathematical operations and bound indices have their usual meanings. Every other symbol is bound to the statement and proof that introduce it: it can be forgotten afterward. In particular, approximation orders, clocks, source spaces, selected metrics, and numerical ledgers are not additional global notation. Later arguments use the stated conclusions, not private coefficients from their proofs.

**Theorem 1.1** (Compression of the complete learning trajectory)**.** *Under the preceding assumptions, each construction can be chosen so that*

$$\frac{\|f_{\mathrm{model}} - f_n\|_{\mathcal{X}}}{\left\|f_n - \widetilde{f}_n\right\|_{\mathcal{X}}} \xrightarrow{\mathbb{P}} 0. \tag{7}$$

*Here $f_n$ is the construction's coupled ordinary dense reference, and both norms use the same promised query domain and all physical times, including the fitted endpoint. Sufficient real-coordinate counts are*

$$\text{Legendre moving state:} \quad Cm\left(\frac{m}{\gamma}\right)^3 n^{5/4}[\log(en)]^2 e^{\sqrt{\log(en)}/2}, \tag{8}$$

$$\text{Harmonic total state:} \quad C(m+d)^2 + \left(\frac{C}{d}\right)^{d+1}\left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^{3d+2}, \tag{9}$$

*and the Taylor total state is at most*

$$\begin{aligned} C(m+p)^2\Bigg[\left(\frac{Ym}{\gamma}\right)^4 &[\log(en)]^3 \\ &+ \left(\frac{m}{\gamma}\right)^2 [\log(en)]^2[\log(e+\log(en))]^2\Bigg] + C(m+p)d. \end{aligned} \tag{10}$$

*Legendre additionally retains $(L-1)n^2$ fixed mixer coordinates. Harmonic and Taylor include the data, selected mixers, metrics, initial copies, and working arrays. Their predictions additionally satisfy*

$$\|f_{\mathrm{model}} - f_n\|_{\mathcal{X}} \le Y/n \tag{11}$$

*at every fixed confidence for sufficiently large width, with the same storage orders. One theoretical initializer uses the dense initialization, training data, and finitely many initialization derivatives; no later trained weight or dense trajectory is an input. The efficient alternatives in Proposition 1.2*

*compute disposable numerical continuations from the initial data. Subsequent compressed evolution is an autonomous nonlinear system with the stated retained state in either case.*

*For Legendre and Taylor, the sufficient width is polynomial: for every $0 < \varepsilon, \delta < 1$, the stated sizes and*

$$\|f_{\mathrm{model}} - f_n\|_{\mathcal{X}} \le \varepsilon \left\|f_n - \widetilde{f}_n\right\|_{\mathcal{X}}$$

*hold with probability at least $1 - \delta$ whenever*

$$n \ge \left[c\beta^L(1 + m + d + p + \gamma^{-1} + Y^{-1} + \delta^{-1} + \varepsilon^{-1})\right]^c, \tag{12}$$

*for an absolute numerical constant $c > 0$, local to this width bound. For Legendre take $p = 0$ and omit $Y^{-1}$; Taylor also satisfies* (11) *on this event. This strengthens only the width qualification: the storage formulas, label condition, query domains and physical optimizers are unchanged.*

**Proposition 1.2** (Efficient initialization)**.** *Under Theorem 1.1, Harmonic and Taylor admit the following exact-real initializers, at every fixed confidence and sufficiently large width.*

| *Method* | *Dense reference* | *Setup work* | *Peak setup memory* |
|---|---|---|---|
| *Harmonic* | *Supplied matrices* | $O(n^2[\log(en)]^{3d/2+1})$ | $O(n^2)$ |
| *Taylor* | *Supplied matrices* | $O(n^2[\log(en)]^{5/2})$ | $O(n^2)$ |
| *Harmonic* | *Fresh implicit Gaussian* | $O(n[\log(en)]^{9d/2+3})$ | $O(n[\log(en)]^{3d/2+1})$ |
| *Taylor* | *Fresh implicit Gaussian* | $O(n[\log(en)]^5)$ | $O(n[\log(en)]^{5/2})$ |

*All four initializers preserve the corresponding retained-state bound, the complete-trajectory error $Y/n$, and* (7)*. The constants in this table may depend on the separately fixed $m, d, p, L, \beta, \gamma, Y$ and confidence. The implicit version is coupled to a fresh ordinary Gaussian dense reference; it is not a subquadratic conversion of supplied matrices or a prescribed entrywise seed.*

The operation model charges scalar arithmetic, comparisons, square roots, elementary functions, real activation values and Gaussian samples as primitives. Nonprimitive evaluators add their actual work, description and scratch. Activation envelopes and a positive population-gap certificate are supplied; computing them is not included. No high activation derivatives or trained-reference values are inputs. These initializers compute through $T = 32(m/\gamma)\log(en)$, not merely a short physical-time prefix, then discard their working structures. The compressed model starts at time zero. The original initialization-jet compiler remains an alternative without an efficient work bound. The proof in Section 8 and finite-order ledgers in Section 8.4 separate all parameter dependencies, retained storage and setup memory. Taylor uses the polynomial source and accuracy gates above without an additional qualitative source threshold; Harmonic retains its eventual source-width qualification.

These are sufficient sizes, not optimality claims. Dense itself has $(L-1)n^2 + n(d+1)$ weight coordinates. Data storage is additional to the Legendre moving-state count. Coordinate counts alone do not bound precision or preprocessing time; Proposition 1.2 supplies separate arithmetic and peak-memory bounds, not a finite-precision guarantee. The finite-panel theorem makes no accuracy promise on inputs first supplied after initialization. Zero labels give the stationary zero predictor instead. The sharp literal-array cost of a frozen-top neural tangent hierarchy is treated separately in Section 11, under a growing-data, deep-linear regime.

## 1.1 The compressed dynamics are nonlinear and feature-learning

A moving kernel alone does not distinguish nonlinear learning from a deep linear network. We therefore separate two comparisons. In this subsection, $f_{\mathrm{NTK},n}$ is the predictor obtained by freezing

the coupled dense reference's initial tangent kernel. Since $w(0) = 0$, this is exactly its readout-only flow with all hidden features fixed at initialization.

**Theorem 1.3** (Compression preserves nonlinear feature learning)**.** *Under the compression theorem's assumptions, suppose additionally that every activation is nonaffine and $|x_a^\top x_b| < d$ for $a \neq b$. For Taylor, augment $\mathcal{X}$ by at most four deterministic passive witness inputs, declared before initialization without their labels. There are $c_*, t_* > 0$, depending on the fixed problem but not on width, such that for every fixed $0 < t \leq t_*$, each compressed predictor satisfies, with probability tending to one as $n \to \infty$,*

$$\inf_{a \in \mathbb{R}^d,\, b \in \mathbb{R}} \sup_{x \in \mathcal{X}} |f_{\mathrm{model}}(t, x) - a^\top x - b| \geq c_* t, \tag{13}$$

$$\sup_{x \in \mathcal{X}} |f_{\mathrm{model}}(t, x) - f_{\mathrm{NTK},n}(t, x)| \geq c_* t^3. \tag{14}$$

*The storage orders and approximation conclusions are unchanged; Taylor's bound uses the augmented panel count, at most $p + 4$ instead of $p$.*

*In the dense reference, every hidden layer satisfies*

$$\left( \frac{1}{n} \sum_{a=1}^{m} \|h^{(\ell)}(t, x_a) - h^{(\ell)}(0, x_a)\|_2^2 \right)^{1/2} \geq c_* t^2 \qquad (1 \leq \ell \leq L) \tag{15}$$

*with the same probability convention. In the first layer, the nonlinear part of the feature increment itself has positive mean-square size of order $t^4$, in the precise sense of* (180)*.*

The first inequality excludes every deep linear predictor, even with biases, an arbitrary optimizer and time-dependent weights. The second excludes the reference network's frozen-feature training. The internal statement further rules out a frozen nonlinear feature component accompanied only by an affine feature change. It does not identify the compressed coordinates with dense neurons. The additional hypotheses belong only to this activity theorem; the broader compression theorem does not require them.

The mechanism has three time scales. The initially zero readout first grows at order $t$; hidden features move at order $t^2$; the output departs from frozen-feature training at order $t^3$. The leading departure in the label direction is a sum of squared hidden-layer accelerations, so contributions from different layers cannot cancel. Separately, the initial feature kernel has nonlinear input components that no nonzero label vector can cancel completely on the sphere. Both signals persist while compression error vanishes. The proof in Section 9 uses only initial Gaussian conditioning and real short-time estimates, not a width-independent complex-time radius or a trained population-flow theorem. These are early-time certificates, not endpoint or test-risk comparisons.

**Proof architecture.** First establish fitting and finite residual activity without a source approximation. A common probabilistic source lemma then controls forward and backward responses on the two analytic domains. Legendre approximates their history integrals. The other two methods share one geometry-preserving selection and one corrected nonlinear optimizer; only their finite source bases differ. A last-layer innovation and a deterministic analytic estimate give a dense lower bound at a training input. Section 10 combines these ingredients. A quantitative source argument and a third-order Legendre history estimate supply the polynomial width refinement. Numerical source construction and a two-direction Gaussian sampler prove Proposition 1.2. Every dependency is supplied below; the older decoder and empirical material remain outside this manuscript.

## 2 Initialization, fitting, and a signed perturbation estimate

**Lemma 2.1** (Initialization and real fitting)**.** *For this lemma put*

$$H = \max_{0\le j\le L} \sqrt{Q_{11}^{(j)}}, \qquad \mathsf{H}(t) = [h^{(L)}(t, v_1), \dots, h^{(L)}(t, v_m)].$$

*For the parameter tuple $\theta = (W^{(1)}, \dots, W^{(L)}, w)$ use the norm*

$$\|\theta\|_{\mathrm{par}}^2 = \|W^{(1)}\|_F^2/n + \sum_{j=2}^{L} \|W^{(j)}\|_F^2 + \|w\|^2/n.$$

*The initialized training Grams satisfy*

$$\left(\frac{h_0^{(j)}(v_a)^\top h_0^{(j)}(v_b)}{n}\right)_{a,b=1}^{m} \xrightarrow{\mathbb{P}} Q^{(j)} \qquad (1 \le j \le L). \tag{16}$$

*With probability tending to one,*

$$\begin{gathered} \frac{\|W_0^{(1)}\|_{\mathrm{op}}}{\sqrt{n}} \le 8, \qquad \max_{2\le j\le L} \|W_0^{(j)}\|_{\mathrm{op}} \le 8, \\ \sup_{\|v\|=1,\, 1\le j\le L} \frac{\|h_0^{(j)}(v)\|}{\sqrt{n}} \le \frac{3H}{2}, \qquad \frac{\mathsf{H}_0^\top \mathsf{H}_0}{mn} \succeq \frac{\gamma}{2m} I_m. \end{gathered} \tag{17}$$

*On this event the flow is global, converges in parameters, fits the labels, and satisfies, for all $t \ge 0$,*

$$\begin{gathered} \frac{\|W^{(1)}(t)\|_{\mathrm{op}}}{\sqrt{n}} < 9, \qquad \max_{2\le j\le L} \|W^{(j)}(t)\|_{\mathrm{op}} < 9, \\ \sup_{\|v\|=1,\, 1\le j\le L} \frac{\|h^{(j)}(t, v)\|}{\sqrt{n}} < 2H, \qquad \frac{\mathsf{H}(t)^\top \mathsf{H}(t)}{mn} \succeq \frac{\gamma}{4m} I_m \end{gathered} \tag{18}$$

$$\begin{gathered} \frac{\|r(t)\|_2}{\sqrt{m}} \le Y e^{-\gamma t/(2m)}, \qquad \frac{\|w(t)\|}{\sqrt{n}} \le 2Y\sqrt{\frac{m}{\gamma}}, \\ \int_t^\infty \|\dot\theta(u)\|_{\mathrm{par}}\, du \le \frac{2\|r(t)\|_2}{\sqrt{\gamma}}. \end{gathered} \tag{19}$$

*The output converges uniformly on the sphere, with*

$$\sup_{\|v\|=1} |f_n(\infty, v) - f_n(t, v)| \le C \frac{m}{\gamma} \frac{\|r(t)\|_2}{\sqrt{m}}. \tag{20}$$

*These assertions hold jointly for all rectangular cavities deleting any fixed bounded number of coordinates per layer, with the original width-$n$ variances, flow factors, and denominators unchanged.*

*Proof.* All auxiliary coefficients in this proof are local. Put $\lambda = \gamma/m$, $\rho = \|r\|_2/\sqrt{m}$,

$$s = \max\left\{1, \max_j \sup_{|\operatorname{Im} z| \le a/2} |\phi_j'(z)|\right\}, \qquad b = \max_j |\phi_j(0)|,$$

and retain $H$ as defined in this lemma's statement. Thus $|\phi_j(z)| \le b + s|z|$ on the half strip. Define

$$\begin{aligned} &D_j = s(9s)^{L-j}, && U_1 = D_1, && U_j = 2HD_j \quad (j \ge 2), \\ &F_1 = sU_1, && F_j = s(2HU_j + 9F_{j-1}), && F = F_L. \end{aligned} \tag{21}$$

Equal input norms give $Q_{aa}^{(j)} = Q_{11}^{(j)}$, hence $\lambda \le H^2$. The coefficients satisfy

$$H \le (2\beta)^L, \qquad F = s^2 \left[ (9s)^{2L-2} + 4H^2 \sum_{j=0}^{L-2} (9s)^{2j} \right], \qquad 8H\sqrt{F} \le \beta^{5L}. \tag{22}$$

The first inequality follows from Minkowski's inequality and $\sqrt{Q_{11}^{(j)}} \le b + s\sqrt{Q_{11}^{(j-1)}}$. For the last, summing the geometric series gives $F \le (21/20)H^2 s^2 (9s)^{2L-2}$, and then $8H\sqrt{F} \le H^2(9s)^L \le 36^L \beta^{3L} \le \beta^{5L}$. Consequently the assumed cap $Y \le \lambda\beta^{-30L}$ implies $Y \le \lambda/(8H\sqrt{F})$, which will close the real-flow estimates.

A maximal 1/4-separated subset of the unit sphere in $\mathbb{R}^k$ has at most $9^k$ points: disjoint radius-1/8 balls fit inside the radius-9/8 ball. Approximating each side of a maximizing bilinear pair gives $\|M\|_{\mathrm{op}} \le 2 \max |u^\top M v|$ over these nets. Each pairing for $W_0^{(\ell)}$, $\ell \ge 2$, or $W_0^{(1)}/\sqrt{n}$ is $N(0, 1/n)$. The exponential Gaussian tail and a union bound prove the operator assertions, with failure bounded by $2(L-1)e^{-(8-2\log 9)n} + 2e^{-(8-\log 9)n + d\log 9}$.

On any fixed finite query list, condition on the preceding initialized layers. Each next empirical Gram is an average of independent Gaussian-row feature products, with conditional mean $\Psi_\ell(C)_{ab} = \mathbb{E}\phi_\ell(Z_a)\phi_\ell(Z_b)$, $Z \sim N(0, C)$. On bounded covariance sets the conditional variances are bounded, by linear growth and Gaussian fourth moments. Moreover $\Psi_\ell$ is continuous on the entire positive semidefinite cone: write $Z = C^{1/2}G$ with a standard Gaussian vector $G$ of the matching dimension, use continuity of the positive square root, and dominate by a fixed Gaussian polynomial. Conditional Chebyshev and induction therefore prove convergence of each empirical Gram to $Q^{(\ell)}$, including when intermediate covariances are singular. Apply this to the training inputs and a fixed sphere net of mesh $h \le H/[4(8s)^L]$. On the operator event the initialized layer-$\ell$ feature map, in neuron RMS, is $(8s)^\ell$-Lipschitz. Its norm at net points is eventually at most $\sqrt{5/4}H$, with probability tending to one; the off-net increment is at most $H/4$. This proves the sphere bound and the training Gram assertion in (17).

Here is also a uniform justification for the cavity assertion, without a union bound over deletion sets. On a sphere net of mesh $n^{-2}$, the preceding RMS bounds and conditional Gaussian tails give $\max_{i,\ell,v} |h_{0,i}^{(\ell)}(v)| = O(\sqrt{\log(en)})$ with probability tending to one. The net has only a fixed power of $n$ points; the coordinate Lipschitz bound is at most $\sqrt{n}(8s)^\ell$, so the same bound holds off the net. Zero-extend each cavity's features into $\mathbb{R}^n$. At a deleted layer, at most a fixed number of original coordinates are removed. At an undeleted coordinate the activation difference is at most $s$ times the preactivation difference. Thus successive RMS differences are bounded by $8s$ times the preceding difference plus $O(\sqrt{\log(en)/n})$. At fixed depth their supremum, simultaneously over all such cavities and sphere queries, tends to zero. Submatrix operator norms do not increase. Taking strictly smaller initial feature and Gram margins before this comparison gives (17) for every cavity. This also shows directly that each cavity's training Gram has the same limit $Q^{(L)}$.

Stop any one of these flows at the first failure of (18). Backward propagation gives $\|\delta_a^{(\ell)}\|/\sqrt{n} \le D_\ell \|w\|/\sqrt{n}$. In Euclidean mobility coordinates $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$, the vector field is minus the gradient of $\rho^2$. Consequently

$$-\partial_t \rho^2 = \|\dot\theta\|_{\mathrm{par}}^2 \ge \lambda \rho^2.$$

The last inequality uses only the readout contribution and the stopped Gram gap. For any stopped interval $[t, T]$, Cauchy–Schwarz gives

$$\left( \int_t^T \|\dot\theta\|_{\mathrm{par}} \right)^2 \le \left( \int_t^T \frac{\|\dot\theta\|_{\mathrm{par}}^2}{\rho} \right) \left( \int_t^T \rho \right) \le 2\rho(t)\, \frac{2\rho(t)}{\lambda}.$$

If a residual vanishes, all velocities vanish, so the estimate extends without division by zero. In particular $\|w\|/\sqrt{n} \le 2Y/\sqrt{\lambda}$. Sample Cauchy–Schwarz bounds each normalized hidden-block velocity by $2\rho U_\ell \|w\|/\sqrt{n}$. Integration and forward subtraction give

$$\frac{\|W^{(1)} - W_0^{(1)}\|_F}{\sqrt{n}} \le \frac{8U_1 Y^2}{\lambda^{3/2}}, \qquad \|W^{(\ell)} - W_0^{(\ell)}\|_F \le \frac{8U_\ell Y^2}{\lambda^{3/2}},$$
$$\sup_v \frac{\|h^{(\ell)} - h_0^{(\ell)}\|}{\sqrt{n}} \le \frac{8F_\ell Y^2}{\lambda^{3/2}}. \tag{23}$$

Indeed each later feature difference is bounded by $s$ times $2H\|W^{(\ell)} - W_0^{(\ell)}\|_F$ plus nine times the preceding feature difference, which is exactly the recursion defining $F_\ell$. The label restriction implies $8FY^2/\lambda^{3/2} \le \sqrt{\lambda}/(8H^2) \le 1/(8H)$. Thus the operators remain below $8 + 1/8$, the features below $3H/2 + H/8$, and the smallest singular value of $\mathsf{H}/\sqrt{mn}$ above $(1/\sqrt{2} - 1/8)\sqrt{\lambda} > \sqrt{\lambda}/2$. Every stop has a strict margin. At fixed width the bounded parameters and locally Lipschitz vector field give global continuation. The finite parameter length proves convergence, and residual decay proves fitting. The argument uses only the retained dimensions and the original denominators $n$, and hence also proves cavity fitting.

Finally the same forward recursion gives

$$\|\dot{w}\|/\sqrt{n} \le 4H\rho, \qquad \sup_v \|\dot{h}^{(L)}(v)\|/\sqrt{n} \le 2F\rho\|w\|/\sqrt{n}.$$

Therefore

$$\sup_{\|v\|=1} |f_n(\infty, v) - f_n(t, v)| \le \frac{16}{\lambda}\left(H^2 + \frac{FY^2}{\lambda}\right)\rho(t).$$

The coefficients $H, F$ depend only on activations and depth, and the headline cap with $\lambda \le 1$ gives $Y^2/\lambda \le 1$. This proves (20), including its uniform endpoint. □

**Lemma 2.2** (Signed perturbation estimate)**.** *Let a differentiable prediction map $F$ take Euclidean parameter space to sample space with inner product $\langle u, v\rangle_m = m^{-1}u^\top v$. Write $r = F(\theta) - y$, $r' = F(\theta') - y$, $\rho = \|r\|_m$, $\rho' = \|r'\|_m$, and $J = DF(\theta)$, $J' = DF(\theta')$. Suppose $\dot\theta = -2J^* r + \mathcal{E}$ and $\dot\theta' = -2J'^* r'$. Here $J^*$ and $J'^*$ are adjoints for the stated parameter and sample inner products. If a constant $K \ge 0$ satisfies, for $e = \|\theta - \theta'\|$,*

$$\|r - r' - J'(\theta - \theta')\|_m \le Ke^2/2, \qquad \|(J - J')(\theta - \theta')\|_m \le Ke^2,$$

*then, on every finite interval,*

$$e(t) \le \left(e(0) + \int_0^t \|\mathcal{E}(u)\|\, du\right) \exp\left\{K\int_0^t (\rho' + 3\rho)\, du\right\}. \tag{24}$$

*Proof.* Put $d = \theta - \theta'$ and $R = r - r' - J'd$. Exact subtraction gives

$$\tfrac{1}{2}\partial_t e^2 = -2\|r - r'\|_m^2 + 2\langle r - r', R\rangle_m - 2\langle r, (J - J')d\rangle_m + \langle d, \mathcal{E}\rangle \le K(\rho' + 3\rho)e^2 + e\|\mathcal{E}\|.$$

Apply the integrating factor to $\sqrt{e^2 + \epsilon^2}$ and let $\epsilon \downarrow 0$. This proves the assertion also through zeros of the parameter discrepancy. □

## 3 One source argument on two analytic domains

The following interface supplies the analytic approximation domains and real-time bounds used by the compression constructions. Its proof contains the stopped-cavity argument and all its auxiliary coefficient estimates.

**Proposition 3.1** (Analytic sources and real-time response bounds)**.** *Fix the finite query lists below before initialization. Eventually at each fixed confidence, the following hold on one event. In this proposition put*

$$T = 32\frac{m}{\gamma}\log(en), \qquad r_t = \frac{\gamma}{\beta^{30L}Y^2 m\sqrt{(d+3)\log(en)}}, \qquad r_q = \frac{1}{\beta^{3L}\sqrt{(d+3)\log(en)}}.$$

*Write $k^{(L)} = w$, $k^{(j)} = W^{(j+1)\top}\delta^{(j+1)}$ $(j < L)$ for the backward responses before the activation derivative. The four source families are*

$$h^{(j)}, \qquad W^{(j)}(0)h^{(j-1)}, \qquad \delta^{(j)}, \qquad W^{(j+1)}(0)^\top\delta^{(j+1)}, \tag{25}$$

*omitting the forward image for $j = 1$ and reverse image for $j = L$.*

*(i) The forward and backward fields are holomorphic on a neighborhood of*

$$([-r_t, T + r_t] + i[-r_t, r_t]) \times \{v \in \mathbb{C}^d : v^\top v = 1, \|\operatorname{Im} v\|_2 \le \sinh r_q\}.$$

*Query holomorphy is relative to the quadric; for $d = 1$ use $\{-1, 1\}$. Here and for $t \in [0, \infty]$ over the real sphere, the first-layer operator divided by $\sqrt{n}$ and all hidden operators have norm at most ten, and*

$$\frac{\|h^{(j)}\|_2}{\sqrt{n}} \le \beta^{3L}, \qquad \frac{\max\{\|\delta^{(j)}\|_2, \|k^{(j)}\|_2\}}{\sqrt{n}} \le 16\beta^{4L}\frac{Ym}{\gamma}. \tag{26}$$

*Each source coordinate has modulus at most $\beta^{6L}\sqrt{n}$ on the complex domain.*

*(ii) On each fixed finite query list, the predictions are holomorphic near $[-r, T + r] + i[-r, r]$, where*

$$r = \frac{m}{\gamma\sqrt{\log(en)}}, \qquad |f_n(t, \sqrt{d}\, v)| \le 16\beta^{6L}\frac{Ym}{\gamma}. \tag{27}$$

*On the entire real training trajectory,*

$$\sup_{t\in[0,\infty]} \max_{a,j,i} |k_{a,i}^{(j)}(t)| \le 32\beta^{21L}\frac{Ym}{\gamma}\sqrt{\log(en)}. \tag{28}$$

*(iii) For a declared finite panel containing the training inputs, there are $0 = t_0 < \cdots < t_J = T$ and $r_b > 0$, $0 \le b < J$, with $t_{b+1} - t_b \le r_b/2$. The sources are holomorphic near $|t - t_b| \le r_b$, with the operator, RMS and coordinate bounds in (i): forward queries range over the panel, backward queries over training inputs. The partition and radii depend deterministically only on $n, m, \gamma, Y$, activations and depth, and*

$$J \le 1 + \beta^{30L}\left(\frac{Ym}{\gamma}\right)^2\sqrt{\log(en)} + \beta^{16L}\frac{m}{\gamma}\log\left(1 + \frac{4\log(en)}{\log 2}\right). \tag{29}$$

*Proof.* We prove stronger intermediate domains with explicitly calculated bounds, then compare them with the displayed interface. All symbols introduced from here through the end of this proof are local to this argument. The quantitative width and initialization proofs explicitly recall the coefficient recurrences they additionally use; other constructions use the proposition's stated bounds. All probability statements concern one sufficiently large width, with the data, depth, activations and confidence fixed. Write $\lambda = \gamma/m$, $Y = \|y\|_2/\sqrt{m}$, $S = 16Y/\lambda$, and $\ell = \log(en)$. We treat $Y > 0$; zero labels give the stationary zero predictor. For each full or cavity trajectory write $\rho = \|r\|_2/\sqrt{m}$, also at complex times using the Euclidean norm of the complex residual vector. Write $W_0^{(j)} = W^{(j)}(0)$ for initialized matrices. Put

$$b = \max_j |\phi_j(0)|, \qquad s = \max\{1, \sup_{j,|\operatorname{Im} z|\le a/2} |\phi_j'(z)|\}, \qquad t_2 = \max\{1, \sup_{j,|\operatorname{Im} z|\le a/2} |\phi_j''(z)|\}.$$

Thus $1 + b, s, t_2, 16/a \le \beta$, and the bound used for activation values is $|\phi_j(z)| \le b + s|z|$, not boundedness. Write $v = x/\sqrt{d}$. Backward fields at a passive input are defined by the same algebraic recursion as at a training input,

$$k^{(L)}(t, v) = w(t), \qquad \delta^{(j)}(t, v) = \phi_j'(z^{(j)}(t, v)) \odot k^{(j)}(t, v), \qquad k^{(j)} = W^{(j+1)\top}\delta^{(j+1)}.$$

This definition uses no passive label and does not add a training term.

**1. Local coefficient estimates and working domains.** The following scalars are bounds, not coordinates of a model:

$$\begin{aligned}
&H_1 = \max(1, b + 20s), && H_j = \max(1, b + 10sH_{j-1}), && \\
&k_L = H_L, && k_j = 10sk_{j+1}, && \tau_j = sk_j, \\
&P_1 = 3, && P_j = H_{j-1} + 10sP_{j-1} + 1, && f_j = sP_j, \\
&A_* = 2sP_L + 2s^2 \sum_{j=2}^{L} k_j P_{j-1}, && D_* = t_2 \sum_{j=1}^{L} P_j^2, && H_* = A_* + t_2 \sum_{j=1}^{L} P_j^2 k_j.
\end{aligned} \tag{30}$$

Set $f_* = \max_j f_j$, $k_* = \max_j k_j$, $H_{\max} = \max_j H_j$, and define

$$\begin{aligned}
&E_* = t_2 \sum_{j=1}^{L} (10s)^{2(j-1)} k_j, && T_* = 2H_*^2 + 4A_*^2(e^2 - 1), \\
&D_0 = (1 + b + s)(1 + T_* + E_* + s^2 k_*^2), && D_1 = 576e^3(1 + b + s)D_*^3, \\
&C_F = s(8f_*^2 + H_{\max}^2 + 1), && C_{\mathrm{abs}} = 8(D_0 + 1), \\
&V_1^0 = \tau_1, && V_j^0 = \tau_j H_{j-1}^2 + 10sV_{j-1}^0, \\
&G_L^0 = sH_L + 14t_2 V_L^0 + s, && G_j^0 = 10sG_{j+1}^0 + s^3 k_{j+1}^2 H_j + 14t_2 V_j^0 + s, \\
&W_G = 128(1 + H_{\max} + s \max_j V_j^0 + \max_j G_j^0). &&
\end{aligned} \tag{31}$$

Choose the exponential-budget constants in this order:

$$\eta_* = \min\{1, (1024 C_{\mathrm{abs}} W_G)^{-1}\}, \qquad \mathcal{B} = 1024e^2 L, \qquad \Lambda_* = \log(e + \mathcal{B}) + \log(1/\eta_*).$$

$$\begin{aligned}
S_* = \min\Bigg\{ & 1, (4A_*)^{-1}, \sqrt{\frac{\eta_*}{4000 D_*}}, \sqrt{\frac{\eta_*}{16eD_*\sqrt{2\mathcal{B}}}}, \\
& \sqrt{\frac{\eta_*}{\Lambda_*}}, \left(\frac{\eta_*^3}{D_1 \mathcal{B}}\right)^{1/4}, (8D_0 C_F)^{-1/2} \Bigg\}.
\end{aligned} \tag{32}$$

These quantities depend only on the activation bounds and depth. Further put

$$
\begin{aligned}
&C_G = 32\max(1, H_{\max}, \max_j \tau_j), \quad K_{\mathrm{src}} = 16C_{\mathrm{abs}}(1+C_G),\\
&g_* = \tau_1 + \sum_{j=2}^{L} \tau_j H_{j-1}, && r_j^* = P_j g_*, && q_j^* = f_j g_*,\\
&j_j^* = 20(10s)^{j-1}, && b_j^* = s j_j^*,\\
&e_1 = t_2 r_1^* P_1, && e_j = t_2 r_j^* P_j + s(q_{j-1}^* + \tau_j H_{j-1} f_{j-1} + 10 e_{j-1}),\\
&a_1^* = t_2 j_1^* P_1 + 2s, && a_j^* = t_2 j_j^* P_j + s(b_{j-1}^* + 10 a_{j-1}^*),\\
&T_Q = 8 f_* \max_j (f_j H_* + e_j), && T_J = 8 f_* \max_j a_j^*.
\end{aligned} \tag{33}
$$

For a Gaussian multiplier $G$, define

$$
\begin{aligned}
U_1(G) &= 4sK_{\mathrm{src}},\\
U_j(G) &= 2\{sK_{\mathrm{src}}(H_{j-1}^2 + f_{j-1}^2 + ST_Q + S^2 H_{j-1} q_{j-1}^*) + G q_{j-1}^* + 1\},\\
V_1(G) &= 2(2G + 2sK_{\mathrm{src}} S^2 + 1),\\
V_j(G) &= 2\{G b_{j-1}^* + sK_{\mathrm{src}} S^2 (T_J + H_{j-1} b_{j-1}^*) + 1\}.
\end{aligned} \tag{34}
$$

Use $U = \max_j U_j(16\sqrt{d+3})$, $V = \max_j V_j(16\sqrt{d+3})$ for the sphere, and $U_{\mathrm{fin}} = \max_j U_j(64)$ for a fixed finite list of inputs. Finally let

$$
\mathcal{K} = H_L^2 + S^2 \left( \tau_1^2 + \sum_{j=2}^{L} \tau_j^2 H_{j-1}^2 \right), \qquad D_W = \max\{\tau_1, \max_{j \geq 2} \tau_j H_{j-1}\}. \tag{35}
$$

**Claim 1** (Local power bounds)**.** *If $Y/\lambda \leq \beta^{-30L}$, then $S \leq \beta^{-26L} \leq S_*$. Moreover*

$$
H_j \leq 3\beta^{2j}, \quad P_j \leq \beta^{3j-2}, \quad k_j \leq 3\beta^{4L-2j}, \quad \tau_j \leq 3\beta^{4L-2j+1}, \quad K_{\mathrm{src}} \leq \beta^{21L}, \tag{36}
$$

$$
U \leq \beta^{26L}\sqrt{d+3}, \qquad V \leq \beta^{2L}\sqrt{d+3}, \qquad U_{\mathrm{fin}} \leq \beta^{26L}, \qquad \mathcal{K} \leq \beta^{14L}. \tag{37}
$$

*Proof.* Unroll the affine recurrences using $10s \leq \beta^2$, $4j \leq \beta^j$, $L, L+1 \leq \beta^{L-1}$, and $\sum_{j=1}^{L} \beta^{cj} \leq \beta^{cL}/(1-\beta^{-c})$. They give $H_j \leq 22\beta(10\beta)^{j-1} \leq 3\beta^{2j}$ and $P_j \leq 4j(10\beta)^{j-1} \leq \beta^{3j-2}$. Substitution gives the following bounds; both columns are included to account for the numerical factors.

| quantity | intermediate bound | power bound |
|---|---|---|
| $A_*$ | $7\beta^{5L-3}$ | $\beta^{5L-2}$ |
| $D_*$ | $2\beta^{6L-3}$ | $\beta^{6L-2}$ |
| $H_*$ | $4\beta^{8L-3}$ | $\beta^{8L-2}$ |
| $E_*$ | $4\beta^{6L-3}$ | $\beta^{6L-2}$ |
| $T_*$ | $3\beta^{16L-4}$ | $\beta^{16L-3}$ |
| $D_0$ | $8\beta^{16L-3}$ | $\beta^{16L-2}$ |
| $D_1$ | $9216e^3\beta^{18L-8}$ | $\beta^{18L-2}$ |
| $C_F$ | $9\beta^{6L-1}$ | $\beta^{6L}$ |
| $C_{\mathrm{abs}}$ | $65\beta^{16L-3}$ | $\beta^{16L-1}$ |

For instance, the summands of $A_*$'s mixed part, $H_*$'s curvature part, and $E_*$ are at most $6\beta^{4L+j-3}, 3\beta^{4L+4j-3}, 3\beta^{4L+2j-3}$. The inner sum in $D_0$ is at most $1 + 3\beta^{16L-4} + 4\beta^{6L-3} + 9\beta^{8L-2} \leq 4\beta^{16L-4}$, and $9216e^3 < \beta^6$. Unrolling the two activity recurrences gives

$$
V_j^0 \leq 27j\beta^{4L+2j-3}, \quad \max_j V_j^0 \leq \beta^{7L-2}, \quad G_j^0 \leq 66L\beta^{8L-2j-1}, \quad W_G \leq \beta^{9L+1}.
$$

In the reverse recurrence the propagated terminal, mixer and curvature contributions are bounded respectively by $380L\beta^{8L-2j-2}, 27L\beta^{8L-2j-1}, 379L\beta^{8L-2j-6}$; the constant contribution is at most $2\beta^{2L-2j-1}$. Hence $\eta_*^{-1} \le \beta^{25L+4}$, $\mathcal{B} \le \beta^{L+3}$, and $\Lambda_* \le \beta^{2L}$. The six nonconstant entries of (32) are at least $\beta^{-c}$, with exponents

$$5L-1, \quad (31L+6)/2, \quad (63L+12)/4, \quad (27L+4)/2, \quad (94L+13)/4, \quad (22L-1)/2.$$

All are at most $26L$. Also $16\beta^{-30L} \le \beta^{-26L}$. The remaining recurrences give

$$g_* \le \beta^{5L-1}, \qquad r_j^* \le \beta^{5L+3j-3}, \qquad q_j^* \le \beta^{5L+3j-2},$$
$$e_j \le 2\beta^{5L+6j-4}, \qquad a_j^* \le 3\beta^{5j-2}, \qquad T_Q \le \beta^{14L-3}, \qquad T_J \le \beta^{8L-1}.$$

For $e_j$, its three forcing exponents are at most $5L+6j-4, 5L+3j-4, 4L+3j-4$, with coefficient nine in the last term; geometric propagation by $\beta^2$ gives the displayed factor two. The same calculation gives the factor three for $a_j^*$. Further $K_{\mathrm{src}} \le 1552\beta^{20L-2} \le \beta^{20L+2}$. Now $ST_Q, S^2H_{j-1}q_{j-1}^* \le 1$, and the bracket in (34) is at most $11\beta^{6L-8}$. Thus for $j \ge 2$

$$U_j(16\sqrt{d+3}) \le 22\beta^{26L-5} + 32\sqrt{d+3}\beta^{8L-5} + 2, \quad U_j(64) \le 22\beta^{26L-5} + 128\beta^{8L-5} + 2.$$

The first-layer bound is $4\beta^{20L+3}$. These give the asserted bounds on $U, U_{\mathrm{fin}}$. In $V_j$, the term containing $S^2$ is at most one, giving $V_j \le 64\sqrt{d+3}\beta^{2L-2} + 4$ for $j \ge 2$; the first value is at most $67\sqrt{d+3}$. Finally substitute (36) in (35); $S \le 1$ and the same geometric sum give $\mathcal{K} \le \beta^{14L}$. Every comparison above holds already at $L = 2, \beta = 10$, and improves as either increases. □

**Working whole-sphere estimates.** The estimates below in fact require only the conclusion of the fitting lemma and $S \le S_*$, both already implied by the headline assumptions. For the remainder of this proof redefine the working radii by

$$T = 32\ell/\lambda, \quad c_t = \frac{a}{64YSU}, \quad c_q = \min\{1/8, a/(8V)\}, \quad r_t = c_t/\sqrt{\ell}, \quad r_q = c_q/\sqrt{\ell}.$$

For every fixed confidence, eventually with that confidence all forward and backward fields extend holomorphically to a neighborhood of

$$([-r_t, T+r_t] + i[-r_t, r_t]) \times \mathcal{Q}_{r_q}, \qquad \mathcal{Q}_r = \{v \in \mathbb{C}^d : v^\top v = 1, \|\operatorname{Im} v\|_2 \le \sinh r\}.$$

For $d = 1$, replace $\mathcal{Q}_r$ by the two points $\{-1, 1\}$. Holomorphy in the query is relative to the complex quadric. Normalized first-layer and hidden operator norms are at most ten, and

$$\|h^{(j)}\|_2/\sqrt{n} \le H_j, \qquad \|k^{(j)}\|_2/\sqrt{n} \le Sk_j, \qquad \|\delta^{(j)}\|_2/\sqrt{n} \le S\tau_j.$$

For training points, $\max|z_{a,i}^{(j)}| \le K_{\mathrm{src}}\sqrt{\ell}$ and $\max|k_{a,i}^{(j)}| \le SK_{\mathrm{src}}\sqrt{\ell}$. In mobility coordinates $\Theta = (W^{(1)}, \sqrt{n}W^{(2)}, \dots, \sqrt{n}W^{(L)}, w)$, put $F_a = nf_n(v_a)$. Then

$$\max_{a,j,i,v} |(D_\Theta z^{(j)}(v)\nabla_\Theta F_a)_i| \le SU\sqrt{\ell}, \qquad \max_{j,i} |\partial_\zeta z_i^{(j)}(u\cos\zeta + v\sin\zeta)| \le V\sqrt{\ell}$$

for real orthonormal frames $(u, v)$ in the query tube. All four source families

$$h^{(j)}, \qquad W_0^{(j)}h^{(j-1)}, \qquad \delta^{(j)}, \qquad W_0^{(j+1)\top}\delta^{(j+1)},$$

omitting boundary images, have coordinate modulus at most $M_0\sqrt{n}$, where $M_0 = 10\max_j(H_j, \tau_j)$. The real fitting trajectory and its endpoint are those of Lemma 2.1. The proof below also establishes the finite-query and expanding-domain variants. Its probability step is shared; the domain verifications are given last.

**2. Independent stops and deterministic differentiation.** Delete a fixed number $u$ of neurons at one layer, preserving the original normalization $n$ in every rectangular matrix. For each deleted interior neuron, its initialized incoming row transpose $y_i$ and outgoing column $x_i$ are independent $N(0, I/n)$, independent of the retained initialization and other omitted pairs. At the first layer the incoming row instead has covariance $I_d$ and evaluates only its own initial preactivation; at the top there is no outgoing root. Use a separate initial test for each cavity, and set its reference coefficient paths identically zero if it fails. All such paths are defined before integrating an omitted root.

For a nested domain $D_s$, stop each cavity at its own first exit level. Stop every training sample's budget

$$\mathcal{H}_a = \frac{1}{n}\sum_{j,i} \exp\{\eta_*(Z_{a,i}^{(j)} + K_{a,i}^{(j)}/S)\}, \quad Z_{a,i}^{(j)} = \sup_{D_s} |z_{a,i}^{(j)}|, \quad K_{a,i}^{(j)} = \sup_{D_s} |k_{a,i}^{(j)}|, \tag{38}$$

at $\mathcal{B}$ for the full network and $2\mathcal{B}$ for cavities. Also stop operators at ten, contour activity at $2\int \rho|dt| = S$, and imaginary preactivations at $3a/8$ and $7a/16$, respectively. Impose provisional coordinate and response maxima twice the claimed full bounds and four times the cavity bounds, as well as the passive size and row bounds used below. These have strict margins on the successful prefix.

For rectangles use $D_s = sD_1$ and coordinate clipping $P_s$. For the query tube also clip its imaginary great-circle coordinate, retaining every real frame. If a cavity stops at $s_c$, extend a coefficient by $b_c(P_{s_c} z)$. With $a_0(z) = \mathrm{clip}_{[0,T]} \operatorname{Re} z$,

$$P_{s_c} a_0(z) = a_0(P_{s_c} z), \qquad b_c(P_{s_c} z) - b_c(P_{s_c} a_0(z)) = b_c(P_{s_c} z) - b_c(a_0(P_{s_c} z)). \tag{39}$$

Clipping is measurable in retained initialization, is Lipschitz, and fixes the successful prefix. Its real anchors traverse a monotone interval and then freeze. Different cavities are extended separately before subtraction. The expanding domain below has the same properties with a uniformly bounded Lipschitz constant. These extensions need not be holomorphic outside their own domains.

Write $\Theta_0 = \Theta(0)$, $\bar{u} = (\Theta - \Theta_0)/S$, $\bar{F}_a = F_a(\Theta_0 + S\bar{u}, e_a)/S$, $\bar{r}_a = \bar{F}_a/n - y_a/S$, and put bars on backward fields divided by $S$. Here $e_a$ is an additive forward port. The two actual ports of a deletion are $e_a = \sum_i W_{:,i}^{(j+1)} h_{a,i}^{(j)}$ and $\bar{q}_a = \sum_i W_{i,:}^{(j)\top} \bar{\delta}_{a,i}^{(j)}$. The exact retained equation is

$$\dot{\bar{u}} = -\frac{2}{m}\sum_a \bar{r}_a \left(\nabla_{\bar{u}} \bar{F}_a + D_{\bar{u}} h_a^{(j-1)\top} \bar{q}_a\right). \tag{40}$$

At the top add $n^{-1}\sum_i \bar{u}_{w,i} h_{a,i}^{(L)}$ to $\bar{r}_a$, multiplying both terms. The reverse port does not enter the forward residual. The gradient blocks of $\bar{F}_a$ are $S\bar{\delta}_a^{(1)} v_a^\top$, $S\bar{\delta}_a^{(j)} h_a^{(j-1)\top}/\sqrt{n}$, and $h_a^{(L)}$.

The physical recurrences and linear activation growth give (30)'s RMS bounds whenever the operator and activity stops hold. For augmented mobility/port directions $U$, write $Z_a^{(j)} U = Dz_a^{(j)}[U]$. The exact derivative recursion is

$$\begin{aligned} Z_a^{(1)} U &= U_{W^{(1)} v_a} + U_{e^{(1)}}, \\ Z_a^{(j)} U &= U_{\sqrt{n} W^{(j)} h_a^{(j-1)}/\sqrt{n}} + W^{(j)} \operatorname{diag}(\phi'_{j-1}) Z_a^{(j-1)} U + U_{e^{(j)}}. \end{aligned} \tag{41}$$

It is bounded by $P_j$; its activated version is bounded by $f_j$. The Hessian, with all activations

evaluated at the reference state, is

$$\begin{aligned} D^2F_a[U,V] &= U_w^\top \operatorname{diag}(\phi_L')Z_a^{(L)}V + V_w^\top \operatorname{diag}(\phi_L')Z_a^{(L)}U \\ &\quad + \sum_j (Z_a^{(j)}U)^\top \operatorname{diag}(k_a^{(j)}\phi_j'')Z_a^{(j)}V \\ &\quad + \sum_{j\ge 2} \frac{\delta_a^{(j)\top}}{\sqrt{n}}\{U_{\sqrt{n}W^{(j)}} \operatorname{diag}(\phi_{j-1}')Z_a^{(j-1)}V + (U \leftrightarrow V)\}. \end{aligned} \tag{42}$$

For a rectangular matrix let $\|B\|_{p,n} = n^{-1/p}(\sum_i \sigma_i(B)^p)^{1/p}$. The budget implies $\|k_a^{(j)}\|_p/n^{1/p} \le (S/\eta_*)p(2\mathcal{B})^{1/p}$, because $x^p \le (p/e)^p e^x$. Each noncurvature Hessian block factors through at most $n$ coordinates, and each curvature block contains one carrier diagonal. Matrix ideal inequalities in (42) therefore give

$$\begin{aligned} &\|D^2F_a\|_{p,n} \le A_* + (SD_*/\eta_*)p(2\mathcal{B})^{1/p}, \qquad \|D^2F_a\|_{\mathrm{HS}}/\sqrt{n} \le H_*, \\ &\|D^2F_a\|_{\mathrm{op}} \le A_* + (SD_*/\eta_*)\log(2n\mathcal{B}). \end{aligned} \tag{43}$$

These bounds also cover $D_\Theta\delta$, since $\partial_{e^{(j)}}F_a = \delta_a^{(j)}$. Residual mixtures preserve the bounds by $\sum_a |r_a|/(m\rho) \le 1$.

Suppose for now that products of the negative-Gram propagators on ordered disjoint contour pieces have norm at most two, and $\rho/S \le \lambda/8$. The variational generator is

$$-\frac{2}{mn}\sum_a g_a g_a^\top - \frac{2}{m}\sum_a r_a D^2F_a, \qquad g_a = \nabla_\Theta F_a.$$

Its full propagator obeys

$$\|J(t,s)\| \le 2\exp\{SA_* + (S^2D_*/\eta_*)\log(2n\mathcal{B})\} \le Cn^\kappa, \qquad \kappa = 1/4000. \tag{44}$$

Here and only in qualitative local estimates, $C$ and powers of $\ell$ may depend on the fixed problem and deletion count. The coefficient of $\log n$ is bounded by (32); it has not been hidden in a width threshold.

**Claim 2** (Local maps, remainders, and passive observables)**.** *On the independent stops just specified, with contours of length $O(\ell)$, all first insertion maps have norm $n^\kappa \operatorname{poly}(\ell)$. Their dependence on the supremum norm of scalar controls is Lipschitz with that bound, and their raw terminal/time/frame derivatives are bounded by $\sqrt{n}n^\kappa \operatorname{poly}(\ell)$. On the uniform Gaussian event proved below, the retained full/cavity differences in preactivations, scaled carriers, responses and angular derivatives are at most $n^{-1/30}$ per coordinate and at most $2n^{1/100}$ in Euclidean norm, for every fixed deletion count.*

*Proof.* For deterministic controls $\alpha_{a,i}, b_{a,i}$, use ports $e_a = \sum_i x_i\alpha_{a,i}$, $\bar{q}_a = \sum_i y_i b_{a,i}$. The three port derivatives of (40) at zero ports are

$$-\frac{2}{mn}\nabla_{\bar{u}}\bar{F}_a\bar{\delta}_a^{(j+1)\top}, \qquad -\frac{2}{m}\bar{r}_a(D_{\bar{u}}\bar{\delta}_a^{(j+1)})^\top, \qquad -\frac{2}{m}\bar{r}_a(D_{\bar{u}}h_a^{(j-1)})^\top.$$

The first term accounts for the changed residual. Integrating them against (44), using gradient norm $O(\sqrt{n})$, response RMS, and contour length $O(\ell)$, gives an insertion-map bound $\sqrt{u}n^\kappa O(\ell^2)$.

The differentiated forward/backward recursions are exactly

$$\begin{aligned}
z_{[1]}^{(j)} &= SV_{\sqrt{n}W^{(j)}}h^{(j-1)}/\sqrt{n} + W^{(j)}h_{[1]}^{(j-1)} + e_{[1]}^{(j)},\\
h_{[1]}^{(j)} &= \phi_j' \odot z_{[1]}^{(j)},\\
\bar{k}_{[1]}^{(j)} &= W^{(j+1)\top}\bar{\delta}_{[1]}^{(j+1)} + SV_{\sqrt{n}W^{(j+1)}}^{\top}\bar{\delta}^{(j+1)}/\sqrt{n} + \bar{q}_{[1]}^{(j)},\\
\bar{\delta}_{[1]}^{(j)} &= \phi_j' \odot \bar{k}_{[1]}^{(j)} + \phi_j'' \odot \bar{k}^{(j)} \odot z_{[1]}^{(j)}.
\end{aligned}$$

At the first layer replace the matrix product by $SV_{W^{(1)}v}$; at the top $\bar{k}_{[1]} = V_w$. A reference carrier costs $O(\ell)$ once in the inhomogeneous last term. Homogeneous propagation costs $10s$. Induction gives the asserted map bounds; subtracting controls gives their Lipschitz bounds because the linear variational equation is linear in these controls. Differentiating these displayed recursions introduces only the terms $\phi'' z_t$, $\phi''' z_t z_{[1]}\bar{k} + \phi'' z_{[1]}\bar{k}_t + \phi''\bar{k}z_{[1],t}$, and ordinary product derivatives. There is at most one raw reference coordinate derivative, bounded by $\sqrt{n}\,\mathrm{poly}(\ell)$. This proves the derivative claim without a width power depending on depth. Cauchy's formula on the strict strip bounds $\phi'''$ by $4\beta^2$.

Here are the nonlinear estimates needed to justify this linearization. Decompose the augmented displacement into $(V, e) + (U, 0)$, and let $N$ bound its linear Euclidean variations, $d_0$ their coordinate maxima, and $u_0$ bound $\|U\|$. Put

$$R = d_0 N + d_0 u_0 + u_0^2, \qquad P = (N + u_0)^2/\sqrt{n}.$$

Assume $N \geq 1$, $d_0, u_0 \leq 1$, $N + u_0 \leq \sqrt{n}$, and $u_0 + R + P \leq 1$. If $E_g = g_1 - g_0 - Dg_0(V + U, e)$ and $g_i = \phi'(z_i)$ in the third line below, the exact identities are

$$\begin{aligned}
E_{z^{(j)}} &= W_1^{(j)} E_{h^{(j-1)}} + S\Delta\bar{u}_{\sqrt{n}W^{(j)}} h_{[1]}^{(j-1)}/\sqrt{n},\\
E_{\bar{k}^{(j)}} &= W_1^{(j+1)\top} E_{\bar{\delta}^{(j+1)}} + S\Delta\bar{u}_{\sqrt{n}W^{(j+1)}}^{\top}\bar{\delta}_{[1]}^{(j+1)}/\sqrt{n},\\
E_{\bar{\delta}} &= g_1 \odot E_{\bar{k}} + (g_1 - g_0) \odot \bar{k}_{[1]} + \bar{k}_0 \odot (g_1 - g_0 - \phi''(z_0) z_{[1]}).
\end{aligned} \tag{45}$$

The first and top boundary remainders vanish. Forward and backward first derivatives are bounded by $J_f = \beta^{10L}$ and $J_b = \beta^{20L}(1 + M)$, where $M \geq 1$ is a common provisional bound for the absolute preactivations and scaled carriers of the reference and full paths, including the deleted-neuron values that define the scalar controls. Such an envelope is $\mathrm{poly}(\ell)$ under the stated stops. The localized product estimate is $\|z_{[1]}^{\odot 2}\|_2 \leq 3J_f^2 R$. Insert $z_0 + z_{[1]}$ into activation subtraction. Taylor's formula and (45), summed by geometric propagation, give

$$\max(\|E_z\|_2, \|E_h\|_2) \leq \beta^{30L}(R + P), \qquad \max(\|E_{\bar{k}}\|_2, \|E_{\bar{\delta}}\|_2) \leq \beta^{60L}(1 + M)(R + P). \tag{46}$$

Indeed the two gate cross terms cost at most $\beta[3J_f J_b R + \beta^{30L}(R + P)(d_0 + J_b u_0)]$ and $M[\beta\|E_z\| + \beta^2\|z_{[1]}^{\odot 2}\|/2]$; the matrix cross term costs $J_b P$. No changed carrier is used as a presumed reference bound.

The hidden gradient remainder is

$$\frac{S}{\sqrt{n}}[E_{\bar{\delta}} h_1^{\top} + \bar{\delta}_{[1]}(h_1 - h_0)^{\top} + \bar{\delta}_0 E_h^{\top}],$$

bounded by $\beta^{70L}(1 + M)(R + P)$. The Hessian identity along partial joining segments bounds the residual difference by $\beta^{13L}(N + u_0)/\sqrt{n}$ and its linearization error by $\beta^{87L}(1 + M)(N + u_0)^2/(2n)$. For a product of residual and gradient, its exact remainder is $\bar{r}_0 E_g + (\Delta\bar{r} - D\bar{r}_0\Delta)g_0 + \Delta\bar{r}(g_1 - g_0)$.

For the reverse force one additionally needs a lower adjoint probe. For an omitted root $y_i$, include all derivatives of $y_i^\top h^{(j-1)}$ in the Gaussian coordinate event. For $q = \sum_i y_i b_{a,i}$, let $Q = 2u\beta M$, $P_q = u\beta M d_0$. Subtracting its adjoint recursion gives

$$\max_j \|k_{q,1}^{(j)} - k_{q,0}^{(j)}\|_2 \le \beta^{20L}(P_q + Q/\sqrt{n})(N + u_0),$$

since the changed gate multiplies the reference probe maximum, the changed matrix costs $Q(N + u_0)/\sqrt{n}$, and propagation costs $11s$. Integrating the corresponding Hessian gives

$$\|(Dh_1 - Dh_0)^\top q\|_2 \le \beta^{50L}(P_q + Q/\sqrt{n})[N + u_0 + (N + u_0)^2].$$

Combining these estimates in (40) yields the vector-field remainder bound

$$\beta^{240L}(1 + u)(1 + M)\{\bar\rho[R + d_0(N^2 + Nu_0 + u_0^2)] + (1 + \bar\rho)P\}, \qquad \bar\rho = \rho/S. \tag{47}$$

This also holds on the complex joining segments: the sufficient gate $d_0+\beta^{10L}u_0+\beta^{30L}(R+P) < a/32$ keeps the segments and auxiliary Taylor points inside $15a/32$. A first-exit argument proves the gate's use before Taylor expansion; the factor $\beta^{240L}$ pays for the Cauchy third-derivative bound. Scaling introduced no inverse power of $S$: in the effective network with readout $\bar u_w$, the affine change of variables multiplies hidden directions by $S \le 1$.

For every existing outgoing column, and for an incoming row at a layer $j \ge 2$, integrating the omitted-parameter equations gives

$$\|\Delta W_{:,i}^{(j+1)}\| \le S^2(\max_r \tau_r)(b + sM)/\sqrt{n}, \qquad \|\Delta W_{i,:}^{(j)}\| \le S^2 H_{\max} sM/\sqrt{n}.$$

At the first layer the incoming-row estimate is instead $\|\Delta W_{i,:}^{(1)}\| \le sS^2M$. There is no retained reverse port below that layer: this row's motion is included in the deleted neuron's scalar controls, rather than in the learned retained-port remainder. Thus every existing learned port entering the retained equation still induces a force at most $S^2\beta^{260L}(1+u)^2(1+M)^3(1+\bar\rho)/\sqrt{n}$. The top residual offset is at most $uM(b+sM)/n$; its product with both retained terms has the same enclosing bound.

It remains to specify all passive quantities in the local claim. For evaluated input $q$ and training driver $a$, let $C_q^{(j)} = D_\Theta h_q^{(j)}$, $\bar R_{qa}^{(j)} = S^{-1}D_\Theta z_q^{(j)} g_a$, and $\bar Q_{qa}^{(j)} = \phi_j' \odot \bar R_{qa}^{(j)}$. Then

$$\bar R_{qa}^{(1)} = \bar\delta_a^{(1)} v_a^\top q, \qquad \bar R_{qa}^{(j)} = \bar\delta_a^{(j)} c_{aq}^{(j-1)} + W^{(j)}\bar Q_{qa}^{(j-1)}, \qquad c_{aq}^{(j-1)} = h_a^{(j-1)\top} h_q^{(j-1)}/n. \tag{48}$$

Angular fields satisfy $J^{(1)} = W^{(1)}q'$, $J^{(j)} = W^{(j)}I^{(j-1)}$, $I = \phi' \odot J$. Differentiate these product formulas: the four response terms are $\bar\delta_{[1]}c + \bar\delta c_{[1]}$, $SV_{\sqrt{n}W}\bar Q/\sqrt{n}$, and $W\bar Q_{[1]}$, together with the omitted port $\sum_i x_i \bar Q_{qa,i}$. The activated variation adds $\phi''\bar R z_{[1]}$. Angular variations have the same matrix and gate terms, with omitted port $\sum_i x_i I_i$. Below a deletion there is also the direct reverse observable

$$S^{-1}(Q_{qa,\mathrm{full}} - C_q g_a) = C_q C_a^\top \bar q_a. \tag{49}$$

Include the forward images of $C_a^{0\top} y_i$ in the Gaussian event; their coefficient maps are independent of that root.

Product subtraction gives

$$|c_{[1]}| \le \beta^{10L}(N + u_0)/\sqrt{n}, \qquad |E_c| \le \beta^{40L}[(R + P)/\sqrt{n} + (N + u_0)^2/n].$$

The exact response remainder is $E_{\bar\delta}c_1 + \bar\delta_{[1]}(c_1 - c_0) + \bar\delta_0 E_c + W_1 E_{\bar Q} + \Delta W\bar Q_{[1]} + E_{\mathrm{ports}}$; its gate remainder is the third line of (45) with $\bar k$ replaced by $\bar R$. The angular case uses that same identity

with $J$. Reference response RMS cancels the $1/\sqrt{n}$ in $E_c$. For $\xi = C_a^{0\top} y_i$, forward directional subtraction gives $\|(C_q^1 - C_q^0)\xi\| \le \beta^{50L}(d_0 + n^{-1/2})(N + u_0)$. The other changed factor in (49) uses the adjoint-probe estimate above. Their sum is bounded by

$$\beta^{70L}(1+u)(1+M)(d_0+n^{-1/2})[N+u_0+(N+u_0)^2].$$

Missing feature products cost $u(b+sM)^2/n$, and learned ports cost $n^{-1/2}\,\mathrm{poly}(\ell)$. These enumerate all additional remainders. Their time and frame derivatives have the same product structure already checked. The probabilistic and first-exit conclusion follows in the next paragraph. □

**The uniform Gaussian event and closing the local comparison.** For fixed controls, a pairing of an omitted root with a linear variation has the form $g^\top R g$, where $R = (E_i^\top L + L^\top E_i)/2$. Consequently $\|R\| \le \|L\|$, and only its same-root diagonal block has a mean $\operatorname{tr} R/n$. Cross-root forms are centered. On $\|g\| \le 2\sqrt{2u}$, interpolation of a centered form costs at most $10u\|\Delta R\|$, including its trace. All stopped scalar controls have polynomial-logarithmic amplitudes and speeds $\sqrt{n}\,\mathrm{poly}(\ell)$; this follows by differentiating the forward, backward and response recursions above. A scalar control with amplitude $M_c$, speed $D_c\sqrt{n}$ on a contour of length $T_c$ has an $n^{-1/8}$-net of logarithmic cardinality at most

$$(2 + 8T_cD_cn^{5/8})\log(1+8M_cn^{1/8}).$$

Indeed sample it at spacing $n^{-1/8}/(8D_c\sqrt{n})$, round on mesh $n^{-1/8}/4$, and interpolate. Apply this separately to the finitely many real and imaginary controls. They are deterministic one-dimensional histories on a prescribed contour; no arbitrary two-dimensional analytic continuation of controls is assumed.

After eventual absorption of the fixed logarithmic factors, all linear and quadratic maps have norm at most $n^{1/200}$. A linear coordinate has variance at most $n^{-99/100}$. For a real symmetric form, diagonalization and

$$\mathbb{E}e^{t(G^\top RG - \operatorname{tr} R)} = \prod_i e^{-t\lambda_i}(1-2t\lambda_i)^{-1/2}, \qquad -\log(1-v) - v \le \frac{v^2}{2(1-|v|)}$$

give, for $g \sim N(0, I/n)$,

$$\Pr\{|g^\top Rg - \operatorname{tr} R/n| > t\} \le 2\exp\left[-c\min\left\{\frac{n^2t^2}{\|R\|_{\mathrm{HS}}^2}, \frac{nt}{\|R\|_{\mathrm{op}}}\right\}\right]$$

with an absolute $c > 0$, by applying exponential Markov with $|2t\lambda_i| \le 1/2$ and optimizing the parameter. Here $\|R\|_{\mathrm{HS}} \le \sqrt{2un}\|R\|$. At threshold $n^{-1/10}/8$, real/imaginary splitting gives $4e^{-n^{.78}}$ eventually. Root-norm failure is at most $e^{-un}$. The control interpolation error is $n^{-1/8+1/200}\,\mathrm{poly}(\ell) = o(n^{-1/10})$. A terminal/time/frame mesh of spacing $n^{-2}/\,\mathrm{poly}(\ell)$ has polynomial size; the sphere frame has $2d$ ambient coordinates. Together with all coordinates, probes and at most $uLn^u$ deletion sets, the logarithmic union cost is eventually less than $n^{.65}$. Thus the uniform event fails with probability at most $e^{-n^{.7}}$. Only after this event is established are the actual adaptive controls substituted. Root-norm bounds also make the total linear vector variation at most $n^{1/100}$, distributing a fixed share among the finitely many components.

Let $U$ be the retained scaled displacement minus its linear variation. It satisfies Duhamel's formula with (47), the learned ports, and the top offset. Set $N = n^{1/100}$, $d_0 = n^{-1/10}$, $u_0 = n^{-1/25}$. On a first-exit prefix $\|U\| \le u_0$,

$$\|U\| \le n^{\kappa}\,\mathrm{poly}(\ell)[n^{-.08} + u_0^2 + n^{-.48} + n^{-1/2}] = o(u_0). \tag{50}$$

The activity integral bounds the residual-weighted terms and the contour length bounds the others. This strictly improves the stop. The explicit remainders then give coordinate discrepancies $n^{-1/30}$ and vector discrepancies $2n^{1/100}$, including passive responses and angular fields. The joining-strip gate also holds eventually. For every deterministic frozen control history, the retained linear variations are centered Gaussian images conditional on the cavity. After substituting adaptive controls, we use only the uniform bounds already proved; we assert neither Gaussianity nor centering of the resulting variations. Pairing a frozen-control variation again with an omitted root produces a quadratic form. Its nonzero trace is retained before adaptive substitution, as treated next. This completes Claim 2.

**3. Trace absorption and Gaussian moments.**

**Every noncentered contraction.** Throughout this paragraph, conditional means and centered forms refer to deterministic frozen controls, conditional on the cavity. The resulting trace bounds hold uniformly over those controls and are then evaluated at the actual adaptive controls. For one interior omitted neuron work in unscaled mobility coordinates, and set $C_a = D_\Theta h_a^{(j-1)}$, $B_a = D_\Theta \delta_a^{(j+1)}$, $E_a = D_{e_a^{(j+1)}} \delta_a^{(j+1)}$, all at the zero-source cavity. With $\alpha_b = h_{b,i}^{(j)}$, $b_b = \delta_{b,i}^{(j)}$, and $d_b = \delta_b^{(j+1)}$, the linear parameter variation is

$$V(t) = -\frac{2}{m} \sum_b \int_0^t J(t,s)\{g_b(s) d_b(s)^\top x_i \alpha_b(s)/n + r_b(s) B_b(s)^\top x_i \alpha_b(s) + r_b(s) C_b(s)^\top y_i b_b(s)\}\, ds.$$

The scalar expansions are

$$z_{a,i} = y_i^\top h_a^0 + y_i^\top C_a V + (\Delta W_{i,:}) h_a + o(1), \qquad k_{a,i} = x_i^\top \delta_a^0 + x_i^\top B_a V + x_i^\top E_a x_i \alpha_a + (\Delta W_{:,i})^\top \delta_a + o(S).$$

The errors follow in scaled coordinates from Claim 2. The four nonzero conditional means are

$$-\frac{2}{m} \sum_b \int r_b b_b \frac{\operatorname{tr}(C_a J C_b^\top)}{n}\, ds, \qquad -\frac{2}{m} \sum_b \int r_b \alpha_b \frac{\operatorname{tr}(B_a J B_b^\top)}{n}\, ds,$$
$$\alpha_a \operatorname{tr}(E_a)/n, \qquad -\frac{2}{mn^2} \sum_b \int \alpha_b d_b^\top B_a J g_b\, ds. \tag{51}$$

All other forms are centered same-root or independent-root forms covered by the uniform Gaussian event. The last displayed mean is $n^{-1+\kappa} \operatorname{poly}(\ell) = o(1)$, since $\|d_b\|, \|g_b\| = O(\sqrt{n})$.

Expand $J$ around its negative-Gram base. A term with $h$ residual-Hessian insertions has ordered activity integral at most $S^h/h!$. The product of its base propagators costs at most two. Normalized Schatten Hölder, with exponent $h+2$ on both endpoint Hessian blocks and the $h$ insertions, gives

$$\frac{2S^h}{h!}[A_* + (SD_*/\eta_*)(h+2)(2\mathcal{B})^{1/(h+2)}]^{h+2}.$$

The zero-insertion term is at most $2H_*^2$. Split the other powers by $(x+y)^{h+2} \le 2^{h+1}(x^{h+2} + y^{h+2})$. The bounded part sums to $4A_*^2(e^{2SA_*} - 1)$; the carrier part is

$$\frac{8e^2 D_*^2 S^2 \mathcal{B}}{\eta_*^2} \sum_{h \ge 1} (2eD_* S^2/\eta_*)^h (h+2)^2 \le \frac{576 e^3 D_*^3 \mathcal{B} S^4}{\eta_*^3}.$$

Indeed $h! \ge (h/e)^h$, $(1 + 2/h)^h \le e^2$, and $\sum_{h \ge 1} q^h (h+2)^2 \le 36q$ for $q \le 1/2$, by differentiating the geometric series. The allowance gives this last condition. For forward endpoints give one exponent

two, the other infinity, and the $h$ Hessians exponent $2h$. The two series are bounded by $e^{2SA_*}-1$ and $\sqrt{2\mathcal{B}}\sum_{h\ge1}(4eD_*S^2/\eta_*)^h$, each at most one by (32). Including the zero term,

$$|\operatorname{tr}(C_aJC_b^\top)|/n \le 8f_*^2. \tag{52}$$

Reciprocal exponents sum to one, giving normalization exactly $n^{-1}$, independent of parameter-space dimension. The direct port Hessian is $E_a=\sum_{r>j}T_{r,j+1}^\top \operatorname{diag}(k_a^{(r)}\phi_r'')T_{r,j+1}$, with $\|T_{r,j+1}\|\le(10s)^{r-j-1}$; hence carrier RMS gives $|\operatorname{tr}E_a|/n\le SE_*$.

Let $G_{h,a,i}$, $G_{\delta,a,i}$ denote suprema of the corresponding cavity Gaussian pairings. Set $Z_i=(m^{-1}\sum_a Z_{a,i}^2)^{1/2}$ and $U_i=(m^{-1}\sum_a(K_{a,i}/S)^2)^{1/2}$. At every driving time Cauchy–Schwarz gives $m^{-1}\sum_b|r_bh_{b,i}|\le\rho(b+sZ_i)$ and $m^{-1}\sum_b|r_b\delta_{b,i}|\le\rho sSU_i$. Integrating the rank-one updates and pairing retained factors in RMS, the learned outgoing column costs $s^2k_*^2S^3$ times its driving amplitude, and the learned incoming row costs $sS^2U_iH_{j-1}^2$; at the first layer the latter is $sS^2U_i$. Thus (51) gives

$$\begin{aligned} K_{a,i}/S &\le G_{\delta,a,i}/S+(D_0+D_1\mathcal{B}S^4/\eta_*^3)(1+Z_{a,i}+Z_i)+o(1),\\ Z_{a,i}&\le G_{h,a,i}+C_FS^2U_i+o(1). \end{aligned} \tag{53}$$

The direct external trace keeps the evaluated sample's amplitude $Z_{a,i}$; only integrated terms use the driving-sample RMS. At the first layer $G_{h,a,i}=|W_{0,i}^{(1)}v_a|$; at the top $\sup|w_i|/S\le b+sZ_i$ and there is no outgoing root. These endpoint formulas obey the same enclosing inequalities.

Put $\overline{G}_h=(m^{-1}\sum_a G_{h,a,i}^2)^{1/2}$ and $\overline{G}_\delta=(m^{-1}\sum_a(G_{\delta,a,i}/S)^2)^{1/2}$. The backward coefficient is at most $2D_0$ and $D=C_FS^2\le1/(8D_0)$. Taking sample RMS gives

$$U_i\le\overline{G}_\delta+2D_0(1+2Z_i)+o(1),\qquad Z_i\le\overline{G}_h+DU_i+o(1).$$

Since $4D_0D\le1/2$, solve these two inequalities and substitute back sample by sample to obtain

$$Z_{a,i}+K_{a,i}/S\le C_{\mathrm{abs}}(1+G_{h,a,i}+G_{\delta,a,i}/S+\overline{G}_h+\overline{G}_\delta)+o(1). \tag{54}$$

This coefficient is independent of sample and deletion counts.

**Real reference moments.** Parameterize a real stopped cavity by its own normalized activity $u=2\int_0^t\rho/S\in[0,1]$, freezing after its endpoint. For $v=|u-u'|$, integrating the updates gives

$$\begin{aligned} \|\Delta w\|_2/\sqrt{n}&\le H_LSv,\qquad \|\Delta W^{(1)}\|_F/\sqrt{n}\le\tau_1S^2v,\\ \|\Delta W^{(j)}\|_F&\le\tau_jH_{j-1}S^2v,\qquad \|\Delta z_a^{(j)}\|_2/\sqrt{n}\le V_j^0S^2v. \end{aligned}$$

The feature difference has the last bound multiplied by $s$. Split a reference carrier at $SR$. Its high part satisfies $\|k\mathbf{1}_{|k|>SR}\|_2/\sqrt{n}\le(4S/\eta_*)\sqrt{2\mathcal{B}}e^{-\eta_*R/4}$, by the budget and $x^2e^{-x/2}\le16$. Choose $R=(4/\eta_*)\log[8\sqrt{2\mathcal{B}}/(\eta_*v)]$ when $v>0$. The high changed-gate part is at most $sSv$; the low part is at most $t_2V_j^0S^3Rv$. Since $R\le\eta_*^{-1}(10\Lambda_*+4\log(1/v))$, $S^2\Lambda_*/\eta_*\le1$, and $v\log(1/v)\le2\sqrt{v}/e$, the low part is bounded by $14t_2V_j^0S\sqrt{v}$. At the top add $sH_LSv$; below it add the changed-matrix term $s^3k_{j+1}^2H_jS^3v$ and propagated difference. This proves

$$\|\delta_a^{(j)}(u)-\delta_a^{(j)}(u')\|_2/(S\sqrt{n})\le G_j^0\sqrt{|u-u'|}. \tag{55}$$

The top carrier was included; no coordinate readout bound was used.

Conditioned on the cavity, a Gaussian reference has bounded initial variance and this square-root activity modulus. Grids of spacing $4^{-q}$ have at most $2 \cdot 4^q$ increments of standard deviation at most $2G2^{-q}$. Gaussian tails at thresholds proportional to $2^{-q}\sqrt{z^2 + 4(q+1)}$, summed over levels and the initial value, give

$$\Pr\{X > W_G(1+z) \mid \text{cavity}\} \le 4e^{-z^2}, \qquad \mathbb{E}[e^{X^2/(16W_G^2)} \mid \text{cavity}] < 2.$$

The factor 128 in $W_G$ bounds the geometric threshold sums. For the moment, $(1+z)^2 \le 2(1+z^2)$ and tail integration give $\mathbb{E}e^{Z^2/8} \le 1 + 4/7$, with $e^{1/8}(1+4/7) < 2$. Jensen gives the same bound for sample RMS: $e^{\overline{X}^2/(16W_G^2)} \le m^{-1}\sum_a e^{X_a^2/(16W_G^2)}$, without sample independence. Completing the square and Hölder for the four terms in (54) give the one-neuron moment base

$$2\exp\{qC_{\text{abs}} + 64q^2C_{\text{abs}}^2W_G^2\} < 4, \qquad 0 \le q \le 8\eta_*. \tag{56}$$

The chosen $\eta_*$ verifies the last inequality.

**Claim 3** (Local Gaussian rectangle bound)**.** *Let a deterministic complex coefficient map on a rectangle of sides at most $H$ satisfy $\sup_z \|b_z\| \le D$ and $\|b_z - b_{z'}\| \le K\|z - z'\|$. For a standard real Gaussian vector $G$, put $X = \sup_z |G^\top b_z|$. Then*

$$M = 256D\sqrt{\log(e + HK/D)}, \qquad \log \mathbb{E}e^{qX} \le qM + 2q^2D^2 \quad (q \ge 0).$$

*For sample RMS with the same bounds, replace $M$ by $M + 2D$. If $D = 0$, the process is zero. For covariance $I/n$, divide coefficient norms by $\sqrt{n}$.*

*Proof.* Use dyadic coefficient meshes with spacing proportional to $D2^{-j}/K$, having at most $4(1 + HK/D)^2 4^j$ points. Connect each point to the preceding mesh; increment standard deviations are at most $3D2^{-j}$. Gaussian exponential moments give $\mathbb{E}\max_{r \le N}|Z_r| \le \sigma\sqrt{2\log(2N)}$ when standard deviations are at most $\sigma$. Summing uses $\sum_{j\ge1} 2^{-j} = 1$, $\sum_{j\ge1} 2^{-j}\sqrt{j} \le 2$. A factor 128 covers the real process, and real/imaginary splitting gives 256.

An $a$-Lipschitz Gaussian function satisfies $\log \mathbb{E}e^{t(F - \mathbb{E}F)} \le a^2t^2/2$ and $\operatorname{Var} F \le a^2$. Here is a derivation. For $P_tg(x) = \mathbb{E}g(e^{-t}x + \sqrt{1 - e^{-2t}}G)$, differentiating invariant entropy and Gaussian integration by parts give

$$\operatorname{Ent}_\gamma(g) = \int_0^\infty \mathbb{E}\frac{|\nabla P_t g|^2}{P_t g}\,dt \le \tfrac{1}{2}\mathbb{E}\frac{|\nabla g|^2}{g}.$$

Use $\nabla P_t g = e^{-t}P_t\nabla g$ and weighted Cauchy–Schwarz for the inequality. Apply it to $g = e^{uF}$ and integrate the resulting derivative inequality for $u^{-1}\log \mathbb{E}e^{uF}$. Truncation and smoothing extend the calculation to Lipschitz functions; their linear growth supplies exponential integrability. Differentiating at zero gives the variance. Here $X$ is at most $2D$-Lipschitz, proving the estimate. The sample RMS is also $2D$-Lipschitz; its mean is at most $M + 2D$ by the variance bound and Cauchy–Schwarz. Independence among sample processes is unnecessary. □

**Initial cavities and maximum improvements.** The initialization event of Lemma 2.1 applies jointly to fixed-size rectangular cavities. To see why weak covariance estimates need not be union-bounded over deletion sets, zero-embed their features. On the full initial operator event, conditional fresh-row Gaussian tails give initial training maxima $Z_0 = C\sqrt{\ell}$ with probability tending to one. Deleting $u$ coordinates changes each later initialized feature vector by at most

$$D_{\text{init}} = (8s)^L\sqrt{u}(b + sZ_0) = O_u(\sqrt{\ell}).$$

Normalized feature and singular-value changes therefore vanish uniformly over deletion sets, transferring the strict fitting margins. For coordinate localization above a deletion, project the complete lower feature difference onto this deterministic ball before pairing with the independent current row. At threshold $n^{-1/10}$, the tail $2\exp[-n^{4/5}/(2D_{\mathrm{init}}^2)]$ pays for the $O_u(n^{u+1})$ tests. A cavity's covariance retains normalization $n$; it is not identified with the full covariance. Initial budgets are below $\mathcal{B}/2$ with probability tending to one: their conditional means are below $2e^{\eta_*^2 H_{\max}^2/2} < 3$, their second exponential moments are bounded, and conditional Chebyshev applies layer by layer. The coordinate-small deletion comparison transfers this event to all cavities. Their own initial tests remain part of each globally extended reference's definition.

Under the provisional stops, Gaussian coefficient RMS is bounded by $H_{\max}$ or $S\max_j \tau_j$. A mesh of spacing $n^{-2}$ has polynomial size, with vanishing off-grid error from the normalized coefficient derivative bounds. Complex Gaussian splitting has tail $4e^{-u^2/(4\sigma^2)}$. The multiplier $C_G$, followed by (54), improves the full training stops to

$$\max|z_{a,i}^{(j)}| \le K_{\mathrm{src}}\sqrt{\ell}, \qquad \max|k_{a,i}^{(j)}| \le SK_{\mathrm{src}}\sqrt{\ell}. \tag{57}$$

There are at most $n^7$ tests eventually on the original rectangle and $n^{10}$ on the expanding finite-panel rectangle; the factor 32 leaves strict margin. This step uses Gaussian tails on a mesh, not already removed budgets or Gaussian supremum moments.

For response and angular stops, differentiate $Q_{qa} = C_q g_a$: $DQ_{qa} = C_q Dg_a + (DC_q)[\cdot]g_a$. The first term has normalized Hilbert–Schmidt norm at most $f_j H_*$. Curvature, changed matrix, changed feature and propagation in the second give respectively $t_2 r_j^* P_j, sq_{j-1}^*, s\tau_j H_{j-1} f_{j-1}, 10se_{j-1}$. These are the recurrence for $e_j$. Angular fields give the recurrence for $a_j^*$, including the first-layer map $U_{W^{(1)}}q'$ of normalized norm at most two. Apply (52) with these endpoints. The four noncentered response terms are exactly the direct feature pairing, the direct reverse term (49), the integrated mixed endpoint, and the learned incoming row. After division by $S$, their coefficients are

$$sK_{\mathrm{src}}H_{j-1}^2, \quad sK_{\mathrm{src}}f_{j-1}^2, \quad sK_{\mathrm{src}}ST_Q, \quad sK_{\mathrm{src}}S^2 H_{j-1}q_{j-1}^*.$$

The centered incoming pairing contributes $Gq_{j-1}^*$. Angular terms are $Gb_{j-1}^*$ and $sK_{\mathrm{src}}S^2(T_J + H_{j-1}b_{j-1}^*)$. Thus the factor two in (34) strictly improves these stops. For sphere frames, at most $n^{4d+10}$ mesh points require $G = 16\sqrt{d+3}$; polar normalization of nearby frames gives bounded frame derivatives for interpolation. A fixed finite input list has no frame mesh and at most $n^{10}$ tests eventually; $G = 64$ gives failure $4n^{10}e^{-1024\ell}$. The number of passive inputs affects only the qualifying width.

Use the provisional passive-preactivation cap $2(GH_{\max} + 1 + U_0S^2 + a)\ell^2$, where $U_0$ is the applicable response bound. Its initial maximum is $(GH_{\max}+1)\sqrt{\ell}$, by the same fresh-row argument; real activity adds at most $U_0S^2\sqrt{\ell}$. The domain checks below control the complex displacement and improve this stop without a passive exponential budget.

**Claim 4** (Local time derivative under provisional stops)**.** *Let $U_0 = U$ or $U_{\mathrm{fin}}$, as appropriate, and put $N_* = \max_j\{r_j^*, 4U_0\}$, $B_n = 1 + (S^2/\eta_*)\log(e+\ell)$. Define*

$$J_L = 2sH_L + 4et_2N_*, \qquad J_j = 10sJ_{j+1} + 2s^3k_{j+1}^2H_j + 4et_2N_*, \qquad J_* = \max_j J_j.$$

*For $\ell \ge \max(e^2, 2\mathcal{B})$, on each independently stopped cavity,*

$$\|\dot{h}_a^{(j)}\|_2/\sqrt{n} \le 2\rho Sq_j^*, \qquad \|\dot{\delta}_a^{(j)}\|_2/\sqrt{n} \le J_*B_n\rho.$$

*These bounds use the provisional response stops, not their improvement.*

*Proof.* The exact response identity gives $\|\dot z\|_2/\sqrt{n} \le 2\rho S r_j^*$, $\|\dot z\|_\infty \le 8\rho S U_0\sqrt{\ell}$. Interpolate to exponent $2v/(v-2)$ and use the carrier $v$-norm supplied by (38). For $v \ge 4$,

$$\|k \odot \dot z\|_2/\sqrt{n} \le (2\rho S^2 N_*/\eta_*) v (2\mathcal{B}\ell)^{1/v}.$$

Choose $v = \max(4, \log(2\mathcal{B}\ell))$. The exponential factor is at most $e$, and $v \le 2\log(e+\ell)$. Backward differentiation now gives the readout term $2sH_L$, propagated derivative, changed-matrix term $2s^3 k_{j+1}^2 H_j$, and changed-gate term $4et_2N_*$. They are the displayed recursion. Forward differentiation gives the first bound. This growing real exponent is a deterministic budget inequality and does not use growing deletion sets. □

**Removing budgets on complete independently stopped paths.** The domain checks below will supply a bounding rectangle of side $O(\ell)$, uniformly Lipschitz retractions commuting with real anchors, and normalized complex-minus-real coefficient radius $D_n = O(B_n/\sqrt{\ell})$ with global modulus $O(B_n)$. Claim 3 then gives mean $O((\log(e+\ell))^{3/2}/\sqrt{\ell})$ and vanishing tail radius. Every fixed exponential moment of these corrections tends to one. Fixed squared-exponential moments do also, by integrating the sub-Gaussian tail; sample RMS uses Jensen. Notice that the modulus here is polylogarithmic, not an arbitrary polynomial in $n$.

For singleton/common-cavity differences, first extend each path by its own retraction, then project their difference onto the deterministic ball of radius $4n^{1/100}$. Both paths omit the paired root. Projection preserves this independence and equals the actual difference on the successful prefix. Its normalized Gaussian radius is $4n^{-49/100}$, and its modulus is at most a fixed power of $n$, by the already proved derivative bounds. The Gaussian rectangle claim again gives moments tending to one at every fixed order. The two coefficient paths need not be independent of each other.

Fix a positive integer moment order $u$, a sample and a layer. Distinct roots in an empirical $u$th moment are independent given their common cavity. Apply Cauchy–Schwarz to separate products of main real references from products of corrections, and then Hölder over the finitely many roots and four reference families for the corrections. Main moments use exponent at most $8\eta_*$ in (56); corrections use fixed orders proportional to $u$ and tend to one. The limiting per-root base is therefore below sixteen. Drop full-event indicators only after dominating by these nonnegative, globally defined cavity bounds. A tuple with $k < u$ distinct roots has normalized multiplicity $O_u(n^{k-u})$. On (57), repeated weights cost at most $e^{2u\eta_* K_{\mathrm{src}}\sqrt{\ell}} = n^{o(1)}$, so collision contributions vanish. Local exceptional events are superpolynomially small, and the stopped budgets are bounded.

Cavities cannot stop first on the successful full prefix: their coordinate differences are $n^{-1/30}$, maxima and poles have strict transferred margins, and

$$\mathcal{H}_a^{\mathrm{cavity}} \le e^{2\eta_* n^{-1/30}} \mathcal{H}_a + O_u(1/n) < 2\mathcal{B}.$$

Operator and activity margins are established in the domain checks. If a full budget is hit, some sample/layer mean is at least $\mathcal{B}/L$. Markov's inequality and the fixed sample/layer union give

$$\limsup_{n\to\infty} \Pr\{\text{a full budget is hit}\} \le mL(16L/\mathcal{B})^u. \tag{58}$$

Take the width limit at each fixed $u$, then the infimum over positive integers. Since $16L/\mathcal{B} < 1$, the result is zero. The label allowance precedes the moment order and does not shrink with confidence or sample count. This is the shared probability step.

**4. Verification of the working domains.**

*Verification of the working whole-sphere estimates.* Lemma 2.1 independently supplies the real trajectory, strict operator caps below nine, and real activity at most $S/2$. The same holds for the independently tested cavities. Impose the eventual deterministic gates

$$n^{-1} \le \min(Y, S, 1), \qquad \ell \ge \max(e^2, 2\mathcal{B}), \qquad \sqrt{\ell} \ge c_t \max\{8, \lambda, 4\mathcal{K}/\log 2, 32YSD_W\}. \tag{59}$$

Follow the real solution to its nearest anchor in $[0, T]$, then at most two short contour pieces of total length $2r_t$. The algebraic residual Gram and the negative-Gram variational generator, divided by their factor two, have norm at most $\mathcal{K}$, by the explicit gradient blocks. No complex positivity is used. Residual growth and products of the base propagators cost at most $e^{4\mathcal{K}r_t} \le 2$. Additional activity is at most $8Yr_t \le S/2$; normalized hidden increments are at most $8YSD_W r_t \le 1/4$. Thus operators have strict margins inside ten and $\rho/S \le \lambda/8$. The total contour length is $O(\ell)$.

These are the premises of the local insertion and trace argument, which improves maxima and response/angular caps. The exact velocity $\partial_t z^{(j)} = -(2/m)\sum_a r_a R_a^{(j)}$ now bounds the complex-time preactivation displacement by $8c_tYSU = a/8$. Every query in the intrinsic tube has the form $u\cosh h + iv\sinh h$ for a real orthonormal frame and $|h| \le r_q$. Its additional displacement is at most $c_qV \le a/8$. Thus the full pole stop improves strictly; the local coordinate comparison transfers this margin to cavities. The passive size stop also improves as already shown.

Under the provisional response stops, Claim 4 bounds normalized complex-minus-real reference radii by $Cr_tB_n = O(B_n/\sqrt{\ell})$, and their global time moduli by $CB_n$. This holds on complete independently stopped paths: the clipped arguments commute with the real-anchor map as in (39). Their parameter rectangle has side $O(\ell)$. The budget-removal argument therefore applies, with probability of failure tending to zero. The logical order is provisional deterministic bounds, local insertion, maximum/response improvement, strict pole/cavity transfer, and then complete-path Gaussian moments and budget removal. None of these steps conditions an omitted Gaussian root on full-network survival.

Before each stop, finite-width bounded parameters and strict separation from the activation singularities give local holomorphic ODE continuation. Compactness and uniqueness glue the local continuations on the nested rectangles. At a first exit the strict derivative strip allows one-sided estimates; the strict improvements continue the solution across the closed boundary. Query holomorphy follows by forward composition on the quadric. Passive backward fields are then holomorphic by their finite downward recursion, with RMS bounds $Sk_j, S\tau_j$, using the same readout and operator bounds; they require no new probabilistic budget. Initialized operator norm eight proves the four-family coordinate bound. All failures tend to zero, hence each prescribed fixed confidence is attained eventually. □

**Claim 5** (Working finite-query time domain and all-time carriers)**.** *Under $Y/\lambda \le \beta^{-30L}$, for any fixed finite list of real unit inputs, eventually with every fixed confidence its predictions are holomorphic on a neighborhood of*

$$[-r, T + r] + i[-r, r], \qquad T = 32\ell/\lambda, \qquad r = 1/(\lambda\sqrt{\ell}),$$

*and $|f_n(t, v)| \le SH_L^2$ there. Two independent copies thus satisfy $|f_n - \widetilde{f}_n| \le 32\beta^{6L}Y/\lambda$ on their common event. In particular the Bernstein ellipse of parameter two on $[0, r]$ is contained in this rectangle. On the real all-time trajectory, also*

$$\sup_{t\in[0,\infty]} \max_{a,j,i} |k_{a,i}^{(j)}(t)| \le 2SK_{\mathrm{src}}\sqrt{\ell}. \tag{60}$$

*The fitted endpoint is included.*

*Proof.* Use the same proof without angular fields or the frame mesh, replacing the Gaussian multiplier by 64. The finite-list response bound is $SU_{\mathrm{fin}}\sqrt{\ell}$. The time-radius coefficient may be any $c \le a/(64YSU_{\mathrm{fin}}) = a/(4\lambda S^2 U_{\mathrm{fin}})$. By Claim 1,

$$\frac{a}{4S^2U_{\mathrm{fin}}} \ge \frac{\beta^{34L-1}}{64} > 1,$$

so $c = 1/\lambda$ is permitted. All other rectangle gates are eventual. Cauchy–Schwarz gives $|f_n| \le SH_L^2$; use $H_L \le \beta^{3L}$ for the difference bound. The parameter-two ellipse on $[0, r]$ has real range $[-r/8, 9r/8]$ and imaginary height $3r/8$, hence lies inside the stated rectangle eventually.

It remains to extend the carrier maximum beyond $T$; no complex late-time extension is required. Let $\|\Delta\theta\|$ denote the physical parameter norm with blocks $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$. Forward subtraction in the real fitting tube gives feature RMS changes bounded by $P_j\|\Delta\theta\|$. Backward subtraction then gives

$$\max_{a,j} \|k_a^{(j)}(t) - k_a^{(j)}(T)\|_\infty \le C_k n \|\theta(t) - \theta(T)\|,$$

where $C_L^k = 1$, $C_j^k = S\tau_{j+1} + 10sC_{j+1}^k + 10t_2SP_{j+1}k_{j+1}$, and $C_k = \max_j C_j^k \le \beta^{20L}$. The terms are the changed mixer, propagated upper difference, and changed activation gate; RMS-to-coordinate conversions account for the two factors $\sqrt{n}$. The real segment has bounded operators and real activations, so this subtraction needs no complex strip hypothesis. Lemma 2.1 gives $\|\theta(t) - \theta(T)\| \le 2Ye^{-16\ell}/\sqrt{\lambda}$. Consequently the last display is eventually at most $SK_{\mathrm{src}}\sqrt{\ell}$, including at $t = \infty$. Add (57) at $T$. □

**Claim 6** (Working residual-adapted finite-panel domain)**.** *Fix the training inputs and any finite list of passive inputs before initialization. Use the same allowance as the working whole-sphere estimates, and let*

$$\nu = \lambda/4, \quad T = 32\ell/\lambda, \quad r_n = \frac{a}{64YSU_{\mathrm{fin}}\sqrt{\ell}},$$
$$h(t) = \frac{1}{2\mathcal{K}}\log(1 + 2\mathcal{K}r_n e^{\nu t}), \quad d_h = \frac{\nu}{2\mathcal{K}}, \quad T_+ = T + h(T) + r_n.$$

*Eventually with every fixed confidence, the required forward and backward fields are holomorphic on a neighborhood of*

$$D_1 = \{x + iy : -h(0) \le x \le T_+, \quad |y| \le h(\max(x, 0))\},$$

*with the same operator, RMS, training-maximum, and four-family coordinate bounds as in the working whole-sphere estimates; the response bound is $SU_{\mathrm{fin}}\sqrt{\ell}$. In particular it suffices to retain forward families on all declared inputs and backward families only on training inputs. For $0 \le t \le T$, the disk of radius $\varrho(t) = h(t)/(2(1 + d_h))$ about $t$ lies in $D_1$. The adaptive subdivision $t_0 = 0$, $t_{b+1} = \min(T, t_b + \varrho(t_b)/2)$ has at most*

$$J \le 1 + 40960\frac{U_{\mathrm{fin}}}{a}\left(\frac{Y}{\lambda}\right)^2\sqrt{\ell} + 80\frac{\mathcal{K}}{\lambda}\log\left(1 + \frac{4\ell}{\log 2}\right) \tag{61}$$

*panels. Here $U_{\mathrm{fin}}/a \le \beta^{27L}$ and $\mathcal{K} \le \beta^{14L}$, independently of the input dimension and the size of the fixed passive list.*

*Proof.* Population RMS and the Gram gap give $\lambda \le H_L^2 \le \mathcal{K}$. Thus $0 \le h' \le d_h \le 1/8$, $h(0) \le r_n$, and $h(t) \le r_n + d_h t$. Define the nested domains by

$$D_s = \{x + iy : -sh(0) \le x \le sT_+, \quad |y| \le sh(\max(x, 0))\}, \quad 0 \le s \le 1.$$

For $X_s(x) = \operatorname{clip}_{[-sh(0), sT_+]} x$, their retractions are

$$P_s(x + iy) = X_s(x) + i \operatorname{clip}_{[-sh(X_s(x)_+), sh(X_s(x)_+)]} y.$$

With $a_0(z) = \operatorname{clip}_{[0,T_+]} \operatorname{Re} z$, $P_s a_0 = a_0 P_s$, and $\operatorname{Lip}(P_s) \le 1 + d_h$. Indeed $|\Delta X_s| \le |\Delta x|$ and the imaginary increment is at most $|\Delta y| + d_h|\Delta x|$; the shear has norm at most $1 + d_h$. Integrate derivatives along the image under $P_s$ of a straight segment to get global coefficient Lipschitz bounds. Convexity of $D_s$ is unnecessary. The real anchors remain monotone and freeze, as required by the moment proof.

We verify the base-propagator premise before Gaussian insertion. Under each cavity's provisional response stop, use Claim 4 with $U_0 = U_{\mathrm{fin}}$. For $G = (mn)^{-1/2}[\nabla_\Theta F_1, \dots, \nabla_\Theta F_m]$, the explicit gradient blocks give

$$\|G\| \le \sqrt{\mathcal{K}}, \qquad \|\dot{G}\| \le C_g B_n \rho, \qquad C_g = J_* \left(1 + \sum_{j=2}^{L} H_{j-1}\right) + 2S^2 \sum_{j=2}^{L} \tau_j q_{j-1}^* + 2S q_L^*. \tag{62}$$

For the second estimate differentiate each column block $m^{-1/2}\delta_a^{(1)} v_a^\top/\sqrt{n}$, $m^{-1/2}\delta_a^{(j)} h_a^{(j-1)\top}/n$, and $m^{-1/2} h_a^{(L)}/\sqrt{n}$, then bound Frobenius norm. The division by $\sqrt{m}$ prevents a sample factor.

The exact residual equation is $\dot{r} = -2G^\top G r$. On a vertical segment based at real $t \ge 0$, norm differentiation and real fitting give

$$\rho(t \pm iv) \le Y e^{-\nu t} e^{2\mathcal{K}v}, \qquad \int_0^{h(t)} \rho(t \pm iv)\, dv \le Y r_n. \tag{63}$$

The second bound is the defining identity for $h(t)$. It also gives $\rho \le Y + 2\mathcal{K} Y r_n \le 2Y$ eventually. The base variational equation is $\dot{P} = -2GG^\top P$, with algebraic transposes. At a real anchor, $A_t = 2G(t)G(t)^\top$ is real symmetric, so $-iA_t$ generates unitary motion. Freeze this generator on the vertical segment and use (62):

$$\|2G(t \pm iv)G(t \pm iv)^\top - A_t\| \le 4\sqrt{\mathcal{K}} C_g B_n \int_0^v \rho(t \pm iu)\, du.$$

Conjugating by the frozen unitary bounds its propagator by

$$\exp\left(4\sqrt{\mathcal{K}} C_g B_n \int_0^{h(t)} (h(t) - v)\rho(t \pm iv)\, dv\right) \le \exp\left(\frac{2C_g}{\sqrt{\mathcal{K}}} B_n Y r_n\right).$$

For the last inequality, integrate the exponential bound in (63); the double integral is at most $Y r_n/(2\mathcal{K})$. Products on ordered disjoint subsegments obey the same bound, since their growth integrals add. Forward real segments contract. The negative-real corner has total exceptional length at most $2r_n$, costing at most $e^{4\mathcal{K} r_n}$. Hence impose the directly checkable eventual gate

$$4\mathcal{K} r_n + (2C_g/\sqrt{\mathcal{K}}) B_n Y r_n < \log(3/2). \tag{64}$$

It supplies the factor-two base premise, without treating a complex Gram matrix as positive. Its left side tends to zero because $B_n r_n = O(\log(e + \ell)/\sqrt{\ell})$.

Also impose (59) with $c_t = a/(64YSU_{\mathrm{fin}})$. Additional activity is at most $8Yr_n$, and normalized parameter increments are at most $8YSD_W r_n \le 1/4$. Thus operators, activity and $\rho/S \le \lambda/8$ meet the shared premises. All canonical anchor contours have length $O(\ell)$, and the bounding rectangle $[-h(0), T_+] + i[-h(T_+), h(T_+)]$ has sides $O(\ell)$. The local map/remainder argument is therefore unchanged: the one-dimensional control entropy is $n^{5/8}\,\mathrm{poly}(\ell)$, endpoint meshes are polynomial, and the same $n^{\cdot 78}$ Gaussian tail dominates them. After the improved response bound, vertical displacement is at most $2YSU_{\mathrm{fin}}\sqrt{\ell}r_n = a/32$; the negative corner costs at most $a/8$. The pole and passive-size stops improve, and cavity transfer is strict.

Finally Claim 4, integrated over these segments, gives the complete-path normalized correction radius

$$D_n = 4Yr_n \max\{2S \max_j q_j^*, J_* B_n/S\} = O(\log(e+\ell)/\sqrt{\ell}).$$

The commuting retractions preserve this estimate and give global modulus $O(B_n)$. Thus the shared Gaussian moment and budget argument applies. Local holomorphic continuation glues because the domain is vertically fibered, retracts onto its real interval, and is simply connected; strict stopped margins continue across its compact boundary. This proves the source event and its confidence.

For the disk claim, if its real coordinate is nonnegative, then $h(x) \ge h(t) - d_h \varrho(t) \ge \varrho(t)$. If it is negative, $t < \varrho(t)$ and $h(t) \le h(0) + d_h t$ imply $\varrho(t) < h(0)$. The right endpoint is below $T + h(T) < T_+$. The radius is nondecreasing with Lipschitz constant $C_r = d_h/[2(1+d_h)]$. Each complete adaptive panel contributes at least $1/(2+C_r)$ to $\int dt/\varrho(t)$, by bounding the radius on that panel by $(1+C_r/2)\varrho(t_b)$. Consequently $J \le 1 + 5\int_0^T dt/h(t)$. Split at $2\mathcal{K}r_n e^{\nu t} = 1$. Before this point use $\log(1+x) \ge x/2$; afterward the logarithm grows at least with slope $\nu/2$. This bounds the integral by

$$\frac{2}{\nu r_n} + \frac{4\mathcal{K}}{\nu} \log\left(1 + \frac{\nu T}{2\log 2}\right).$$

Substitute $\nu = \lambda/4$, $S = 16Y/\lambda$ and $T$ to obtain (61). The coefficient bounds follow from Claim 1 and $a^{-1} \le \beta/16$. □

**5. Reduction to the public bounds.** We now compare the stronger working estimates with the public statement. The quantitative width and initialization proofs explicitly recall the coefficient recurrences they additionally use. The working whole-sphere time radius is

$$\frac{a}{64YSU\sqrt{\ell}} = \frac{a\lambda}{1024Y^2U\sqrt{\ell}}.$$

Since $U \le \beta^{26L}\sqrt{d+3}$ and $a^{-1} \le \beta/16$,

$$\frac{1024U}{a} \le 64\beta^{26L+1}\sqrt{d+3} \le \beta^{30L}\sqrt{d+3}.$$

The last inequality uses $64 \le \beta^{4L-1}$, valid for $L \ge 2$, $\beta \ge 10$. This gives at least the public time radius. Likewise $V \le \beta^{2L}\sqrt{d+3}$ gives

$$\max\{8, 8V/a\} \le \max\{8, \tfrac{1}{2}\beta^{2L+1}\sqrt{d+3}\} \le \beta^{3L}\sqrt{d+3},$$

so the working query tube contains the stated tube.

The local power bounds yield, for every layer,

$$H_j \le 3\beta^{2j} \le \beta^{3L}, \qquad \max(k_j, \tau_j) \le 3\beta^{4L-1} \le \beta^{4L}.$$

Multiplication by $S = 16Ym/\gamma$ gives the backward bounds. The four-family coordinate envelope obeys

$$M_0 = 10 \max_j (H_j, \tau_j) \le 30\beta^{4L-1} \le \beta^{6L}.$$

For all real times, the fitting statement bounds normalized operators by nine and $\|w\|_2/\sqrt{n} \le 2Y/\sqrt{\lambda}$. To check the remaining population comparison directly, Minkowski and linear activation growth give $\sqrt{Q_{11}^{(j)}} \le b + s\sqrt{Q_{11}^{(j-1)}}$, starting at $Q_{11}^{(0)} = 1$. The local recursion for $H_j$ dominates this recursion, so $\lambda \le Q_{11}^{(L)} \le H_L^2$. Hence $2Y/\sqrt{\lambda} \le SH_L$. The same forward and backward recursions with operator bound nine give $H_j, Sk_j, S\tau_j$ on the whole real trajectory. Their parameter limits give the endpoint bounds as well. This proves (26) in both stated domains.

The working finite-query claim has exactly the public time radius. Its prediction envelope is $SH_L^2 \le 16\beta^{6L}Ym/\gamma$, proving (27). Its all-time training-carrier envelope is $2SK_{\mathrm{src}}\sqrt{\ell} \le 32\beta^{21L}(Ym/\gamma)\sqrt{\ell}$. Substituting the explicit definition of $k^{(j)}$ proves (28), without exporting that auxiliary notation.

For the finite panel, choose $r_b = \varrho(t_b)$ in the working domain's adaptive partition. Its disk containment and half-radius intervals are exactly those in (iii), and its source envelopes have just been bounded in canonical quantities. Its defining coefficients depend only on the width, $m, \gamma, Y$, activations and depth, so these choices are deterministic. Finally

$$40960\frac{U_{\mathrm{fin}}}{a} \le 40960\beta^{27L} \le \beta^{30L}, \qquad 80\mathcal{K} \le 80\beta^{14L} \le \beta^{16L},$$

because $40960 \le \beta^{3L}$ and $80 \le \beta^{2L}$. Applying these inequalities to (61) proves (29). For any requested finite collection of the source domains, split the fixed failure allowance and intersect their events. This supplies the single event asserted in the proposition and completes its proof. □

# 4 Legendre history compression

**Lemma 4.1** (Projection identities)**.** *Let $A > 0$ and let $q \ge 1$ be an integer. Set $p_j^A(\xi) = P_j(2\xi/A - 1)$, where $P_j$ is the Legendre polynomial, and let $\Pi_q^A$ project onto degrees below $q$ in $L^2([0, A])$. For an absolutely continuous Hilbert-valued history $u$,*

$$\|(I - \Pi_q^A)u\|_{L^2}^2 \le \frac{1}{q(q+1)} \int_0^A \xi(A-\xi)\|u'(\xi)\|^2 \, d\xi, \tag{65}$$

$$\|u(A) - (\Pi_q^A u)(A)\| \le \sqrt{A/(3q)}\, \|u'\|_{L^2}, \qquad \|(\Pi_q^A u)(A)\| \le 64\sqrt{q}\, \|u\|_{L^\infty}. \tag{66}$$

*For a fixed absolutely continuous history $g$, the growing-interval error obeys*

$$\frac{d}{dA} \int_0^A \|g - \Pi_q^A g\|^2 \, d\xi = \|g(A) - (\Pi_q^A g)(A)\|^2. \tag{67}$$

*Proof.* Rodrigues' formula and integration by parts give $\int_0^A p_i^A p_j^A = A\mathbf{1}_{i=j}/(2j+1)$ and $-[\xi(A-\xi)(p_j^A)']' = j(j+1)p_j^A$. The derivative polynomials are orthogonal for the weight $\xi(A-\xi)$. Bessel's inequality followed by the ordinary expansion therefore proves (65); polynomial approximation and increasing finite-dimensional projections give the stated absolutely continuous, Hilbert-valued version. Integration against each lower polynomial also gives $P_j' = \sum_{i<j,\ j-i \text{ odd}} (2i+1)P_i$. Thus the endpoint kernel is

$$K_q^A(A, \xi) = \frac{1}{A} \sum_{j<q} (2j+1)p_j^A(\xi) = \tfrac{1}{2}[(p_q^A)'(\xi) + (p_{q-1}^A)'(\xi)].$$

Its integral is one, and integration by parts gives $u(A) - (\Pi_q^A u)(A) = \frac{1}{2}\int_0^A (p_q^A + p_{q-1}^A)u'$. The squared $L^2$ norm of this multiplier is $A[(2q+1)^{-1} + (2q-1)^{-1}]/4 \le A/(3q)$.

For completeness the needed kernel $L^1$ estimate has a short one-dimensional proof. Put $v(\theta) = \sqrt{\sin\theta} P_k(\cos\theta)$ and $a_k = (k+1/2)^2 + (4\sin^2\theta)^{-1}$. The Legendre equation implies $v'' + a_k v = 0$, whence $\partial_\theta(v^2 + v'^2/a_k) = -a_k' v'^2/a_k^2 \ge 0$ on $(0, \pi/2)$. At the midpoint use $P_{2j}(0) = (-1)^j \binom{2j}{j}/4^j$, $P_{2j+1}'(0) = (2j+1)P_{2j}(0)$, and $(\binom{2j}{j}/4^j)^2 \le 1/(j+1)$; the last inequality follows by induction. The midpoint energy is at most $4/k$. Differentiating $P_k = v/\sqrt{\sin\theta}$ now yields

$$|\partial_\theta P_k(\cos\theta)| \le 3\sqrt{k}(\sin\theta)^{-1/2} + 2k^{-1/2}(\sin\theta)^{-3/2}.$$

On $[1/k, \pi/2]$ its integral is at most $3\pi\sqrt{k} + 8$, using $\sin\theta \ge 2\theta/\pi$. On $[0, 1/k]$ use $|P_k'| \le k(k+1)/2$ to obtain at most $1/2$. This derivative bound follows from the derivative expansion and $|P_j| \le 1$, the latter following from the integral formula $P_j(\cos\theta) = \pi^{-1}\int_0^\pi (\cos\theta + i\sin\theta\cos\psi)^j \, d\psi$ (expand its generating function). Reflection gives total variation at most $36\sqrt{k}$. The endpoint kernel consequently has $L^1$ norm at most $36\sqrt{q} \le 64\sqrt{q}$; $q = 1$ is the averaging kernel. Finally differentiation of the squared projection error proves (67): the interior term vanishes because $\partial_A \Pi_q^A g$ is still a polynomial of degree below $q$. □

**Proposition 4.2** (Legendre approximation at the prescribed size)**.** *Eventually at each fixed confidence, there is an initialization-only Legendre model with*

$$q = \left\lceil C\left(\frac{m}{\gamma}\right)^3 n^{1/4}[\log(en)]^2 e^{\sqrt{\log(en)}/2} \right\rceil \tag{68}$$

*such that*

$$\sup_{t\in[0,\infty],\ \|x\|=\sqrt{d}} |f_{\mathrm{Leg}}(t,x) - f_n(t,x)| \le \frac{CY}{\sqrt{n}[\log(en)]^3}. \tag{69}$$

*Its moving storage is at most*

$$Cm(m/\gamma)^3 n^{5/4}[\log(en)]^2 e^{\sqrt{\log(en)}/2};$$

*the fixed storage is the additional* $(L-1)n^2$.

*Proof.* **Local notation and construction.** All auxiliary quantities below belong to this proof. Put

$$\lambda = \gamma/m, \qquad H = \max_{0\le j\le L}\sqrt{Q_{11}^{(j)}}, \qquad s = \max\left\{1, \max_j \sup_{|\operatorname{Im} z|\le a/2} |\phi_j'(z)|\right\}, \qquad b = \max_j |\phi_j(0)|,$$

and define $D_j = s(9s)^{L-j}$. These definitions are made here; no coefficient from the proof of Lemma 2.1 is imported. In particular $H \le (2\beta)^L$, by the affine population-RMS recursion, and $s, b \le \beta$.

The Legendre model constructed here shares the realized dense initialization and physical time. Here the whole-sphere version of (6) is abbreviated by

$$\|g\|_* = \sup_{t\in[0,\infty],\,\|x\|=\sqrt{d}} |g(t,x)|,$$

including the fitted limit at $t = \infty$. Write $\widehat{f} = f_{\mathrm{Leg}} = n^{-1}\widehat{w}^\top \widehat{h}^{(L)}$, $\widehat{r}_a = \widehat{f}(x_a) - y_a$ and $\widehat{\rho} = \|\widehat{r}\|_2/\sqrt{m}$. We first construct the model for an arbitrary integer $q \ge 1$ and then take the order in the statement.

At each hidden interface $\ell = 2, \dots, L$, sample $a$, and order $j = 0, \dots, q-1$, retain $\bar{h}_{a,j}^{(\ell-1)}, \bar{\delta}_{a,j}^{(\ell)} \in \mathbb{R}^n$. Together with $\widehat{W}^{(1)}$, $\widehat{w}$, and a scalar $\tau$, these are the moving state. Reconstruct

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{mn\tau} \sum_{a=1}^{m} \sum_{j=0}^{q-1} (2j+1) \bar{\delta}_{a,j}^{(\ell)} \bar{h}_{a,j}^{(\ell-1)\top}. \tag{70}$$

Use these matrices in the ordinary forward pass and in $\widehat{k}_a^{(L)} = \widehat{w}$, $\widehat{\delta}_a^{(\ell)} = \phi'_\ell(\widehat{z}_a^{(\ell)}) \odot \widehat{k}_a^{(\ell)}$, $\widehat{k}_a^{(\ell)} = \widehat{W}^{(\ell+1)\top} \widehat{\delta}_a^{(\ell+1)}$. The autonomous physical-time equations are

$$\begin{aligned}
\dot{\widehat{W}}^{(1)} &= -\frac{2}{m} \sum_a \widehat{r}_a \widehat{\delta}_a^{(1)} v_a^\top, & \dot{\widehat{w}} &= -\frac{2}{m} \sum_a \widehat{r}_a \widehat{h}_a^{(L)}, \qquad \dot{\tau} = \widehat{\rho}, \\
\dot{\bar{h}}_{a,j}^{(\ell-1)} &= \widehat{\rho} \widehat{h}_a^{(\ell-1)} - \frac{\widehat{\rho}}{\tau} \left( j \bar{h}_{a,j}^{(\ell-1)} + \sum_{i<j} (2i+1) \bar{h}_{a,i}^{(\ell-1)} \right), \\
\dot{\bar{\delta}}_{a,j}^{(\ell)} &= \widehat{r}_a \widehat{\delta}_a^{(\ell)} - \frac{\widehat{\rho}}{\tau} \left( j \bar{\delta}_{a,j}^{(\ell)} + \sum_{i<j} (2i+1) \bar{\delta}_{a,i}^{(\ell)} \right).
\end{aligned} \tag{71}$$

Initially $\tau = 1$, $\widehat{W}^{(1)} = W_0^{(1)}$, $\widehat{w} = 0$, $\bar{h}_{a,0}^{(\ell-1)} = h_{0,a}^{(\ell-1)}$, and all other moments vanish. No equation divides by a residual. At $\widehat{\rho} = 0$ all velocities vanish, and the vector field is locally Lipschitz for $\tau > 0$. The exact inventory is

$$S_{\text{moving}} = n(d+1) + 1 + 2(L-1)mnq, \qquad S_{\text{fixed}} = (L-1)n^2. \tag{72}$$

The latter entries are the initialized mixers. Data storage is additional; reconstructed matrices need not be retained independently.

**Moment identities and the physical defect.** For use in this proof, let $p_j^A(\xi) = P_j(2\xi/A - 1)$ and let $\Pi_q^A$ be the degree-below-$q$ orthogonal projection in $L^2([0, A])$, as in Lemma 4.1. The histories in the proof are specified by $h_a(\tau(t)) = \widehat{h}_a(t)$ and $b_a(\tau(t)) = (\widehat{r}_a/\widehat{\rho})\widehat{\delta}_a(t)$. On the prefix $[0, 1]$ they equal the initialized forward feature and zero, respectively. The backward history is continuous at the join because the initial readout is zero. The moments in (71) are exactly $\int_0^\tau p_j^\tau h_a$ and $\int_0^\tau p_j^\tau b_a$; differentiating these integrals proves the equations, including their factors $j$ and $2i+1$. Differentiating their bilinear projection pairing gives the exact defect

$$\dot{\widehat{W}}^{(\ell)} = -\frac{2}{mn} \sum_a \widehat{r}_a \widehat{\delta}_a^{(\ell)} \widehat{h}_a^{(\ell-1)\top} + \mathcal{E}_\ell, \qquad \mathcal{E}_\ell = \frac{2\widehat{\rho}}{mn} \sum_a e_{b,a}^{(\ell)} e_{h,a}^{(\ell-1)\top}, \tag{73}$$

where $e_g = g(\tau) - (\Pi_q^\tau g)(\tau)$. Indeed the derivative of the pairing of the two projected histories equals the full endpoint product minus the product of their endpoint errors, by orthogonality. These identities hold until a residual zero and extend there by stationary continuation. Histories are proof objects, not extra state.

**Claim 7** (Order-independent physical bounds)**.** *On* (17)*, under $Y \le \lambda \beta^{-30L}$, every finite order is globally defined, has convergent physical parameters and moments, and fits the labels. Its operator and feature bounds are those of* (18)*, and*

$$\widehat{\rho}(t) \le Y e^{-\lambda t/4}, \qquad \tau(t) \le 1 + 4Y/\lambda, \qquad \|\widehat{w}(t)\|/\sqrt{n} \le 4Y/\sqrt{\lambda}. \tag{74}$$

*Verification of the local claim.* Let $\sigma = 8Y/\lambda$, $R = \sigma\sqrt{\lambda}$, and $\kappa = \lambda/4$. Define, within the present construction proof,

$$C_1 = 4s^2 D_1^2, \quad C_\ell = 3s^2(64H^4 D_\ell^2 + 82C_{\ell-1}), \quad E_* = 130\sum_{\ell=2}^{L} D_\ell\sqrt{C_{\ell-1}}.$$

It suffices that

$$\sigma \le \min\left\{1, \frac{1}{64H^2 D_1}, \frac{1}{\sqrt{8\sqrt{C_L}}}, \left(\frac{1}{8H^2 D_1 E_*}\right)^{2/7}\right\}. \tag{75}$$

The present label cap implies this: $D_\ell \le \beta^{2L}$, $\sqrt{C_\ell} \le \beta^{8L}$, and $E_* \le \beta^{12L}$. For the middle estimate unroll $C_\ell = 192s^2H^4D_\ell^2 + 246s^2C_{\ell-1}$, bound $192s^2, 246s^2 \le \beta^5$, $H^4D_\ell^2 \le \beta^{10L}$, and sum at most $L$ terms. For the last use $130L \le (\beta^L)^2$. Substitution gives a lower bound $\beta^{-8L}$ for the minimum in (75), whereas $\sigma \le 8\beta^{-30L}$.

Stop at clock activity $\sigma$, readout RMS $R$, operator norm nine, or feature RMS $2H$. Before the stop, $\|\widehat{\delta}_a^{(\ell)}\|/\sqrt{n} \le D_\ell R$. Projection contraction, sample Cauchy–Schwarz, and the zero backward prefix give

$$\|\widehat{W}^{(\ell)} - W_0^{(\ell)}\|_F \le 4HD_\ell R\sqrt{(1+\sigma)\sigma}, \qquad \|\widehat{W}^{(1)} - W_0^{(1)}\|_F/\sqrt{n} \le 2D_1 R\sigma.$$

These are strictly below one by (75) and $\sqrt{\lambda} \le H$. Let $Z_\ell$ be the supremum, over queries and stopped horizons, of $n^{-1}\int_1^{\tau(t)} \|\partial_\xi h^{(\ell)}(\xi, v)\|^2 d\xi$. The polynomial endpoint norm on $L^2([0, A])$ is $q/\sqrt{A}$, since the sum of squared normalized basis values at $A$ is $\sum_{j<q}(2j+1)/A = q^2/A$. Thus the aggregate backward endpoint error is at most $(q+1)D_\ell R$. By (67), (65), and (73),

$$\int_0^t \widehat{\rho}\|\mathcal{E}_\ell/\widehat{\rho}\|_F^2 du \le 2(1+\sigma)^2 D_\ell^2 R^2 Z_{\ell-1}.$$

Here the forward prefix error is zero, $\xi(A-\xi) \le A^2/4$, and $(q+1)/q \le 2$. The first-layer chain rule and the three later derivative terms (gradient update, defect, and propagated feature derivative) now give

$$Z_1 \le 4\sigma s^2 D_1^2 R^2, \qquad Z_\ell \le 3s^2\{64\sigma H^4 D_\ell^2 R^2 + [81 + 32H^2 D_\ell^2 R^2]Z_{\ell-1}\} \le C_\ell \sigma R^2.$$

The last step uses $32H^2D_\ell^2R^2 < 1$. Consequently every feature moves in RMS by at most $\sqrt{\sigma Z_\ell} \le \sqrt{C_L}\sigma^2\sqrt{\lambda} \le \sqrt{\lambda}/8$. This excludes the feature boundary and preserves the top Gram gap $\lambda/4$ by singular-value perturbation.

For the pointwise defect use the sharper bounds (66). The two endpoint errors are at most $65\sqrt{q}D_\ell R$ and $\sqrt{(1+\sigma)/(3q)}\sqrt{Z_{\ell-1}}$, respectively. Hence

$$\sum_{\ell=2}^{L} \|\mathcal{E}_\ell\|_F \le E_*\sqrt{\sigma}R^2\widehat{\rho}. \tag{76}$$

The prediction Jacobian applied to this hidden defect has sample RMS at most $2HD_1R\sum\|\mathcal{E}_\ell\|_F \le \lambda\widehat{\rho}/4$, by the last entry in (75). Let $J$ be the Jacobian of the training prediction vector in mobility coordinates $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$, evaluated at the reconstructed state, and put $\Gamma = JJ^\top/m$. The mean full tangent Gram then gives $\dot{\widehat{r}} = -2\Gamma\widehat{r} + J\mathcal{E}$ with $\Gamma \succeq \lambda I/4$. Thus $\dot{\widehat{\rho}} \le -\kappa\widehat{\rho}$ and $\int_t^T \widehat{\rho} \le \widehat{\rho}(t)/\kappa$ on every stopped interval. The clock activity is at most $\sigma/2$.

Let $\mathcal{F}$ be the dense gradient vector field at the reconstructed parameters, in the norm of Lemma 2.1. Its squared norm is $4\widehat{r}^\top\Gamma\widehat{r}/m \geq \lambda\widehat{\rho}^2$, and the defect estimate implies $-\partial_t\widehat{\rho}^2 \geq \|\mathcal{F}\|_{\mathrm{par}}^2/2$. Multiplying by $e^{\kappa t}$, integrating by parts, and using $\widehat{\rho} \leq Ye^{-\kappa t}$ gives $\int e^{\kappa t}\|\mathcal{F}\|_{\mathrm{par}}^2 dt \leq 4Y^2$. Weighted Cauchy–Schwarz then gives $\int \|\mathcal{F}\|_{\mathrm{par}} \leq 2Y/\sqrt{\kappa} = R/2$. The readout has no defect, so its boundary is excluded as well. Every stopped inequality improves strictly. The integral formulas bound all moments at finite $n, q$, proving global continuation. Uniqueness excludes a finite first residual zero when $Y > 0$, since zero residual is an equilibrium of the entire stored-state ODE. The integrable gradient and defect lengths yield physical limits; the moment equations also have integrable velocities. This proves (74) and fitting at the limit. This finishes the local fitting claim. The coefficients $C_j$ just defined remain private to this construction proof.

**Comparison with dense training.** Intersect the initialization event and the all-time training-carrier bound (28) of Proposition 3.1. Set $X = \beta^L$, $z = Y/\lambda$, $\chi = 1 + \lambda^{-1}$, $u = \sqrt{\log(en)}$, and $M = 32X^{21}zu$. Write $\rho_D = \|r\|_2/\sqrt{m}$ for the dense residual RMS. For the backward comparison at any parameter state, define its pre-gated training responses by $k_a^{(L)} = w$ and $k_a^{(j)} = W^{(j+1)\top}\delta_a^{(j+1)}$ for $j < L$. Only actual dense training carriers, not closure carriers or passive query carriers, are bounded by $M$. Throughout this proof $z \leq X^{-30}$, $X \geq 100$. Use the Euclidean mobility coordinates $(W^{(1)}/\sqrt{n}, W^{(2)}, \ldots, W^{(L)}, w/\sqrt{n})$, whose Euclidean norm is the parameter norm of Lemma 2.1. Let $e$ be the Euclidean discrepancy and $D$ the sum of its $L+1$ block norms, so $D \leq \sqrt{L+1}e$.

We first verify the signed comparison with no reciprocal-gap factor inserted into its Jacobian coefficient. Put $T_0 = 9s$, $R = 8z\sqrt{\lambda}$, $B_\delta = sT_0^{L-1}R$, and define

$$
\begin{gathered}
Q_1 = b + 9s, \quad Q_\ell = b + 9sQ_{\ell-1}, \quad Q = \max(1, Q_1, \ldots, Q_L), \\
F_{\mathrm{seg}} = QT_0^{L-1}, \quad P = 1 + Q(L-1), \\
d_0 = sT_0^{L-1}(1 + B_\delta), \quad d_1 = Lt_2F_{\mathrm{seg}}T_0^{L-1}, \\
K_0 = \sqrt{L+1}\{Pd_0 + [1 + (L-1)B_\delta]sF_{\mathrm{seg}}\}, \qquad K_1 = \sqrt{L+1}Pd_1,
\end{gathered}
$$

where $t_2 = \max\{1, \max_\ell \sup_{|\operatorname{Im} z| \leq a/2} |\phi_\ell''(z)|\} \leq \beta$. On the straight parameter segment from dense to closure, matrix and readout norms keep their bounds by convexity; linear growth gives feature RMS at most $Q$. Forward subtraction therefore bounds the preactivation difference at segment fraction $\vartheta \in [0,1]$ by $F_{\mathrm{seg}}\vartheta D$. In backward subtraction use

$$
\delta_\vartheta - \delta_D = \phi'(z_\vartheta) \odot (k_\vartheta - k_D) + [\phi'(z_\vartheta) - \phi'(z_D)] \odot k_D.
$$

Propagation costs $T_0$, a changed mixer acts on a response of RMS at most $B_\delta$, and the changed gate uses only the dense carrier $M$. Induction gives response difference at most $(d_0 + d_1M)\vartheta D$. Writing $f_a(\theta) = f_\theta(x_a)$, the gradient blocks of a training output are $\delta_a^{(1)}v_a^\top/\sqrt{n}$, $\delta_a^{(\ell)}h_a^{(\ell-1)\top}/n$, and $h_a^{(L)}/\sqrt{n}$. Subtract each hidden product as $(\delta_\vartheta - \delta_D)h_\vartheta^\top + \delta_D(h_\vartheta - h_D)^\top$. Their block bounds give $\|\nabla f_a(\theta_\vartheta) - \nabla f_a(\theta_D)\| \leq (K_0 + K_1M)\vartheta e$. Segment integration proves both hypotheses of Lemma 2.2 with $K = K_0 + K_1M$. The residual integrals are at most $2z$ and $4z$, so for every finite horizon

$$
\sup_{t \leq T} D(t) \leq \mathcal{A}_n\mathcal{D}_T, \quad \mathcal{A}_n = \sqrt{L+1}e^{14(K_0+K_1M)z}, \quad \mathcal{D}_T = \int_0^T \sum_{\ell=2}^{L} \|\mathcal{E}_\ell\|_F dt. \tag{77}
$$

In particular the carrier contribution is additive in $K$, not multiplied once per layer.

Here are coefficient bounds needed to control the forcing. Their dependence on the gap and labels is displayed before any simplification. Put $a_0 = 4z$, $\kappa = \lambda/4$, $F_z = 2HT_0^{L-1}$, $P_0 = 1 + 2H(L-1)$,

retain the local coefficients $C_\ell$ defined above, and define

$$\begin{aligned}
B_d &= sT_0^{L-1}(1+B_\delta) + Lt_2F_zT_0^{L-1}M,\\
V_z &= 2F_zB_\delta P_0,\\
V_\delta &= T_0^{L-1}[4sH + 4sH(L-1)B_\delta^2 + Lt_2MV_z],\\
T_1 &= 2D_1R, \qquad T_\ell = 4HD_\ell R + 130D_\ell\sqrt{8zC_{\ell-1}}R^2,\\
V_1 &= sT_1, \qquad V_\ell = s(2HT_\ell + 9V_{\ell-1}), \qquad V_h = \max_\ell V_\ell,\\
C_c &= 4\left(4H^2 + D_1^2R^2 + 4H^2R^2\sum_{\ell=2}^{L} D_\ell^2\right) + \lambda/2,\\
F_h &= V_ha_0\sqrt{(1+a_0)/2},\\
b_0 &= \sqrt{1+a_0}(Y/\kappa)V_\delta + 2B_\delta\sqrt{a_0}, \qquad b_1 = 2\sqrt{1+a_0}\,C_cB_\delta/\kappa,\\
C_f &= 2H + RsF_z.
\end{aligned}$$

The meanings follow directly from the recursions: $B_dD$ bounds the response discrepancy between the two actual trajectories; $V_z\rho_D$ and $V_\delta\rho_D$ bound dense preactivation and response speeds; $V_h\widehat{\rho}$ bounds closure feature speeds, using (76). Finally $\|\partial_t(\widehat{r}/\widehat{\rho})\|_m \le C_c$ follows by differentiating the normalized residual and bounding its tangent Gram by its trace and its defect by $\lambda\widehat{\rho}/4$.

We spell out sufficient power bounds to fix the exponential rate. The preceding finite recursions imply

$$\begin{aligned}
&K_0 \le X^9, \quad K_1 \le X^{13}, \quad 14(K_0+K_1M)z \le X^{10}z + X^{36}z^2u,\\
&V_h \le X^{20}R, \quad C_c \le X^{11}, \quad B_d \le X^{33}(1+zu), \quad V_\delta \le X^{40}(1+zu),\\
&F_h/Y \le X^{22}z\sqrt{\chi}, \quad b_0 \le X^{42}z(1+zu), \quad b_1 \le X^{16}z\sqrt{\chi}, \quad C_f \le X^7.
\end{aligned} \tag{78}$$

To check the first line, $Q \le X^2$, $F_{\mathrm{seg}} \le X^4$, $d_0 \le 2X^3$, $d_1 \le X^8$, $P \le X^4$ give the first two bounds; insert $M \le 32X^{21}zu$ and use $448 \le X^2$. For the second line, $R \le X^{-27}$, $\sqrt{C_\ell} \le X^8$, and $D_\ell \le X^2$ give $T_\ell \le X^{12}R$. The $V_\ell$ recursion has forcing at most $2X^{15}R$, propagation at most $T_0^L \le X^2$, and at most $L \le X$ terms. The trace defining $C_c$ gives its stated bound. Also $F_z \le X^5$, $P_0 \le X^4$, and $V_z \le 2X^{12}R$. The non-carrier and carrier terms of $V_\delta$ are at most $4LX^5$ and $64LX^{36}Rzu$, respectively; those of $B_d$ are at most $2X^3$ and $32LX^{29}zu$. For the final line retain the normalization: $F_h/Y \le 32X^{20}z/\sqrt{\lambda}$, $b_1 \le 96X^{14}z/\sqrt{\lambda}$, and $b_0 \le 6X^{40}z(1+zu) + 32X^5z^{3/2}$. These inequalities imply (78) because $X \ge 100$, $z \le 1$, and $\lambda^{-1/2} \le \sqrt{\chi}$. In particular

$$\mathcal{A}_n \le eXe^u, \qquad \log\mathcal{A}_n = O_{\mathrm{problem}}(\sqrt{\log(en)}). \tag{79}$$

The universal coefficient of $u$ follows from $X^{36}z^2 \le 1$; no extra power of $m/\gamma$ occurs in that exponent.

By (65), the forward history tail on $[0,A]$, $A \le 1+a_0$, has aggregate RMS $L^2$ norm at most $F_h/q$: use its clock derivative bound $V_h$ and $\int_1^A \xi(A-\xi)d\xi \le A(A-1)^2/2$. For the backward history compare it with $\widetilde{b}_a(\tau(t)) = (\widehat{r}_a/\widehat{\rho})\delta_{D,a}(t)$, with zero prefix. Its physical derivative has aggregate RMS at most $C_cB_\delta + V_\delta\rho_D$. Freeze it after $t_q = 2\log q/\kappa$ (or the current horizon if earlier). Since $A - \tau(t) \le \widehat{\rho}(t)/\kappa$, the weighted derivative energy cancels the inverse clock speed and is at most

$$\frac{1+a_0}{\kappa}\left[2t_qC_c^2B_\delta^2 + 2V_\delta^2\int_0^{t_q}\rho_D(t)^2dt\right].$$

The frozen tail is therefore at most $(b_1\sqrt{\log q}+\sqrt{1+a_0}(Y/\kappa)V_\delta)/q$. The freezing error is at most $2B_\delta\sqrt{a_0}e^{-\kappa t_q/2}=2B_\delta\sqrt{a_0}/q$. The difference between closure and dense response histories contributes at most $B_d\sqrt{a_0}\sup_{t\le T}D(t)$; the normalized residual has sample norm one. All aggregate history norms here use the factor $(mn)^{-1/2}$.

Use Cauchy–Schwarz in samples and then in $dA=\widehat{\rho}\,dt$ in (73). The initial projection errors vanish, so (67) gives

$$\mathcal{D}_T\le 2(L-1)F_h\left[\frac{b_0+b_1\sqrt{\log q}}{q^2}+\frac{B_d\sqrt{a_0}\sup_{t\le T}D(t)}{q}\right]. \tag{80}$$

For $q\ge q_{\mathrm{abs}}:=4(L-1)F_hB_d\sqrt{a_0}\mathcal{A}_n$, (77) absorbs the last term with factor at most one half. Whole-sphere forward/readout subtraction costs $C_fD$. The coefficients do not depend on $T$, so increasing horizons and the already constructed physical limits yield

$$\|f_{\mathrm{Leg}}-f_n\|_*\le\frac{4(L-1)C_f\mathcal{A}_nF_h}{q^2}(b_0+b_1\sqrt{\log q}).$$

By the ledger its numerator divided by $Y$ is at most $eX^{75}z^2\chi(1+zu)e^u\sqrt{\log(eq)}$. The cap and $\lambda\le H^2\le X^4$ imply $Y\le X^{-26}\le 1$, so $z\le\lambda^{-1}\le\chi$. Consequently

$$\|f_{\mathrm{Leg}}-f_n\|_*\le CY\chi^6e^{\sqrt{\log(en)}}\frac{\sqrt{\log(en)\log(eq)}}{q^2}\qquad(q\ge q_{\mathrm{abs}}). \tag{81}$$

This enlargement preserves the displayed sample/gap power without spending the label cap to reduce it. All coefficients except $\mathcal{A}_n$ grow at most polynomially in $u$, hence $q_{\mathrm{abs}}=n^{o(1)}$ for the fixed problem. The prescribed order (68) eventually exceeds it. Since $m/\gamma\ge 1$, $\chi\le 2m/\gamma$, and $\log(eq)\le C\log(en)$ eventually, substitution proves (69). The count follows from (72); $nd$ and rounding are absorbed only after taking sufficiently large width for the fixed problem. One event supports every order used here, so no order-dependent failure union is needed. All initialization data are the specified dense initialization and initialized training features. □

# 5 An actual dense-trajectory fluctuation

**Lemma 5.1** (Derivative to trajectory)**.** *Let $r>0$ and $M\ge 0$. If the scalar function $g$ is holomorphic on a neighborhood of the filled parameter-two Bernstein ellipse for $[0,r]$ and bounded there by $M$, then for every integer $N\ge 1$,*

$$\sup_{0\le t\le r}|g(t)|\ge\frac{r}{2N^2}|g'(0)|-24M2^{-N}. \tag{82}$$

*Proof.* Set $t=r(x+1)/2$. The Joukowski map $x=(z+z^{-1})/2$ and Cauchy's coefficient formula give Chebyshev coefficients $|a_k|\le 2M2^{-k}$. The degree-$N$ truncation has uniform tail at most $2M2^{-N}$; its physical derivative tail at zero is at most $4Mr^{-1}2^{-N}(N^2+4N+6)$, by summing $\sum_{k>N}k^22^{-k}$. For a degree-$N$ polynomial, including complex coefficients, $|P'(-1)|\le N^2\|P\|_{[-1,1]}$. Indeed interpolate at $x_j=\cos(j\pi/N)$: the endpoint derivative weights have alternating signs, and their absolute sum equals $|T_N'(-1)|=N^2$ by evaluating on $T_N(x_j)=(-1)^j$. The sign assertion follows directly from the products defining the Lagrange basis; the endpoint weight is $\sum_{j<N}(-1-x_j)^{-1}<0$. Apply this inequality to the truncation, include the physical factor $2/r$, and add the two tails. Rearranging gives error coefficient $4+8/N+12/N^2\le 24$, as required. □

**Proposition 5.2** (Positive-time dense variability)**.** *Under the assumptions of Section 1, for every fixed $0 < \delta < 1$, there is a constant $c_{\phi,L,\delta} > 0$, depending only on the activations, depth and this confidence, such that all sufficiently large individual widths satisfy*

$$\Pr\left\{\|f_n - \widetilde{f}_n\|_{\mathcal{X}} \geq c_{\phi,L,\delta}\frac{Y\sqrt{\gamma}}{\sqrt{n}[\log(en)]^{5/2}}\right\} \geq 1 - \delta. \tag{83}$$

*This holds for either the whole sphere or any declared panel containing the training inputs. The witness is at a positive training time; no endpoint lower bound is asserted.*

*Proof.* All moments and fluctuation variables introduced here are private to this proof. Let $G \sim N(0,1)$, define

$$\mu_{2,j} = Q_{11}^{(j)} \quad (j = L-1, L), \qquad \mu_4 = \mathbb{E}\phi_L(\sqrt{\mu_{2,L-1}}G)^4,$$

and write $\Phi$ for the standard normal distribution function. Let $Z \sim N(0, Q^{(L-1)})$, $U_a = \phi_L(Z_a)$ and $Q = \mathbb{E}UU^\top = Q^{(L)}$. We first establish the innovation and conditional fluctuation needed below, then convert an initial derivative into a positive-time trajectory separation.

**Claim 8** (A final-layer innovation)**.** *For $m \geq 2$ and $Y > 0$, some deterministic training index $a_*$ satisfies*

$$\mathrm{Var}\{U_{a_*}(U^\top y)\} \geq \frac{\gamma^3\mu_{2,L}}{16\mu_4}Y^2 > 0. \tag{84}$$

*Verification.* All fourth moments are finite by linear growth. The positive gap gives $\mu_{2,L} > 0$ and $\mu_4 > 0$. Put $c = Qy$, $A = \|c\|$, $T = \operatorname{tr} Q$, $M_4 = \mathbb{E}\|U\|^4$, and $D = T - c^\top Qc/A^2 > 0$. The last inequality holds because $m \geq 2$ and $Q \succeq \gamma I$. For $V = \mathbb{E}\|U(U^\top y) - c\|^2$, the squared-area identity gives

$$A^2\|U\|^2 - (c^\top U)^2 \leq \|U(U^\top y) - c\|^2\|U\|^2.$$

Splitting at $\|U\| = R_0$ yields $A^2D \leq R_0^2V + A^2M_4/R_0^2$. Choose $R_0^2 = 2M_4/D$ to obtain $V \geq [y^\top Q^2(TI - Q)y]^2/(4M_4\|Qy\|^2)$. Here $M_4 \leq m^2\mu_4$ and $T = m\mu_{2,L}$. Every eigenvalue $x$ of $Q$ lies in $[\gamma, T - (m-1)\gamma]$. Thus, for $N = y^\top Q^2(TI - Q)y$,

$$N \geq (m-1)\gamma\|Qy\|^2, \qquad N \geq \gamma^2(T - \gamma)\|y\|^2.$$

The second uses $x(T - x) - \gamma(T - \gamma) = (x - \gamma)(T - \gamma - x) \geq 0$ and then $x \geq \gamma$. Multiplying these lower bounds gives

$$V \geq \frac{(m-1)\gamma^3(m\mu_{2,L} - \gamma)}{4m^2\mu_4}\|y\|^2 \geq \frac{\gamma^3\mu_{2,L}}{16\mu_4}\|y\|^2,$$

because $\gamma \leq \mu_{2,L}$ and $(m-1)^2/m^2 \geq 1/4$. At least one of the $m$ coordinate variances is at least $V/m$, which proves (84).

**Claim 9** (Conditional scalar fluctuation)**.** *For the index in the preceding local claim, put $g_n(t) = f_n(t, x_{a_*}) - \widetilde{f}_n(t, x_{a_*})$, where the two dense initializations are independent. There is a fixed $s_* > 0$ with*

$$s_*^2 \geq \frac{\gamma^3\mu_{2,L}Y^2}{2m^2\mu_4}, \qquad \liminf_{n\to\infty}\Pr\{\sqrt{n}|g_n'(0)| \geq us_*\} \geq 2[1 - \Phi(u)] \quad (u > 0), \tag{85}$$

*where $\Phi$ is the standard normal distribution function.*

*Verification.* Let $\mathcal{F}_n$ contain both initializations through layer $L-1$. Their training empirical Grams converge to $Q^{(L-1)}$ by (16) of Lemma 2.1, including when this matrix is singular. Conditionally on $\mathcal{F}_n$, the last-layer preactivation rows of each run are independent Gaussian vectors. Applying $\phi_L$ gives independent feature vectors, generally non-Gaussian; the two groups are also independent. For either run write $U_{i,a} = h_{0,i}^{(L)}(v_a)$ for its initialized top-layer feature entries. The zero readout gives exactly

$$\dot{f}_n(0, x_a) = \frac{2}{mn} \sum_{i=1}^{n} U_{i,a}(U_i^\top y).$$

Write $v_* = \mathrm{Var}[U_{a_*}(U^\top y)] > 0$ and $s_*^2 = 8v_*/m^2$. Conditional variances of the row products converge to $v_*$: couple their Gaussian vectors with the continuous covariance square root and dominate the fourth-degree products by Gaussian polynomials. Their centered absolute third moments are uniformly bounded on bounded covariance sets, since linear growth requires only Gaussian sixth moments.

For a centered row product $X$, Taylor's remainder gives $|\mathbb{E}(e^{itX/\sqrt{n}} \mid \mathcal{F}_n) - 1 + t^2\mathbb{E}(X^2 \mid \mathcal{F}_n)/(2n)| \le |t|^3\mathbb{E}(|X|^3 \mid \mathcal{F}_n)/(6n^{3/2})$. Raise this identity to the $n$th power in each of the two independent groups. It proves that the conditional law of $Z_n = \sqrt{n}(g_n'(0) - \mathbb{E}[g_n'(0) \mid \mathcal{F}_n])$ converges weakly in probability to $N(0, s_*^2)$. To justify this random-law statement, every subsequence has a further subsequence on which both covariance matrices converge almost surely; the preceding characteristic-function calculation and the continuity theorem then apply deterministically on almost every such outcome. For a continuous limiting CDF, weak convergence is uniform in its argument: choose finite grids with small Gaussian CDF increments and small tails, and use monotonicity between grid points. Therefore the conditional Kolmogorov distance $d_n$ tends to zero in probability, and $\mathbb{E}d_n \to 0$ since $0 \le d_n \le 1$.

For any $\mathcal{F}_n$-measurable shift $b_n$, a centered Gaussian has its largest interval probability at the interval centered at zero. Consequently

$$\Pr\{|Z_n + b_n| \le us_* \mid \mathcal{F}_n\} \le 2\Phi(u) - 1 + 2d_n.$$

The Gaussian assertion follows by differentiating the interval mass with respect to its center and using the even, decreasing Gaussian density. Take $b_n = \sqrt{n}\,\mathbb{E}[g_n'(0) \mid \mathcal{F}_n]$ and average. This proves (85); previous layers may produce an arbitrary shift and cannot increase the limiting small-ball probability. No recursive matrix central limit theorem is needed.

**From the derivative to the trajectory.** Use the finite-query conclusion (27) of Proposition 3.1 for the two runs and the finite training query list. Put $\lambda = \gamma/m$, $\ell = \log(en)$ and $r = 1/(\lambda\sqrt{\ell})$. Each run is bounded there by $16\beta^{6L}Y/\lambda$, so their difference has twice this bound. The parameter-two ellipse of $[0, r]$ has real projection $[-r/8, 9r/8]$ and imaginary projection $[-3r/8, 3r/8]$, and hence lies inside the supplied rectangle. On its event, $g_n$ is bounded by $32\beta^{6L}Y/\lambda$. Choose $N = \lceil 2\ell/\log 2 \rceil \le 4\ell$. Lemma 5.1 gives

$$\sup_{0 \le t \le r} |g_n(t)| \ge \frac{|g_n'(0)|}{32\lambda\ell^{5/2}} - \frac{768\beta^{6L}Y}{\lambda n^2}.$$

For fixed $u > 0$, the second term is eventually at most half the first on $|g_n'(0)| \ge us_*/\sqrt{n}$. The two source failures tend to zero; no independence from the derivative event is required. Thus the conditional fluctuation claim gives a limiting lower probability at least $2[1 - \Phi(u)]$ for threshold $us_*/(64\lambda\sqrt{n}\ell^{5/2})$. Insert the lower bound on $s_*$, use $\lambda = \gamma/m$ and $1/(64\sqrt{2}) \ge 1/128$, and choose $u = \Phi^{-1}(1/2 + \delta/4)$. The limiting probability is at least $1 - \delta/2$, leaving a strict margin to obtain $1 - \delta$ eventually. Since $g_n(0) = 0$, a positive threshold on the compact witness interval requires a strictly positive time. That training query belongs to each promised domain. At its fitted endpoint

the predictions equal the same label, consistently with the asserted scope. The asserted constant can therefore be chosen as the following local quantity:

$$c_{\phi,L,\delta} = \frac{\Phi^{-1}(1/2+\delta/4)}{128}\sqrt{\frac{\mu_{2,L}}{\mu_4}} > 0.$$

Its moment ratio depends only on the activations and depth, as required. □

# 6 Polynomial sufficient widths for Legendre and Taylor

The sufficient widths for Legendre and finite-panel Taylor can be made polynomial in the exposed problem parameters while preserving their storage bounds. This requires a finite source-probability argument and a stronger Legendre approximation estimate. The logarithmic relative-error estimate alone would not give polynomial dependence on a requested error.

Use the model, initialization, activations, label condition and trajectory norm in the common setup. All auxiliary notation in this technical section is local to it. Put

$$X = \beta^L, \qquad \lambda = \gamma/m \in (0,1], \qquad \chi = 1+\lambda^{-1}, \qquad S = 16Y/\lambda, \qquad \ell_n = \log(en).$$

Thus $X \geq 100$, $Y/\lambda \leq X^{-30}$ and $S \leq X^{-26}$. The scalar $S$ is a normalization, not additional state. For $0 < \rho < 1/4$ and a declared list of $N \geq m$ inputs define

$$u_\rho = \max\left\{1, \left\lceil \frac{\log(4emL/\rho)}{\log(64e^2)} \right\rceil \right\}, \qquad A_{N,\rho} = X^{4000}\chi^8(N+d+u_\rho+1)^8,$$

$$N_{\mathrm{src}}(N,\rho) = \left\lceil (A_{N,\rho}/\rho)^{40000} \right\rceil. \tag{86}$$

The large common exponent is convenient for a uniform finite ledger; it does not claim practical sharpness. In particular this gate has no $Y^{-1}$.

**Theorem 6.1** (Polynomial onset with the original storage powers)**.** *For $0 < \delta, \varepsilon < 1$, a sufficient width for*

$$\Pr\left\{ \|f_{\mathrm{model}} - f_n\|_{\mathcal{X}} \leq \varepsilon \|f_n - \widetilde{f}_n\|_{\mathcal{X}} \right\} \geq 1-\delta$$

*is bounded by*

$$\left[ c_{\mathrm{wid}} \beta^L (1+m+d+p+\gamma^{-1}+Y^{-1}+\delta^{-1}+\varepsilon^{-1}) \right]^{c_{\mathrm{wid}}}$$

*with one absolute finite numerical constant $c_{\mathrm{wid}}$, local to this theorem. For Legendre take $p = 0$ and omit $Y^{-1}$; its domain is the whole input sphere. For Taylor, $p$ is the number of passive inputs declared before initialization and its domain is that panel together with the training inputs. The Legendre and Taylor storage bounds in* (8) *and* (10) *hold with activation/depth coefficients bounded by fixed powers of $X$. Taylor also has absolute error $Y/n$.*

*The constructions, physical optimizer, initialization-only provenance and label condition are unchanged. Confidence applies to each individual width. The inverse-label gate for Taylor is needed only to simplify its exact source degree to a multiple of* $\log(en)$*. This result makes no polynomial onset claim for Harmonic, for the nonlinear feature-learning certificate, or for the stronger coefficient in* (83)*, and gives no pointwise-in-time or endpoint denominator bound.*

We prove a finite source lemma, obtain a finite lower bound for actual dense variability, improve the Legendre history defect from order $q^{-2}$ to $q^{-3}$, and then assemble the two approximation estimates. The exact gates at the end make the polynomial assertion explicit.

### 6.1 Finite initialization and analytic source probability

The coefficient recurrences in (30), (31), (32), (33), (34) and (35) are used with precisely their displayed definitions. In this subsection their symbols $H_j, P_j, f_j, k_j, \tau_j, A_*, D_*, \eta_*, \mathcal{B}, \mathcal{K}, D_W, K_{\mathrm{src}}, U_{\mathrm{fin}}$ have those meanings. We quantify the finite-input proof; no angular source or whole-sphere complex domain is used. The scalar envelopes used in those recurrences are, locally,

$$b = \max_j |\phi_j(0)|, \quad s = \max\{1, \sup_{j,|\operatorname{Im} z|\le a/2} |\phi_j'(z)|\}, \quad t_2 = \max\{1, \sup_{j,|\operatorname{Im} z|\le a/2} |\phi_j''(z)|\}.$$

The additional recurrence coefficients, including $C_{\mathrm{abs}}$ and $W_G$, always mean the displayed quantities in (31); they remain local to this argument. The pre-gated carriers are $k_a^{(L)} = w$ and $k_a^{(j)} = W^{(j+1)\top}\delta_a^{(j+1)}$ for $j < L$. In the stopped-flow estimates below, $\rho(t) = \|r(t)\|_2/\sqrt{m}$ denotes residual RMS; the constant $\rho$ in the width and probability gates denotes the failure allowance. The Euclidean mobility coordinates used for real parameter differences are $\theta = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$, with the ordinary Euclidean/Frobenius block norm.

**Lemma 6.2** (Finite training and panel sources)**.** *At $n \ge N_{\mathrm{src}}(N,\rho)$, with probability at least $1-\rho$, the ambient real fitting event holds and the following source conclusions hold. The four families are exactly* (25)*; forward inputs range over the $N$ declared inputs and backward inputs over the $m$ training inputs.*

*For the training list alone one may use the closed rectangle*

$$[-r, T+r] + i[-r, r], \qquad T = 32\ell_n/\lambda, \qquad r = [u_\rho X^{100}(1+\lambda)\sqrt{\ell_n}]^{-1}. \tag{87}$$

*All training sources and the responses $R_{ba}^{(j)} = D_\Theta z_b^{(j)} \nabla_\Theta(nf_a)$, where*

$$\Theta = (W^{(1)}, \sqrt{n}W^{(2)}, \dots, \sqrt{n}W^{(L)}, w),$$

*are holomorphic near this rectangle. The source RMS bounds are* (26)*, their coordinates are bounded by $X^6\sqrt{n}$, and $|f_n| \le 16X^6Y/\lambda$. Moreover*

$$\max_{a,b,j,i} |R_{ba,i}^{(j)}| \le SU_{\mathrm{fin}}\sqrt{\ell_n}$$

*on that rectangle. On either complex source domain the first-layer operator divided by $\sqrt{n}$ and all hidden operators have norm at most ten, the complex training residual RMS is at most $2Y$, and every declared preactivation has imaginary part below $3a/8$. The training coordinate maxima are bounded by $K_{\mathrm{src}}\sqrt{\ell_n}$ for preactivations and $SK_{\mathrm{src}}\sqrt{\ell_n}$ for carriers. These strict strip and coordinate bounds are included in the event, not inferred from an RMS bound. On the whole real training trajectory, including its limit,*

$$\max_{a,j,i} |k_{a,i}^{(j)}| \le 32X^{21}(Y/\lambda)\sqrt{\ell_n}, \qquad \max_{a,b,j,i} |R_{ba,i}^{(j)}| \le YX^{110}\chi^3(1+\sqrt{\ell_n}). \tag{88}$$

*For the $N$-input panel put $a_0 = \min(a,1)$ and*

$$b_n = \log(e+\ell_n), \quad B_n = 1 + (S^2/\eta_*)b_n, \quad r_n = \min\left\{\frac{a_0}{64YSU_{\mathrm{fin}}\sqrt{\ell_n}}, \frac{1}{X^{300}(1+\lambda B_n)\sqrt{b_n}}\right\}. \tag{89}$$

*There is a deterministic disk cover and half-radius panel partition of $[0,T]$, with the same source envelopes, whose number of panels obeys*

$$J \le 1 + X^{400}\left[(Y/\lambda)^2\sqrt{\ell_n} + \lambda^{-1}\log(e+\ell_n)\right]. \tag{90}$$

*Neither this radius nor its panel-count coefficient depends on confidence.*

*Proof.* First quantify initialization. Let $H = \max_{0\le j\le L}\sqrt{Q_{11}^{(j)}} \ge 1$ and let $b, s$ have their meanings in the source proof. Define

$$R_0 = 2H, \quad K_0 = b + sR_0, \quad D_0^{\mathrm{cov}} = 2sb + 4s^2R_0, \quad T_1 = \sum_{i=0}^{L-1} s^i, \quad T_2 = \sum_{i=0}^{L-1} s^{2i},$$

$$C_0^{\mathrm{cov}} = T_2(2K_0 + 1 + D_0^{\mathrm{cov}}T_1), \quad e_0 = \min\{(8T_1)^{-1}, \gamma/(4mC_0^{\mathrm{cov}})\}, \quad h_0 = \min\{1/2, H/[4(8s)^L]\}.$$

The Gaussian Lipschitz inequality proved in the Gaussian proof gives, for an $a$-Lipschitz nonnegative Gaussian function $F$, $\operatorname{Var} F \le a^2$ and $0 \le (\mathbb{E}F^2)^{1/2} - \mathbb{E}F \le a$. Consequently, for $e \ge 2a$,

$$\Pr\{|F - (\mathbb{E}F^2)^{1/2}| > e\} \le 2e^{-e^2/(8a^2)}.$$

For iid Gaussian row pairs with marginal standard deviations at most $R_0$, the empirical RMS of $\phi(Z_a)$ is $sR_0/\sqrt{n}$-Lipschitz. The RMS of $\phi(Z_a) \pm \phi(Z_b)$ is $2sR_0/\sqrt{n}$-Lipschitz, including singular covariance pairs. Polarization therefore gives covariance-entry error at most $(2K_0+1)e_0$ if both such norms have error at most $e_0 \le 1$.

To propagate covariance errors without dividing by a variance, consider two Gaussian pairs with standard deviations $\sigma_a, \sigma_b$ and $\sigma'_a, \sigma'_b$, all at most $R_0$, and set $\Delta = \max_i |\sigma_i - \sigma'_i|$. Shrink both pairs to marginal deviations $\tau_i = \min(\sigma_i, \sigma'_i)$, preserving their correlations. Product subtraction and Cauchy–Schwarz bound the total scaling cost by $2sK_0\Delta$. At these common marginals Gaussian integration by parts gives the correlation derivative $\tau_a\tau_b\mathbb{E}[\phi'(Z_a)\phi'(Z_b)]$. The formula follows by differentiating the nonsingular density and integrating its mixed second derivative; linear growth and bounded derivatives justify the integrations. Continuity covers correlations $\pm 1$ and zero marginal variances. Since

$$\tau_a\tau_b|\rho_c - \rho'_c| \le |C_{ab} - C'_{ab}| + 2R_0\Delta,$$

the transformed entry error is at most $s^2|C_{ab} - C'_{ab}| + D_0^{\mathrm{cov}}\Delta$. The displayed inequality follows by writing the correlation difference over the larger of $\sigma_a\sigma_b, \sigma'_a\sigma'_b$ and using $|C'_{ab}| \le \sigma'_a\sigma'_b$.

Condition at each layer on the preceding initialized layers. On a sphere $h_0$-net of size at most $(1 + 2/h_0)^d$, and the training list, let $\Delta_j$ denote the largest RMS error and $E_j$ the largest training Gram-entry error. The preceding estimates give

$$\Delta_j \le e_0 + s\Delta_{j-1}, \qquad E_j \le (2K_0+1)e_0 + s^2E_{j-1} + D_0^{\mathrm{cov}}\Delta_{j-1}.$$

Thus $\Delta_j \le e_0T_1 \le 1/8$ and $E_j \le e_0C_0^{\mathrm{cov}} \le \gamma/(4m)$. This also keeps all conditional standard deviations below $R_0$. With

$$\Xi = \log\frac{16L(m+1)^2}{\rho_0} + d\log(1 + 2/h_0),$$

a sufficient initialized fitting width at failure $\rho_0$ is

$$\left\lceil \max\left\{1, \frac{\log(8L/\rho_0)}{8-2\log 9}, \frac{d\log 9+\log(8/\rho_0)}{8-\log 9}, \frac{32s^2R_0^2\Xi}{e_0^2}\right\}\right\rceil. \tag{91}$$

Indeed the layer union has at most $3m^2+(1+2/h_0)^d$ norm tests per layer and costs at most $\rho_0/2$. The operator-net tails in the fitting proof cost the other half. The operator event makes the initialized layer-$j$ feature map in neuron RMS $(8s)^j$-Lipschitz; its off-net increment is at most $H/4$. This proves (17). The deterministic fitting proof then applies with the existing label cap. Unrolling the displayed coefficients gives

$$R_0, K_0 \le X^3, \quad D_0^{\mathrm{cov}} \le X^4, \quad T_1 \le X^2,\ T_2 \le X^3,\ C_0^{\mathrm{cov}} \le X^{10}, \quad e_0^{-1} \le X^{11}\chi.$$

Since $32s^2R_0^2 \le X^8$ and $\log(1+2/h_0) \le 3L\log\beta$, the width is at most $\lceil X^{32}\chi^2[dL\log\beta + \log(16L(m+1)^2/\rho_0)]\rceil$. This proof imposes no ordering between $m$ and $d$.

For source initialization choose $\rho_0 = \rho/64$. Conditional Gaussian tails over the declared list give $Z_0 = 2H\sqrt{2\log(64nNL/\rho)}$ as an initial preactivation envelope. Deleting at most $u = u_\rho$ neurons changes each subsequent initialized feature vector in Euclidean norm by at most $D_{\mathrm{init}} = (8s)^L\sqrt{u}(b+sZ_0)$, by zero embedding and forward subtraction. The full training Gram starts above $3\gamma/4$ and the source gate gives $D_{\mathrm{init}}/\sqrt{n} < \sqrt{\lambda}/8$. Thus every cavity's normalized least singular value remains above $(\sqrt{3}/2 - 1/8)\sqrt{\lambda} > \sqrt{\lambda/2}$, preserving a strict initial fitting margin. Pairing a projected lower difference with a fresh row gives tail $2\exp[-n^{4/5}/(2D_{\mathrm{init}}^2)]$ at threshold $n^{-1/10}$; the projection is made before conditioning on that fresh row. There are at most $uLn^uNLn$ such tests. Initially each exponential budget has conditional mean below four and second moment below fifteen: $\mathbb{E}e^{\eta_*|Z|} \le 2e^{1/2}$ and $\mathbb{E}e^{2\eta_*|Z|} \le 2e^2$, since $\eta_* 2H \le 1$. Conditional Chebyshev costs at most $15mL/(16n)$ for the layer averages to remain below eight. Together these initialization failures are below $\rho/4$ at (86). Cavities have their own initial tests; failed tests define zero reference coefficient paths.

We next supply the quantitative map coefficients that replace qualitative polynomials in the source proof. Use its independently stopped normalized equation (40) with $\bar{u} = (\Theta - \Theta_0)/S$ and $\bar{F}_a = nf_a/S$. Zero readout gives $\bar{F}_a = \bar{u}_w^\top h_a^{(L)}$; there is no inverse $S$ in its derivatives. At a cavity, freeze scalar forward and scaled reverse controls and let $g \sim N(0, I_r/n)$, $r \le 2un$, concatenate the omitted interior roots. First-layer incoming rows are represented only in their own uniformly tested scalar controls; there is no retained reverse port below that layer.

Write $\kappa_0 = 1/4000$, $\Gamma = \sqrt{\mathcal{K}}$, $\tau_* = \max_j \tau_j$, $f_* = \max_j f_j$, and $T_0 = 64/\lambda$. The propagator coefficient is

$$J_0 = 2e^{1/4}(2\mathcal{B})^{1/4000}, \qquad \|J(t,s)\| \le J_0 n^{\kappa_0}.$$

Define, in the order displayed,

$$M_0 = \eta_*^{-1}[1+\log(2\mathcal{B})], \quad A_0 = b + sM_0, \quad B_0 = sM_0, \quad E_0 = A_* + SD_*M_0,$$

$$T_L^0 = 2sH_L + 2t_2M_0V_L^0, \quad T_j^0 = 10sT_{j+1}^0 + 2s\tau_{j+1}^2H_j + 2t_2M_0V_j^0,$$
$$D_c = \max\{\lambda s \max_j V_j^0/4, \lambda \max_j T_j^0/8\}.$$

The source budget bounds every scaled carrier by $M_0\ell_n$; the controls have amplitudes $A_0\ell_n, B_0\ell_n$ and speed at most $D_c\sqrt{n}\ell_n$. The first-variation coefficients are

$$v_0 = J_0\{A_0(2T_0\Gamma\tau_* + E_0) + Sf_*B_0\}, \quad z_j^0 = P_j(Sv_0 + A_0), \quad h_j^0 = sz_j^0,$$

$$k_L^0 = v_0, \quad d_j^0 = sk_j^0 + t_2 M_0 z_j^0, \quad k_j^0 = 10 d_{j+1}^0 + S\tau_{j+1} v_0 + B_0 \quad (j < L). \tag{92}$$

They bound the parameter, forward and backward linear maps from $g$ by, respectively, $\sqrt{u} n^{\kappa_0} v_0 \ell_n^2$, $\sqrt{u} n^{\kappa_0} (z_j^0, h_j^0) \ell_n^2$ and $\sqrt{u} n^{\kappa_0} (k_j^0, d_j^0) \ell_n^3$. To verify the parameter coefficient, the three differentiated port forces are

$$-\frac{2}{mn} \nabla_{\bar{u}} \bar{F}_a \bar{\delta}_a^{(j+1)\top}, \quad -\frac{2}{m} \bar{r}_a (D_{\bar{u}} \bar{\delta}_a^{(j+1)})^\top, \quad -\frac{2}{m} \bar{r}_a (D_{\bar{u}} h_a^{(j-1)})^\top.$$

Their norms use $\Gamma, \tau_*, E_0 \ell_n, S f_*$, and Duhamel integration uses $2 \int \rho / S \le 1$ and contour length $T_0 \ell_n$. The differentiated forward/backward identities in the local-map proof then give exactly the remaining recurrences. A reference carrier enters an additive forcing once; homogeneous propagation is $10s$.

For a control difference replace $A_0, B_0$ by one in the forcing, keeping all reference coefficients fixed:

$$v_c = J_0 (2T_0 \Gamma \tau_* + E_0 + S f_*), \quad z_j^c = P_j (S v_c + 1), \quad h_j^c = s z_j^c,$$

$$k_L^c = v_c, \quad d_j^c = s k_j^c + t_2 M_0 z_j^c, \quad k_j^c = 10 d_{j+1}^c + S\tau_{j+1} v_c + 1.$$

The maximum $C_c$ of these coefficients bounds the map difference by $C_c \sqrt{u} n^{\kappa_0} \ell_n^2$ times the supremum control error. Linearity in controls proves the assertion also for enlarged control nets.

For completeness the time-modulus recurrence is explicit. Put

$$a_j^z = \lambda S^2 V_j^0 / 4, \quad a_j^h = s a_j^z, \quad a_j^W = \lambda S^2 \tau_j H_{j-1} / 4 \ (j \ge 2), \quad a_j^\delta = \lambda T_j^0 / 8,$$

$$a_L^k = \lambda H_L / 4, \quad a_j^k = a_{j+1}^W \tau_{j+1} + 10 a_{j+1}^\delta, \quad A_t = 2\mathcal{K} + (\lambda/4) S E_0,$$

$$v_t = A_t v_0 + A_0 (2\Gamma \tau_* + \lambda E_0 / 4) + (\lambda/4) S f_* B_0.$$

Take $t_3 = 4\beta^2$, valid on the joining strip by Cauchy's formula, and define

$$z_1^t = S v_t + D_c, \quad h_1^t = s z_1^t + t_2 a_1^z z_1^0,$$

$$z_j^t = S(H_{j-1} v_t + a_{j-1}^h v_0) + a_j^W h_{j-1}^0 + 10 h_{j-1}^t + D_c, \quad h_j^t = s z_j^t + t_2 a_j^z z_j^0,$$

$$k_L^t = v_t, \quad k_j^t = a_{j+1}^W d_{j+1}^0 + 10 d_{j+1}^t + S\tau_{j+1} v_t + S a_{j+1}^\delta v_0 + D_c,$$

$$d_j^t = s k_j^t + t_2 a_j^z k_j^0 + (t_3 M_0 a_j^z + t_2 a_j^k) z_j^0 + t_2 M_0 z_j^t.$$

The lower-Jacobian probes have coefficients

$$z_1^J = 0, \quad h_1^J = t_2 a_1^z P_1, \quad z_j^J = a_{j-1}^h + a_j^W f_{j-1} + 10 h_{j-1}^J, \quad h_j^J = s z_j^J + t_2 a_j^z P_j.$$

If $C_t$ is the maximum of $v_t, z_j^t, h_j^t, k_j^t, d_j^t, z_j^J, h_j^J$, all terminal derivatives are bounded by $C_t \sqrt{un}\, n^{\kappa_0} \ell_n^4$. These formulas result by differentiating each displayed linearized product; a raw coordinate derivative costs one $\sqrt{n}$, and each homogeneous layer propagation remains $10s$. No derivative of the variational Jacobian is needed in $v_t$, since its equation is differentiated only in

time. For a root block selector $E_i$, the quadratic matrix is $\operatorname{sym}(E_i^\top L)$, so it has the same norm and moduli as $L$. Centered interpolation costs at most $10u\|\Delta L\|$, including the trace change, on $\|g\| \le 2\sqrt{2u}$.

Here is a numerical bound for all these recurrences. The power bounds in Claim 1 and their geometric sums give

| Coefficients | Upper bounds |
|---|---|
| $H_j, P_j, f_j$; $k_j, \tau_j$; $V_j^0$ | $X^3$; $X^4$; $X^7$ |
| $A_*, D_*, \eta_*^{-1}, \mathcal{B}, \Gamma, J_0$ | $X^5, X^6, X^{27}, X^3, X^7, X$ |
| $M_0, A_0, B_0, E_0$ | $X^{31}, X^{33}, X^{32}, X^{38}$ |
| $T_j^0, D_c$ | $X^{45}, \lambda X^{45}$ |
| parameter, forward, backward map coefficients | $X^{80}\chi, X^{85}\chi, X^{125}\chi$ |
| control differences; reference speeds divided by $\lambda$ | $X^{90}\chi$; $X^{47}$ |
| parameter, forward, backward time coefficients | $X^{125}\chi, X^{140}\chi, X^{190}\chi$ |

For example substitute the first two rows in $v_0$; its largest product has exponent less than 80 and at most one $T_0$ factor. The forward recursion costs $P_j \le X^3$. In the backward recursion the extra $M_0$ is additive and its complete propagation costs $(10s)^L \le X^2$, plus at most $L \le X$ summands. In the largest time forcing $t_2 M_0 z_j^t$ the exponents sum to at most $1 + 31 + 140$; the same propagation stays below 190. These checks also cover the lower probes. Thus a common coefficient $C_0 = X^{300}\chi$ suffices.

The response maps require additional algebraic products, not a new flow. For training driver $a$ and declared evaluated input $b$, put $\bar{R} = R/S$ and $\bar{Q} = \phi' \odot \bar{R}$. Then

$$\bar{R}_{ba}^{(1)} = \bar{\delta}_a^{(1)} v_a^\top v_b, \qquad \bar{R}_{ba}^{(j)} = \bar{\delta}_a^{(j)} c_{ab}^{(j-1)} + W^{(j)} \bar{Q}_{ba}^{(j-1)}, \quad c_{ab}^{(j-1)} = h_a^{(j-1)\top} h_b^{(j-1)}/n.$$

Its linearized terms are $\bar{\delta}_{[1]} c + \bar{\delta} c_{[1]}$, $SV_{\sqrt{n}W}\bar{Q}/\sqrt{n}$, $W\bar{Q}_{[1]}$ and the omitted response port, where $c_{[1]} = (h_{a,[1]}^\top h_b + h_a^\top h_{b,[1]})/n$ and $\bar{Q}_{[1]} = \phi' \bar{R}_{[1]} + \phi'' \bar{R} z_{b,[1]}$. Below a deletion retain the direct observable $C_b C_a^\top \bar{q}_a$, $C_a = D_\Theta h_a$, and its forward images of $C_a^\top y_i$. Its subtraction is

$$(C_b^1 C_a^{1\top} - C_b^0 C_a^{0\top}) y_i = (C_b^1 - C_b^0) C_a^{0\top} y_i + C_b^1 (C_a^1 - C_a^0)^\top y_i.$$

The first term uses reference-probe coordinate localization; the second uses the lower-adjoint bound in the local-map proof. Hence both retain the factor $d_0 + n^{-1/2}$. Differentiating the response products introduces $\phi'' \dot{z}_b \bar{R}$; use the coordinate response cap and RMS speed of $z_b$, then convert to a scalar control once. With raw amplitude coefficient at most $X^{80}$ and $U_{\mathrm{fin}} \le X^{26}$, these formulas and geometric propagation give the common envelope

$$X^{100}(1 + X^{80} + 4U_{\mathrm{fin}})^2 (1 + C_0)^2 \le X^{1000}\chi^2 =: \mathcal{C}.$$

The exponent count is $100 + 160 + 600$, with the remaining 140 paying all numerical sums and bounded layer propagation. Thus the map norms, control moduli and actual analytic derivatives are bounded by

$$\mathcal{C}\sqrt{u} n^{\kappa_0} \ell_n^5, \qquad \mathcal{C}\sqrt{u} n^{\kappa_0} \ell_n^4, \qquad \mathcal{C}\sqrt{un} n^{\kappa_0} \ell_n^8. \tag{93}$$

There are at most $4u(N+1)^2$ real scalar controls, of amplitude $M = \mathcal{C}\ell_n$ and speed $D = \mathcal{C}\sqrt{n}\ell_n^3$. Every bound concerns frozen one-dimensional histories until the uniform event is established.

We verify the geometry before forming that event. For the training rectangle, the radius in (87) is at most $\min\{1/8, 1/\lambda, \log 2/(4\mathcal{K}), 1/(32YSD_W)\}$ and at most $a/(64YSU_{\mathrm{fin}}\sqrt{\ell_n})$, by the source

power bounds. The residual, activity, operator and pole estimates in the working-domain verification therefore apply without angular fields. The normalized proof has no $n^{-1} \le Y$ gate.

For the panel set $\nu = \lambda/4$ and

$$h(t) = \frac{1}{2\mathcal{K}} \log(1 + 2\mathcal{K} r_n e^{\nu t}), \quad d_h = \nu/(2\mathcal{K}) \le 1/8, \quad T_+ = T + h(T) + r_n,$$

$$D_s = \{x + iy : -sh(0) \le x \le sT_+,\ |y| \le sh(x_+)\},$$

$$P_s(x + iy) = X_s(x) + i \operatorname{clip}_{[-sh(X_s(x)_+), sh(X_s(x)_+)]} y, \quad X_s(x) = \operatorname{clip}_{[-sh(0), sT_+]} x.$$

The retraction is $(1 + d_h)$-Lipschitz and commutes with real anchors. Its bounding rectangle has sides at most $H_0 = 128\ell_n/\lambda$; canonical real-then-vertical contours have length at most $H_0/2$. The derivative recurrence in Claim 4 gives $J_* \le X^{34}$, and (62) gives $C_g \le X^{38}$; both are below $X^{100}$. The complete propagator gate is the sum

$$4\mathcal{K} r_n + (2C_g/\sqrt{\mathcal{K}}) B_n Y r_n \le 4\mathcal{K}/X^{300} + C_g/(8X^{300}) < 1/100.$$

Here $Y = \lambda S/16$ and $\lambda B_n/(1 + \lambda B_n) \le 1$. As in (63), vertical residual activity is at most $Y r_n$; freezing the real symmetric Gram generator makes its imaginary-time evolution unitary, and the integrated generator difference is the second term of this gate. Forward real segments contract. Thus ordered products of base propagators cost less than two. Additional activity is at most $8Y r_n$, normalized operator motion at most $8YSD_W r_n$, and improved vertical preactivation displacement at most $2YSU_{\mathrm{fin}}\sqrt{\ell_n} r_n \le a_0/32$. The negative corner costs at most $a_0/8$. All stopped margins are strict.

We detail the uniform endpoint event, because an endpoint clamped by a full-network stop would not have frozen Gaussian coefficients. In what follows $A = A_{N,\rho} = \mathcal{C}^4(N + d + u + 1)^8$ and

$$A\ell_n^{16} \le n^{1/1000}, \quad u \le n^{1/8000}, \quad \ell_n \le n^{1/16000}. \tag{94}$$

To verify the first inequality at the source width put $v = \log(A/\rho) \ge 8000 \log 10$. At $\log n = 40000v$, $\log A + 16 \log(1 + 40000v) \le 17v < 40v$; the difference increases thereafter. The other inequalities follow from the same gate.

For a cavity's own stop $\sigma_c$, condition only on its retained initialization. Put $\xi = n^{-2}/(1 + A\ell_n^8)$, $J_s = \lceil 16H_0/\xi \rceil$, $s_j = j/J_s$, and take a rectangular product grid $\mathscr{G}$ with side spacing at most $\xi/8$ on the bounding rectangle. Test all endpoints

$$z_{j,q,c} = P_{\min(s_j, \sigma_c)} q, \qquad q \in \mathscr{G}, \quad 0 \le j \le J_s. \tag{95}$$

Each endpoint is retained-measurable. Its contour is explicitly $\zeta(t) = \operatorname{sgn}(x) \min(t, |x|) + i \operatorname{sgn}(y) \min((t - |x|)_+, |y|)$ on $[0, H_0]$, followed by freezing. It lies in the cavity's own domain. On this contour solve $V' = \zeta'(\mathsf{A}V + \sum_a \mathsf{B}_a^e e_a + \mathsf{B}_a^q \bar{q}_a)$, $V(0) = 0$, where the coefficients are the differentiated port forces above. Bounded measurable histories suffice; this equation is a linear map of the omitted roots. No clamped extension is claimed holomorphic.

For control accuracy $\tau = n^{-1/8}$ sample $[0, H_0]$ in $\lceil 16DH_0/\tau \rceil$ equal intervals, round each real component to mesh $\tau/8$ and interpolate. Every admissible history is within $\tau$ and the logarithmic cardinality is at most

$$4u(N + 1)^2(2 + 16DH_0/\tau) \log(1 + 32M/\tau) \le A\ell_n^8 n^{5/8} \le n^{0.627}.$$

The coefficient is paid because $16384u(N+1)^2\mathcal{C}\lambda^{-1} \le A$. The net uses a loose amplitude bound $2\mathcal{C}\ell_n$. Affine dependence on controls and (93) therefore bound every map on the entire net by $4\mathcal{C}^2\sqrt{u}n^{\kappa_0}\ell_n^5 \le n^{1/800} < n^{1/200}$.

For $g \sim N(0, I_r/n)$, $r \le 2un$, a real symmetric matrix obeys

$$\Pr\{|g^\top R g - \operatorname{tr} R/n| > t\} \le 2\exp\left[-\frac{1}{8}\min\left\{\frac{n^2t^2}{\|R\|_{\mathrm{HS}}^2}, \frac{nt}{\|R\|}\right\}\right].$$

Diagonalization and $-\log(1-v) - v \le v^2/[2(1-|v|)]$, followed by exponential Markov with parameter at most $n/(4\|R\|)$, prove this bound. Since $\|R\|_{\mathrm{HS}}^2 \le 2un\, n^{1/100}$, real/imaginary splitting at threshold $n^{-1/10}/8$ gives at most $4e^{-n^{0.78}}$. The linear-coordinate tail is smaller. The Euclidean image mean is at most $\sqrt{2un}n^{1/200} < n^{1/100}/4$ and Gaussian norm concentration pays its remaining threshold $n^{1/100}/2$. The root-norm failure is at most $e^{-un}$, and control interpolation costs at most $10\mathcal{C}u^{3/2}n^{\kappa_0-1/8}\ell_n^4 < n^{-1/10}/8$.

Now select any common prefix $D_\sigma$, even one depending on all roots. For $z \in D_\sigma$ choose a nearest grid point $q$ and the largest $s_j \le \sigma$. Clipping gives $\sup_z |P_s z - P_t z| \le 3H_0|s-t|$; hence

$$|z - P_{s_j} q| \le (3/16 + 9\sqrt{2}/64)\xi.$$

Its canonical contour lies inside $D_\sigma$, so actual controls may now be substituted. With those actual analytic functions fixed, the linear ODE has a unique holomorphic solution on this simply connected vertically fibered domain. Local power-series uniqueness and continuation along compact paths prove this assertion; contraction to the real interval gives simple connectivity. The image under $P_\sigma$ of the straight joining segment has length below $\xi/2$. Integrating the actual derivative bound in (93) along it costs at most $10u\mathcal{C}\sqrt{u}n^{\kappa_0+1/2}\ell_n^8\xi < n^{-1/10}/8$. Boundary levels follow by continuity; level zero is a tested endpoint. The endpoint family has at most $(2 + 16H_0/\xi)^3 \le n^9$ members. Together with at most $An^3$ maps per endpoint and $uLn^u$ deletion sets, the total logarithmic union cost is below $n^{0.65}$, so its failure is below $e^{-n^{0.7}} < \rho/16$.

For the training rectangle an even simpler grid works: round each real coordinate toward zero to a multiple of $\xi$. The resulting grid point and its straight segment to the target lie in every common rectangle prefix. At each fixed grid point first clamp the canonical contour only by the cavity's own rectangle. The same conditional tests and interpolation then apply, with at most $n^6$ endpoints. Thus the preceding larger probability allowance covers both geometries.

The exact nonlinear identities (45), (46) and (47) now close the comparison quantitatively. Take $N_0 = n^{1/100}$, $d_0 = n^{-1/10}$ and $u_0 = n^{-1/25}$ for the linear Euclidean, linear coordinate and nonlinear remainder stops. Their variables are $R = d_0N_0 + d_0u_0 + u_0^2$ and $P = (N_0+u_0)^2/\sqrt{n}$. The force bound has coefficient $X^{240}(1+u)(1+M)$, and learned ports and the top offset have coefficient $X^{260}(1+u)^2(1+M)^3$. Inserting $M \le X^{80}\ell_n$, the map coefficients above, activity $2\int \rho/S \le 1$ and length $64\ell_n/\lambda$ gives

$$\|U\| \le A\ell_n^8 n^{\kappa_0}[n^{-0.08} + u_0^2 + n^{-0.48} + n^{-1/2}] < u_0/2.$$

The largest resulting power is $n^{-0.07875}$, below $u_0/2$ at the source gate. The joining-strip inequality $d_0 + X^{10}u_0 + X^{30}(R+P) < a/32$ holds at the same gate. This is a first-exit argument. It gives retained coordinate differences below $n^{-1/30}$ and vector differences below $2n^{1/100}$, including the augmented responses. At the first layer the incoming-row increment is bounded by $sS^2M$, with no $n^{-1/2}$; it enters the deleted scalar controls, not a nonexistent retained reverse port.

The four nonzero scalar traces are exactly (51). The normalized Schatten calculation and scalar feedback solution in the trace calculation apply before adaptive control substitution. Their first

three traces keep the original constants; the adaptive-residual trace is at most $A\ell_n^8 n^{-1+\kappa_0}$, and finite remainders are below $n^{-1/40}$. Thus the original $C_{\mathrm{abs}}$ works. A finite Gaussian grid improves training maxima to $K_{\mathrm{src}}\sqrt{\ell_n}$ and $SK_{\mathrm{src}}\sqrt{\ell_n}$. For the response the four deterministic coefficients, after division by $S$, are

$$sK_{\mathrm{src}}H_{j-1}^2, \quad sK_{\mathrm{src}}f_{j-1}^2, \quad sK_{\mathrm{src}}ST_Q, \quad sK_{\mathrm{src}}S^2H_{j-1}q_{j-1}^*.$$

The centered row coefficient is $64q_{j-1}^*$, giving precisely $U_j(64)$. There are at most $mNLn$ response tests times the polynomial endpoint count; Gaussian multiplier 64 pays them at the source gate. Both maximum failures together are below $\rho/16$. The exact velocity $\dot{z}_b = -(2/m)\sum_a r_a R_{ba}$ then improves poles and passive maxima. The real displacement is at most $2S^2U_{\mathrm{fin}}\sqrt{\ell_n}$; no passive budget is required.

It remains to remove budgets at the actual confidence order $u$, rather than take a limit at each fixed order. Complete each common-cavity reference by its own stopped retraction before integrating a root. For its normalized complex-minus-real Gaussian coefficients the panel derivative estimate gives

$$D \le X^{100}\lambda r_n B_n, \quad K \le X^{100}\lambda B_n, \quad H \le 128\ell_n/\lambda.$$

Use the larger radius $\bar{D} = X^{-200}\lambda B_n/[(1+\lambda B_n)\sqrt{b_n}]$. The rectangle lemma Claim 3 bounds the logarithmic moment by $a(M_G + 2\bar{D}) + 2a^2\bar{D}^2$, where $M_G = 256\bar{D}\sqrt{\log(e + HK/\bar{D})}$. Here

$$\log(e + HK/\bar{D}) \le 300\log X + 16b_n + \log(1/\lambda).$$

For $b \ge 1, x \ge 0$, $\min(1, be^{-x})\sqrt{x} \le \sqrt{\log b} + 1$, by splitting at $x = \log b$ and writing the excess as $s$ with $e^{-s}\sqrt{s} \le 1$. Apply this with $b = B_n$, $x = \log(1/\lambda)$ and use $B_n \le 1 + b_n$. Then $M_G + 2\bar{D} \le X^{-190}$, so every such correction has moment at most $e^{X^{-180}}$ for $0 \le a \le 8$, including its sample RMS.

For the smaller training rectangle use instead $D \le X^{40}\lambda B_n r$, $K \le X^{40}\lambda B_n$, $H \le 64\ell_n/\lambda$. Substitution of (87) bounds the mean plus $2D$ by $X^{-50}$. To check uniformity, split the entropy into $\log 128 + \log u + 100\log X + \log\chi + (3/2)\log\ell_n$ and use $u^{-1}\sqrt{\log u} \le 1$, $\lambda\sqrt{\log\chi} \le 2$, and boundedness of $(1+\log(e+\ell_n))(1+\sqrt{\log\ell_n})/\sqrt{\ell_n}$. The numerical bound for the last expression follows from $\log(e+x) \le 2 + \log x$ and maximizing each $(\log x)^a/\sqrt{x}$, $a \in \{0, 1/2, 1, 3/2\}$. Five of the sixty available powers of $X$ absorb the constants. Thus fixed-exponent corrections are small on both domains.

For each distinct omitted root $i$ form its *complete* common-cavity reference weight $A_i$: exponentiate $\eta_* C_{\mathrm{abs}}$ times one plus its four real scalar/sample-RMS references and four complex-minus-real corrections. The real moment (56) and Cauchy–Schwarz give $\mathbb{E}(A_i^2 \mid \text{cavity}) < 3$; Hölder within the four correction terms uses exponent $16\eta_* C_{\mathrm{abs}} < 8$. Different $A_i$ are conditionally independent because they use distinct root pairs of the same cavity.

Only the singleton-to-common-cavity differences require order $u$. Extend both paths by their separate own stops and radially project their whole difference to the ball of radius $4n^{1/100}$. Both omit the paired root. Their Gaussian radius is $v_n = 4n^{-49/100}$ and the same rectangle lemma gives mean at most

$$\mu_n = 256v_n\sqrt{\log(e + X^{400}\chi\ell_n^4/v_n)} + 2v_n.$$

The source gate gives $u\mu_n + u^2v_n^2 < 10^{-3}$. For $B_i^{\mathrm{wt}} = e^{\eta_*(Z_{a,i}+K_{a,i}/S)}$, scalar absorption, Cauchy–Schwarz between main references and corrections, and Hölder over at most four corrections per root give, for distinct $I$, $|I| = k \le u$,

$$\mathbb{E}\left[1_E \prod_{i\in I} B_i^{\mathrm{wt}}\right] \le 3^{k/2}\exp\{kn^{-1/40} + 4k\mu_n + 64k^2 v_n^2\} \le 16^k.$$

The event indicator is dropped only after pathwise domination by complete references. No independence of sample processes or correction paths is used. From the defining constants, $2\eta_* K_{\mathrm{src}} \le 33/4096 < 1/100$, so repeated weights cost at most $W_n = e^{\sqrt{\ell_n}/100}$. Partition an ordered $u$-tuple by its distinct indices. A partition with $j$ repetitions is encoded by $j$ edges from block minima, giving at most $\binom{u}{2}^j/j!$ possibilities. The preceding distinct-product bound therefore yields

$$\mathbb{E}\left[1_E (n^{-1}\sum_i B_i^{\mathrm{wt}})^u\right] \le 16^u \exp\{\tbinom{u}{2} W_n/(16n)\} \le e16^u,$$

where $n \ge 4u^3$ suffices and is implied by the source gate. Markov and the sample/layer union make the budget-hit probability at most $emL(64e^2)^{-u} \le \rho/4$. The transferred cavity budgets obey $\mathcal{H}_a^{\mathrm{cavity}} \le e^{2\eta_* n^{-1/30}}\mathcal{H}_a + u/n < 2\mathcal{B}$. All other stops have the strict margins already checked. For either geometry the four failure allowances are respectively $\rho/4$, $\rho/16$, $\rho/16$ and $\rho/4$. For the simultaneous rectangle and panel event, initialization is shared; sum each of the other three allowances twice. The total is strictly below $\rho$. Holomorphic local existence and uniqueness then continue the solutions through the compact domain boundary. This completes the finite probability calculation.

The all-time real extension needs no new probability. Fitting gives $\|\theta(t)-\theta(T)\| \le 2Y\lambda^{-1/2}e^{-16\ell_n}$ in Euclidean mobility coordinates. The backward-only subtraction in the finite-query proof costs at most $X^{20}n$ and gives the stated carrier cap. Subtract the response recursion with the state at $T$ fixed. Changed gates multiply only reference coordinate caps, while changed mixers and pairings use operator and RMS bounds. A coefficient $X^{100}\chi^2(1+\sqrt{\ell_n})$ bounds this RMS Lipschitz subtraction; one conversion to coordinates leaves a tail at most $YX^{103}\chi^3(1+\sqrt{\ell_n})n^{-31/2}$. The source gate makes it at most $Y\chi\sqrt{\ell_n}$, proving (88).

Finally the panel disk radius is $\varrho(t) = h(t)/[2(1+d_h)]$. As in the panel-domain proof, it lies inside the domain and the partition $t_{b+1} = \min(T, t_b + \varrho(t_b)/2)$ satisfies

$$J \le 1 + 40/(\lambda r_n) + 80\mathcal{K}\lambda^{-1}\log(1 + 4\ell_n/\log 2).$$

The reciprocal minimum in (89) is at most

$$\frac{1024U_{\mathrm{fin}}}{a_0}(Y/\lambda)^2\sqrt{\ell_n} + X^{300}\lambda^{-1}\sqrt{b_n} + X^{300}B_n\sqrt{b_n}.$$

For $x = \log\ell_n \ge 0$, $b_n \le 2+x$ and $(2+x)^{3/2}e^{-x/2} \le 4$. Hence $b_n^{3/2} \le 4\sqrt{\ell_n}$. Use $S^2 = 256(Y/\lambda)^2$, $a_0^{-1} \le \beta$ and the coefficient bounds to obtain (90). This proves the source lemma.

□

## 6.2 A finite lower bound for actual dense variability

For $0 < \tau < 1$, define

$$a_\tau = \frac{\tau\gamma^2}{2^{20}mu_{\tau/8}X^{108}(1+\lambda)},$$

$$N_{\mathrm{low}}(\tau) = \left\lceil \max\left\{ \begin{matrix} N_{\mathrm{src}}(m,\tau/8), \quad 2^{44}m^6X^{96}\gamma^{-12}\tau^{-2}\log(64/\tau), \\ [2^{30}u_{\tau/8}X^{114}m\chi(1+\gamma^{-1})^2/\tau]^2 \end{matrix} \right\} \right\rceil. \tag{96}$$

**Proposition 6.3** (Quantitative trajectory separation)**.** *For each $n \geq N_{\mathrm{low}}(\tau)$, two independent dense runs satisfy*

$$\Pr\left\{\|f_n - \widetilde{f}_n\|_{\mathcal{X}} \geq a_\tau \frac{Y}{\sqrt{n}\ell_n^{5/2}}\right\} \geq 1 - \tau. \tag{97}$$

*The domain may be any promised query set containing the training inputs. The lower coefficient is weaker than that in* (83)*.*

*Proof.* We first prove a shift-uniform finite scalar inequality. For iid centered real variables $V_i$ with variance $\sigma^2 > 0$ and third absolute moment at most $M_3$, put $Z = n^{-1/2}\sum_i V_i$. For every $h > 0$ and real $b$,

$$\Pr\{|Z - b| \leq h\} \leq 3h/\sigma + 4\exp[-9h^2\sigma^4 n/(128M_3^2)]. \tag{98}$$

For an independent copy $V'$, Taylor expansion gives $|\mathbb{E}e^{itV}|^2 = \mathbb{E}\cos(t(V - V')) \leq 1 - \sigma^2 t^2 + (4/3)M_3|t|^3$, because $\mathbb{E}|V - V'|^3 \leq 8M_3$. Therefore $|\mathbb{E}e^{itV}| \leq e^{-\sigma^2 t^2/4}$ for $|t| \leq 3\sigma^2/(8M_3)$. The Gaussian majorant $1_{|z-b|\leq h} \leq \sqrt{e}\, e^{-(z-b)^2/(2h^2)}$ and its Fourier integral bound the probability by

$$\frac{\sqrt{e}\, h}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-h^2t^2/2}|\mathbb{E}e^{itZ}|\, dt.$$

On $|t| \leq 3\sigma^2\sqrt{n}/(8M_3)$ use the preceding Gaussian bound; its full-line integral is below $3h/\sigma$. On the complement use modulus one and $\Pr(|G| > s) \leq 2e^{-s^2/2}$. This proves (98). Fourier integrability justifies the exchange of expectation and integration. The argument works conditionally on any sigma-field supplying conditional iid variables and moment bounds; the shift may be measurable in that sigma-field.

Condition on the first run through layer $L - 1$ and the entire independent second run. Its final feature rows $U_i \in \mathbb{R}^m$ are conditionally iid. The initialized covariance calculation above, now choosing

$$e_1 = \min\{(8T_1)^{-1}, \gamma/[4m\max(C_0^{\mathrm{cov}}, s^2C_0^{\mathrm{cov}} + D_0^{\mathrm{cov}}T_1)]\},$$

gives, except with probability at most $\tau/8$,

$$\bar{Q} := \mathbb{E}(U_iU_i^\top \mid \mathcal{F}) \succeq (\gamma/2)I, \qquad \max_a \|U_{i,a}\|_{L^6(\cdot|\mathcal{F})} \leq X^4.$$

Indeed the conditional last-layer mean entry error is at most $s^2E_{L-1} + D_0^{\mathrm{cov}}\Delta_{L-1} \leq \gamma/(4m)$. Its required width is $32s^2R_0^2e_1^{-2}\log[256L(m+1)^2/\tau]$, which is bounded by the source gate in (96). Linear growth and Gaussian sixth moments prove the $L^6$ bound.

The squared-area calculation in the innovation proof does not require equal diagonal moments. For any random $U$ with $\bar{Q} \succeq gI$, put $c = \bar{Q}y$, $T_q = \operatorname{tr}\bar{Q}$, $M_4 = \mathbb{E}\|U\|^4$, and $V = \mathbb{E}\|U(U^\top y) - c\|^2$. It proves

$$V \geq \frac{[y^\top\bar{Q}^2(T_qI - \bar{Q})y]^2}{4M_4\|\bar{Q}y\|^2} \geq \frac{(m-1)g^3(T_q - g)}{4M_4}\|y\|^2.$$

For clarity the last step multiplies the two spectral lower bounds $(m-1)g\|\bar{Q}y\|^2$ and $g^2(T_q - g)\|y\|^2$ for the numerator before squaring. Every eigenvalue lies in $[g, T_q - (m-1)g]$, which verifies both bounds. With $g = \gamma/2$, $T_q \geq mg$ and $M_4 \leq m^2X^{16}$, at least one coordinate satisfies

$$\mathrm{Var}[U_a(U^\top y) \mid \mathcal{F}] \geq \sigma_0^2 := Y^2\gamma^4/(256X^{16}).$$

Choose the smallest maximizing index, which is $\mathcal{F}$-measurable. The centered product has third absolute moment at most $8m^3Y^3X^{24}$: use $\sum_b |y_b| \le mY$, the conditional $L^6$ bound, and $\mathbb{E}|V-\mathbb{E}V|^3 \le 8\mathbb{E}|V|^3$. Thus $M_3^2/\sigma_0^6 \le 2^{30}m^6X^{96}\gamma^{-12}$. Zero readout gives the exact identity

$$\dot{f}_n(0,x_a) = \frac{2}{mn}\sum_i U_{i,a}(U_i^\top y).$$

The second run and the conditional means are only a measurable shift. Apply (98) with $h = \tau\sigma_0/24$; its scalar failure is at most $\tau/4$ under the second gate of (96). Hence, allowing the covariance failure, with probability at least $1-\tau/2$,

$$\max_a |\dot{f}_n(0,x_a) - \dot{\tilde{f}}_n(0,x_a)| \ge \frac{\tau Y\gamma^2}{192mX^8\sqrt{n}}.$$

Use the training rectangle with failure $\tau/8$ for each run. The parameter-two ellipse on $[0,r]$ lies inside it, and the prediction difference is bounded by $32X^6Y/\lambda$. Taking $K = \lceil 2\ell_n/\log 2\rceil \le 4\ell_n$ in Lemma 5.1 yields

$$\|f_n - \widetilde{f}_n\|_{\mathcal{X}} \ge \frac{\tau Y\gamma^2}{6144mu_{\tau/8}X^{108}(1+\lambda)\sqrt{n}\ell_n^{5/2}} - \frac{768X^6Y}{\lambda n^2}.$$

The last term is at most half the first whenever $n^{3/2} \ge 2^{24}u_{\tau/8}X^{114}m(1+\lambda)\ell_n^{5/2}/(\lambda\tau\gamma^2)$. Since $\ell_n^{5/2} \le 64n^{1/4}$, the third gate in (96) suffices. The probability union costs less than $\tau$ and the stated coefficient is smaller than the remaining leading coefficient. Both predictors start at zero, so the witness time is strictly positive. At training inputs their fitted limits agree with the same label; no endpoint separation follows.

□

### 6.3 The third-order Legendre history estimate

The Legendre model is exactly (70) and (71), with its constant forward and zero backward prefixes. Let

$$q = \left\lceil \lambda^{-3}n^{1/4}\ell_n^2 e^{\sqrt{\ell_n}/2}\right\rceil, \qquad P = X^{500}\chi^{10}. \tag{99}$$

For fixed positive $a, b$, define the numerical logarithm coefficient

$$C(a,b) = \max\{1, e^b(a/(eb))^a\}.$$

Maximizing $x^a e^{-bx}$ over $x \ge 1$ proves $\ell_n^a \le C(a,b)n^b$. Define the structural gate

$$N_{\mathrm{Leg}} = \left\lceil \max\left\{(2e^{100}\lambda^{-3})^8, (4P^3)^4, [8P^4C(2,1/64)]^8, d^4\right\}\right\rceil. \tag{100}$$

**Proposition 6.4** (Improved error without additional state)**.** *On the finite training-source event, for $n \ge N_{\mathrm{Leg}}$,*

$$\|f_{\mathrm{Leg}} - f_n\|_{\mathcal{X}} \le 4YP^5\ell_n^2 n^{-47/64}. \tag{101}$$

*The exact moving and fixed inventories remain $n(d+1)+1+2(L-1)mnq$ and $(L-1)n^2$, respectively.*

*Proof.* Use Euclidean mobility coordinates $\theta = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$. For a finite physical horizon $T_0$ put

$$d_{T_0} = \int_0^{T_0} \sum_{j=2}^{L} \|\mathcal{E}_j(t)\|_F dt, \qquad D_{T_0} = \sup_{t \le T_0} \sum_{j=1}^{L+1} \|\widehat{\theta}_j(t) - \theta_j(t)\|, \qquad A = \tau(T_0).$$

Here $\mathcal{E}_j$ is the exact physical defect in (73). The order-independent fitting lemma gives

$$\widehat{\rho}(t) \le Y e^{-\lambda t/4}, \qquad \int_t^{T_0} \widehat{\rho}(s) ds \le 4\widehat{\rho}(t)/\lambda, \qquad A \le 1 + 4Y/\lambda \le 2.$$

Work first on the bootstrap $D_{T_0} \le Y/\sqrt{n}$. Subtract backward recursions with the dense trajectory as reference, so changed gates multiply only its known coordinate carrier. The all-time response bound (88) and the exact recursion for $R$ then give

$$\max_{a,j} \|\widehat{k}_a^{(j)} - k_a^{(j)}\|_2/\sqrt{n} \le X^{40}(1 + \sqrt{\ell_n}) D_{T_0},$$

$$\max_{a,b,j} \|\widehat{R}_{ba}^{(j)} - R_{ba}^{(j)}\|_2/\sqrt{n} \le X^{130}\chi^4(1 + \sqrt{\ell_n}) D_{T_0}.$$

The nontrivial products can be checked directly: forward differences cost $X^5 D$, feature-pair differences $X^9 D$, and the changed response gate costs at most $Y X^{116}\chi^3(1 + \sqrt{\ell_n})D$ using the reference response maximum. Homogeneous propagation costs $(9s)^L \le X^2$, with at most $L \le X$ terms. Thus the displayed powers include all terms. One RMS-to-coordinate conversion and the bootstrap yield

$$\max |\widehat{k}| \le Y X^{50}\chi(1 + \sqrt{\ell_n}), \qquad \max |\widehat{R}| \le Y X^{135}\chi^4(1 + \sqrt{\ell_n}).$$

Let $\widehat{v}_a = \widehat{r}_a/\widehat{\rho}$ and let $\mathcal{F}$ be the ordinary dense vector field evaluated at the reconstructed closure state. At finite times $\widehat{\rho} > 0$ by uniqueness in the fitting proof; stationary continuation covers residual zeros. The normalized ordinary feature velocity is

$$a_b^{(j)}(t) = D_\theta h_b^{(j)}(\widehat{\theta}(t)) \frac{\mathcal{F}(\widehat{\theta}(t))}{\widehat{\rho}(t)} = -\frac{2}{m}\phi'(\widehat{z}_b^{(j)}) \odot \sum_a \widehat{v}_a \widehat{R}_{ba}^{(j)}.$$

The residual calculation in the Legendre fitting proof gives $\|\widehat{v}\|_2/\sqrt{m} = 1$ and $\|\dot{\widehat{v}}\|_2/\sqrt{m} \le X^{11}$. Differentiating the *actual* closure trajectory, including its defect, gives the following bounds:

$$\begin{aligned}
\max_{b,j} \|\dot{\widehat{z}}_b^{(j)}\|_2/\sqrt{n} &\le Y X^{30}\chi^3\widehat{\rho}, \\
\max_{a,j} \|\dot{\widehat{\delta}}_a^{(j)}\|_2/\sqrt{n} &\le X^{110}\chi^4(1 + \sqrt{\ell_n})\widehat{\rho}, \\
\max_{a,b,j} \|\dot{\widehat{R}}_{ba}^{(j)}\|_2/\sqrt{n} &\le X^{220}\chi^8(1 + \sqrt{\ell_n})\widehat{\rho}, \\
\max_{b,j} (\|a_b^{(j)}\|_2 + \|\dot{a}_b^{(j)}\|_2)/\sqrt{n} &\le Y X^{230}\chi^8(1 + \sqrt{\ell_n}).
\end{aligned} \tag{102}$$

To verify these estimates, (76) bounds the full hidden block speed by $Y X^{20}\chi^3\widehat{\rho}$; forward differentiation costs $X^5$. In $\dot{\delta} = \phi'' \dot{z}\, k + \phi' \dot{k}$, use the coordinate maximum of $k$ and RMS of $\dot{z}$. At the top $\|\dot{w}\|/\sqrt{n} \le X^4\widehat{\rho}$; below it, $\dot{k} = \dot{W}^\top \delta + W^\top \dot{\delta}$. The forcing coefficient is at most $X^{90}\chi^4(1 + \sqrt{\ell_n})$, covered by the second line after bounded layer propagation. Differentiating $R = \delta c + WQ$ adds $\dot{\delta} c$,

$\delta\dot{c}$, $\dot{W}Q$, $W\phi''\dot{z}R$ and $W\phi'\dot{R}$. The curvature product uses the coordinate maximum of $R$ and RMS of $\dot{z}$; all remaining terms use pairing derivatives and operator bounds. The largest forcing is below $X^{180}\chi^8(1+\sqrt{\ell_n})\widehat{\rho}$, giving the third line. Finally differentiate $a_b^{(j)}$: the three terms use $\dot{z}$ times the response maximum, $\dot{\widehat{v}}$ times response RMS, and $\dot{R}$ times $\|\widehat{v}\|_2/\sqrt{m}=1$. Since $\widehat{\rho}\le Y$, they give the fourth line. In particular no coordinate bound on $\dot{z}$ or $\sqrt{n}\|\mathcal{E}\|$ is introduced.

Subtract the accumulated physical feature defect from its history:

$$g_b^{(j)}(\tau(t)) = \widehat{h}_b^{(j)}(t) - \int_0^t D_\theta h_b^{(j)}(\widehat{\theta}(s))\mathcal{E}(s)\,ds,$$

and use its constant initial value on $[0,1]$. Aggregate history norms use $(mn)^{-1/2}$ over samples and neurons. The forward derivative bound gives $\|h-g\|_{L^2}/\sqrt{mn}\le X^{10}d_{T_0}$. Exactly $\partial_\tau g = a$ and $\partial_\tau^2 g = \dot{a}/\widehat{\rho}$ on $(1,A)$. The first derivative vanishes at the prefix join because the initial readout vanishes. Thus no distributional jump occurs there.

For $\mathscr{D}_A g = -\partial_\xi[\xi(A-\xi)\partial_\xi g]$, the inequality $(A-\tau(t))/\widehat{\rho}(t)\le 4/\lambda$ cancels the apparent terminal singularity. By (102),

$$\|\mathscr{D}_A g\|_{L^2}/\sqrt{mn} \le YX^{240}\chi^9(1+\sqrt{\ell_n}).$$

The shifted Legendre modes have eigenvalues $j(j+1)$. Two integrations by parts have zero endpoint flux and no prefix jump; Bessel's inequality then gives $\|(I-\Pi_q^A)g\|_{L^2}\le [q(q+1)]^{-1}\|\mathscr{D}_A g\|_{L^2}$. Projection contraction on $h-g$ yields the forward tail

$$H_{T_0} := \max_j \|(I-\Pi_q^A)h^{(j)}\|_{L^2}/\sqrt{mn} \le YP\ell_n/q^2 + Pd_{T_0}.$$

The backward-history freezing calculation following (78) is unchanged. Its coefficients satisfy $b_0\le X^{42}(Y/\lambda)(1+(Y/\lambda)\sqrt{\ell_n})$, $b_1\le X^{16}(Y/\lambda)\sqrt{\chi}$ and $B_d\sqrt{4Y/\lambda}\le P\ell_n$. Therefore, for the history $b_a = (\widehat{r}_a/\widehat{\rho})\widehat{\delta}_a$,

$$B_{T_0} := \max_j \|(I-\Pi_q^A)b^{(j)}\|_{L^2}/\sqrt{mn} \le YP\ell_n/q + P\ell_n D_{T_0} \le 2YP\ell_n/q \qquad (q\le\sqrt{n}).$$

For each fixed full history the growing-projection identity (67) integrates its squared endpoint error to its final squared tail norm, since initial errors vanish. Applying Cauchy–Schwarz in samples and activity time to the exact defect thus gives $d_{T_0}\le 2(L-1)H_{T_0}B_{T_0}$. The signed physical comparison (77) and its coefficient ledger give

$$D_{T_0}\le \mathcal{A}_n d_{T_0}, \quad \mathcal{A}_n \le eX\exp[X^{36}(Y/\lambda)^2\sqrt{\ell_n}] \le X^2 n^{1/64} \le Pn^{1/64}.$$

For the last inequality use $X^{36}(Y/\lambda)^2\le X^{-24}$ and $c\sqrt{\ell_n}\le \ell_n/64+16c^2$ with $c\le X^{-24}$. Absorb the term linear in $d_{T_0}$ when $q\ge 4P^3\ell_n$, using $Y\le 1$ and $2(L-1)\le P$. This yields

$$d_{T_0}\le 4YP^3\ell_n^2/q^3, \qquad D_{T_0}\le 4YP^4\ell_n^2 n^{1/64}/q^3. \tag{103}$$

The gate (100) ensures $q\le\sqrt{n}$, $q\ge 4P^3\ell_n$, and $8P^4\ell_n^2\le n^{15/64}$. For the first use $\ell_n^2 e^{\sqrt{\ell_n}/2}\le e^{100}n^{1/8}$ and absorb the ceiling by two. The second follows from $n^{1/4}\ge 4P^3$. For the third use $\ell_n^2\le C(2,1/64)n^{1/64}$. Consequently the bootstrap improves to $D_{T_0}\le Y/(2\sqrt{n})$. Continuity excludes a first exit. All constants are independent of $T_0$, so the estimate extends to all times and to the fitted limits. Real whole-sphere prediction subtraction costs at most $X^7 D_{T_0}$, proving (101). The exact inventory is (72); $n\ge d^4$ absorbs the first-matrix term in the original headline count.

□

### 6.4 Taylor accuracy, storage and final width gates

Use the same exact-metric selection and selected dynamics as Proposition 7.2. Its proof needs the real fitting, RMS, carrier, and finite source-approximation bounds; the polynomial source lemma supplies each of them on the capped panel domain. The explicit comparison ledger permits $C_{\mathrm{sel}} = X^{60}$: it gives $F \le X^6$, $K_1 \le X^{18}$, $H_C \le 2X^3$, $F_c \le X^{15}$ and $\mathcal{B}_n/(Y/\lambda) \le X^{22}(1+\sqrt{\ell_n})$. Using $e^v - 1 \le ve^v$, its finite-time and tail coefficients are below $C_{\mathrm{sel}}(Y/\lambda)(1+\lambda^{-1/2})(1+\sqrt{\ell_n})e^{\sqrt{\ell_n}}$ and $C_{\mathrm{sel}}(Y/\lambda)(1+\lambda^{-1/2})$, respectively. Choose exactly the tolerance

$$\eta = \min\left\{1, Y, 16Y/\lambda, \frac{\lambda}{2C_{\mathrm{sel}}n(1+\lambda^{-1/2})(1+\sqrt{\ell_n})e^{\sqrt{\ell_n}}}\right\}.$$

The finite-time error is at most $Y/(2n)$. The fitted-tail contribution is at most $Y/(2n)$ under the label-independent gate

$$n^7 \ge 2C_{\mathrm{sel}}\lambda^{-1}(1+\lambda^{-1/2}). \tag{104}$$

Indeed $ne^{-8\ell_n} \le n^{-7}$. The disk cover strictly contains the real interval in its open union, so some positive closed rectangle lies inside it by compactness. Thus the initialization-only compiler Proposition 7.3 applies without a new setup-cost claim.

Let $k$ be the dimension of the declared input span, $M_n = X^6\sqrt{n}$, and

$$K = \max\{0, \lceil \log_2(8M_n/\eta)\rceil\}, \qquad R = 2m + k + 1 + 2(2m+p)J(K+1).$$

On each half-radius panel the Taylor tail is at most $M_n 2^{-K} \le \eta/8$. Approximate each base coefficient in Euclidean norm to $\eta/128$ and apply its initialized matrix exactly. Geometric summation gives base coefficient error at most $\eta/64$ and image error at most $\eta/8$, since initialized operators have norm eight. This is the same exact-pairing source contract used in the Taylor proof. Input-span reduction preserves the panel trajectory and all Gaussian first-row laws; it uses no passive labels. The exact retained count remains

$$1020(L+1)R^2 + 36R + 10m(k+1) + pk + (m+p) + 3(m+p)d.$$

The additional gate

$$n \ge N_{\mathrm{count}} := [32X^6(1+C_{\mathrm{sel}})\lambda^{-3/2}(1+Y^{-1})]^4 \tag{105}$$

implies $K+1 \le 8\ell_n$. To check it, the logarithm defining $K$ is at most $(3/2)\log n + \sqrt{\ell_n} + \log(1+\sqrt{\ell_n})$ plus the logarithm of the bracket in this gate. The latter is at most $(1/4)\log n$, and the source gate gives $\ell_n \ge 16$, so each intervening term is at most $\ell_n/4$. Division by $\log 2$ and the ceiling leave the stated bound. The branches $\eta = Y$ and $\eta = 1$ are smaller. Squaring the rank bound with (90) and $k \le m+p$ yields

$$R^2 \le X^{810}(m+p)^2\left[(Y/\lambda)^4\ell_n^3 + \lambda^{-2}\ell_n^2 b_n^2\right].$$

The exact inventory is therefore bounded by the Taylor headline with, for example, coefficient $X^{820}$. The inverse-label gate belongs to this simplification only. Without it the exact $K$ displays the extra $\log(1/Y)$ dependence. Dense arrays, jets, coefficients and proof-panel anchors are discarded after initialization, exactly as before.

We finally prove Theorem 6.1. Allocate $\tau = \delta/2$ to the dense lower event and $\rho = \delta/4$ to the relevant source event. On their intersection the Legendre ratio is at most

$$4P^5 a_\tau^{-1} \ell_n^{9/2} n^{-15/64} \le 4P^5 a_\tau^{-1} C(9/2, 7/64) n^{-1/8}.$$

Consequently the following fully specified maximum is a sufficient Legendre width:

$$\left\lceil \max\left\{ N_{\mathrm{src}}(m, \delta/4), N_{\mathrm{low}}(\delta/2), N_{\mathrm{Leg}}, [4P^5 C(9/2, 7/64)/(a_{\delta/2}\varepsilon)]^8 \right\} \right\rceil. \tag{106}$$

For Taylor the absolute error $Y/n$ and the same lower event give ratio at most $a_\tau^{-1}\ell_n^{5/2}/\sqrt{n} \le 64 a_\tau^{-1} n^{-1/4}$. A sufficient width is

$$\left\lceil \max\left\{ N_{\mathrm{src}}(m+p, \delta/4), N_{\mathrm{low}}(\delta/2), N_{\mathrm{count}}, [2C_{\mathrm{sel}}\lambda^{-1}(1+\lambda^{-1/2})]^{1/7}, [64/(a_{\delta/2}\varepsilon)]^4 \right\} \right\rceil. \tag{107}$$

The probability union costs less than $\delta$; no independence between the source and comparison events is required. Since $\lambda^{-1} = m/\gamma$, $u_\rho$ is logarithmic in $mL/\rho$, $L \le X$, and $a_\tau^{-1}$ has only the displayed polynomial factors, these maxima are bounded by the polynomial asserted in the theorem. The proof counts retained real coordinates and exact-arithmetic error; it does not infer efficient preprocessing or finite precision from a polynomial sufficient width.

# 7 One selected nonlinear dynamics for both source constructions

The sparse-metric lemma first supplies exact geometry on any finite source space. The transfer proposition then constructs and analyzes one autonomous selected model, with its optimizer and comparison coefficients local to that proof. Two later propositions supply its source spaces from whole-sphere polynomials or finite-panel Taylor expansions. They use only the transfer statement's error and storage interface.

**Lemma 7.1** (Sparse coordinates with an exact source metric)**.** *Let $E \subset \mathbb{R}^n$ be a linear subspace containing $\mathbf{1}_n$, with $r = \dim E$. There are a set $I$ of at most $9r$ coordinates, a positive diagonal matrix $D$ on $I$, and a positive definite matrix $M$ such that*

$$u_I^\top M v_I = u^\top v/n \quad (u, v \in E), \qquad D/4 \preceq M \preceq D, \qquad \mathbf{1}^\top M \mathbf{1} = 1, \quad \mathbf{1}^\top D \mathbf{1} \le 4.$$

*Write $\|u\|_M = (u^\top M u)^{1/2}$. Consequently a coordinate multiplication operator $B$ satisfies $\|B\|_{M\to M} \le 2\max_i |B_{ii}|$, and an activation with $|\phi'| \le s$, $|\phi(0)| \le b$ satisfies $\|\phi(u) - \phi(v)\|_M \le 2s\|u - v\|_M$ and $\|\phi(u)\|_M \le 2b + 2s\|u\|_M$.*

*Proof.* We give the finite sparsification argument. For vectors $v_i \in \mathbb{R}^r$ with $\sum_i v_i v_i^\top = I$, start with $A = 0$, lower barrier $l = -3r$ and upper barrier $u = 6r$. Maintain

$$lI \prec A \prec uI, \qquad \Phi_l(A) = \mathrm{tr}(A - lI)^{-1} \le \tfrac{1}{3}, \quad \Phi^u(A) = \mathrm{tr}(uI - A)^{-1} \le \tfrac{1}{6}.$$

Set $l' = l + 1$, $u' = u + 2$, and define

$$U(v) = \frac{v^\top (u'I - A)^{-2} v}{\Phi^u(A) - \Phi^{u'}(A)} + v^\top (u'I - A)^{-1} v, \qquad L(v) = \frac{v^\top (A - l'I)^{-2} v}{\Phi_{l'}(A) - \Phi_l(A)} - v^\top (A - l'I)^{-1} v.$$

The lower potential implies that every eigenvalue is at least three above $l$, so both inverse matrices and both denominators are positive. The identity $(B + tvv^\top)^{-1} = B^{-1} - B^{-1}vv^\top B^{-1}/(t^{-1} + v^\top B^{-1} v)$ shows that $U(v) \le t^{-1} \le L(v)$ preserves the two potential inequalities at the shifted barriers. For the upper potential, the positive first term of $U(v)$ also preserves the strict spectral barrier.

Such a nonzero $v_i$ exists. If $a_j$ are the eigenvalues of $A$, then $(u-a_j)^{-1}-(u'-a_j)^{-1}\geq 2(u'-a_j)^{-2}$, hence $\sum_i U(v_i)\leq 1/2+\Phi^{u'}(A)\leq 2/3$. For the lower sum put $a'_j=(a_j-l')^{-1}$, $b'_j=(a_j-l)^{-1}=a'_j/(1+a'_j)$ and $\Delta=\sum_j a'_jb'_j=\Phi_{l'}-\Phi_l$. Weighted Cauchy–Schwarz gives

$$\Delta^2\leq\Big(\sum_j (a'_j)^2b'_j\Big)\sum_j b'_j\leq\sum_j (a'_j)^2-\Delta.$$

Therefore $\sum_i L(v_i)\geq 1-\Phi_l(A)\geq 2/3$. Choose $t>0$ with reciprocal in the resulting nonempty interval and add $tv_iv_i^\top$. After $9r$ iterations the barriers are $6r, 24r$. Dividing all accumulated weights by $6r$ gives nonnegative weights $s_i$, supported on at most $9r$ indices, with $I\preceq\sum_i s_iv_iv_i^\top\preceq 4I$.

Choose $U$ with range $E$ and $U^\top U/n=I$, apply this result to $v_i=U_{i,:}^\top/\sqrt{n}$, and set $P=U_I$, $D=\operatorname{diag}(s_i/n)$, $G=P^\top DP$ and $Z=D^{1/2}P$. Define

$$M=D^{1/2}\{ZG^{-2}Z^\top+I-ZG^{-1}Z^\top\}D^{1/2}. \tag{108}$$

On $\operatorname{ran} Z$ the middle matrix has eigenvalues those of $G^{-1}$, and on its orthogonal complement it is the identity. Thus $D/4\preceq M\preceq D$, while multiplication gives $P^\top MP=I$. The constant source proves the two mass assertions. Finally $\|Bu\|_M\leq\|Bu\|_D\leq\max_i|B_{ii}|\|u\|_D\leq 2\max_i|B_{ii}|\|u\|_M$. Apply this to coordinate divided differences and use $\|\mathbf{1}\|_M=1$ for the activation claims. □

### 7.1 A portable source-to-runtime construction

**Proposition 7.2** (Source-to-runtime transfer)**.** *Work on the fitting and analytic-source event of Lemma 2.1 and proposition 3.1, or its quantitative finite-panel version in Lemma 6.2. Write $W_0^{(j)}=W^{(j)}(0)$ for the initialized matrices in this statement. Fix*

$$T=32\frac{m}{\gamma}\log(en),\qquad 0<\eta\leq\min\{1,Y,16Ym/\gamma\},\qquad q\geq\max\{m,d\}.$$

*For each layer let $E_j\subset\mathbb{R}^n$ have dimension at most $q$ and contain the constant, exact initialized training features and their initialized forward images when present, and (at the first layer) the columns of $W_0^{(1)}$. Suppose these spaces approximate the families* (25) *to coordinate accuracy $\eta$ on $[0,T]$, with forward queries in $\mathcal{X}$ and backward queries at training inputs. Require exact initialized-image pairing: each image approximant is the corresponding initialized map applied to its base approximant. There is an autonomous selected nonlinear predictor satisfying*

$$\begin{aligned}\|f_{\mathrm{model}}-f_n\|_{\mathcal{X}}\leq{}& C\frac{Ym}{\gamma}\left(1+\sqrt{\frac{m}{\gamma}}\right)(1+\sqrt{\log(en)})e^{\sqrt{\log(en)}}\eta\\&+C\frac{Ym}{\gamma}\left(1+\sqrt{\frac{m}{\gamma}}\right)e^{-\gamma T/(4m)}.\end{aligned} \tag{109}$$

*It fits, has a fitted limit, and retains at most*

$$1020(L+1)q^2+36q+10m(d+1) \tag{110}$$

*real coordinates, including data, selected indices, metrics, initial copies and working arrays. Here $C\geq 1$. In particular, the permissible choice*

$$\eta=\min\left\{1,Y,\frac{16Ym}{\gamma},\frac{\gamma}{2Cmn(1+\sqrt{m/\gamma})(1+\sqrt{\log(en)})e^{\sqrt{\log(en)}}}\right\} \tag{111}$$

*gives error at most $Y/n$ eventually. Source bases, dense coefficients and compilation scratch are discarded after the selected initial arrays are formed.*

*Proof.* All auxiliary quantities in this proof belong to this construction. Put $\lambda = \gamma/m$, $z = Y/\lambda$, $\alpha = Y/\sqrt{\lambda}$, $S = 16z$, and $\ell = \log(en)$, so $0 < \lambda \le 1$ and $0 < z \le \beta^{-30L}$. For activation estimates use the envelopes $b = \beta - 1$ and $s = t_2 = \beta$; thus $|\phi_j(0)| \le b$, $|\phi_j'| \le s$ and $|\phi_j''| \le t_2$ on the real line. Write $c_n = (y_a - f_n(t, x_a))_{a=1}^m$ for the dense training deficits and $\rho_n = \|c_n\|_2/\sqrt{m}$ for their RMS. The public source bounds permit the local envelopes

$$u_0 = \beta^{3L}, \qquad \tau = \beta^{4L}, \qquad K_{\mathrm{src}} = \beta^{21L}.$$

Dense feature norms are at most $u_0$ and dense training response norms at most $S\tau$ in neuron RMS. Dense training carriers on $[0, T]$ are the local fields $k_{n,a}^{(L)} = w$ and $k_{n,a}^{(j)} = W^{(j+1)\top}\delta_{n,a}^{(j+1)}$; by (28) they are bounded coordinatewise by $32zK_{\mathrm{src}}\sqrt{\ell}$. Dense vector norms below are normalized by $\sqrt{n}$; selected norms are specified by their metrics. Every query $v$ denotes $x/\sqrt{d}$ for an input $x$ in the promised domain.

**Autonomous selected model.** Apply Lemma 7.1 to obtain $(I_j, M_j, D_j)$ and let $q_j = |I_j| \le 9q$. The selected moving variables are $W_C^{(1)} \in \mathbb{R}^{q_1 \times d}$, $B_C^{(j)} \in \mathbb{R}^{q_j \times q_{j-1}}$ $(2 \le j \le L)$, $w_C \in \mathbb{R}^{q_L}$ and $c_C \in \mathbb{R}^m$. For $v = x/\sqrt{d}$ the forward pass is

$$z_C^{(1)} = W_C^{(1)} v, \qquad z_C^{(j)} = B_C^{(j)} h_C^{(j-1)}, \qquad h_C^{(j)} = \phi_j(z_C^{(j)}).$$

All layer norms and adjoints below use $M_j$; in particular $(B_C^{(j)})^* = M_{j-1}^{-1} B_C^{(j)\top} M_j$. Let $V_C$ have columns $h_C^{(L)}(v_a)/\sqrt{m}$ and define

$$Q_C = V_C^* V_C, \qquad \widehat{w}_C = w_C + V_C Q_C^{-1}\{(y - c_C)/\sqrt{m} - V_C^* w_C\}, \qquad f_C(v) = \langle \widehat{w}_C, h_C^{(L)}(v)\rangle_{M_L}. \tag{112}$$

The statement's predictor is $f_{\mathrm{model}}(t, x) = f_C(t, x/\sqrt{d})$; physical time is suppressed in the local forward formulas. Thus $f_C(v_a) = y_a - c_{C,a}$ exactly. Define training signals by $k_{C,a}^{(L)} = \widehat{w}_C$, $\delta_{C,a}^{(j)} = \phi_j'(z_{C,a}^{(j)}) \odot k_{C,a}^{(j)}$ and $k_{C,a}^{(j)} = (B_C^{(j+1)})^* \delta_{C,a}^{(j+1)}$. The unnormalized sample matrix is

$$\begin{aligned} K_{C,ab} = {} & \langle h_{C,a}^{(L)}, h_{C,b}^{(L)}\rangle_{M_L} + \langle \delta_{C,a}^{(1)}, \delta_{C,b}^{(1)}\rangle_{M_1} v_a^\top v_b \\ & + \sum_{j=2}^{L} \langle \delta_{C,a}^{(j)}, \delta_{C,b}^{(j)}\rangle_{M_j} \langle h_{C,a}^{(j-1)}, h_{C,b}^{(j-1)}\rangle_{M_{j-1}}. \end{aligned} \tag{113}$$

The autonomous optimizer is

$$\begin{aligned} \dot{W}_C^{(1)} &= \frac{2}{m}\sum_a c_{C,a}\delta_{C,a}^{(1)} v_a^\top, & \dot{B}_C^{(j)} &= \frac{2}{m}\sum_a c_{C,a}\delta_{C,a}^{(j)} h_{C,a}^{(j-1)\top} M_{j-1}, \\ \dot{w}_C &= \frac{2}{m}\sum_a c_{C,a} h_{C,a}^{(L)}, & \dot{c}_C &= -2K_C c_C/m. \end{aligned} \tag{114}$$

These are prescribed directions; coordinate gates need not be self-adjoint in $M_j$, and no gradient interpretation of the corrected predictor is used.

For bases $U_j$ of $E_j$ with $U_j^\top U_j/n = I$, set $P_j = (U_j)_{I_j}$. Initialize

$$W_C^{(1)}(0) = (W_0^{(1)})_{I_1}, \quad w_C(0) = 0, \quad c_C(0) = y, \qquad B_C^{(j)}(0) = P_j \frac{U_j^\top W_0^{(j)} U_{j-1}}{n} P_{j-1}^\top M_{j-1}. \tag{115}$$

Since $P_j^\top M_j P_j = I$, these operators have norms at most eight. They reproduce an initialized forward image exactly whenever both members belong to the source spaces; their metric adjoints reproduce

paired reverse images. The exact additions therefore preserve the entire initialized training forward pass and Gram. All bases, dense arrays and source coefficients are discarded after these finite initial arrays are formed.

**Claim 10** (Independent fitting and endpoint tails)**.** *With the notation of this construction, the following local bounds hold. The selected dynamics exist for every $t \geq 0$, have $Q_C \succeq \lambda I/4$, fit the training labels and converge in parameter space. Write*

$$H_1^c = 2b + 16s, \quad H_j^c = 2b + 18sH_{j-1}^c, \quad H_c = H_L^c, \quad d_j^c = 2s(18s)^{L-j}, \quad F_c = (d_1^c)^2 + 4H_c^2 \sum_{j=2}^{L} (d_j^c)^2.$$

*For the selected parameter tuple $\theta_C = (W_C^{(1)}, (B_C^{(j)})_{j=2}^L, w_C)$ use*

$$\|\theta_C\|_{\mathrm{par}}^2 = \mathrm{tr}(W_C^{(1)\top} M_1 W_C^{(1)}) + \sum_{j=2}^{L} \|M_j^{1/2} B_C^{(j)} M_{j-1}^{-1/2}\|_F^2 + \|w_C\|_{M_L}^2.$$

*Then $\|h_C^{(j)}\| \leq 2H_c$, $\|\widehat{w}_C\| \leq 5\alpha$, $\rho_C := \|c_C\|/\sqrt{m} \leq Ye^{-\lambda t/2}$ and $\int_0^\infty \|\dot{\theta}_C\|_{\mathrm{par}} \leq 2\alpha$. There is a coefficient $\mathcal{D} \leq Cz(1 + \lambda^{-1/2})$ such that comparison on $[0, T]$ extends to all times and the fitted endpoint with additional error at most $\mathcal{D}e^{-\lambda T/4}$. Here and below $C$ depends only on activations and depth.*

*Proof of the claim.* Stop at operator norm nine, feature norm $2H_c$, or Gram margin $\lambda/4$. Expanding the velocities, with all cross-sample terms, gives

$$-\frac{d}{dt}\rho_C^2 = 4c_C^\top K_C c_C/m^2 = \|\dot{\theta}_C\|_{\mathrm{par}}^2, \qquad K_C/m \succeq Q_C. \tag{116}$$

Thus $-\dot{\rho}_C \geq \lambda\rho_C/2$ and $\|\dot{\theta}_C\| \leq 2(-\dot{\rho}_C)/\sqrt{\lambda}$. Integration gives the asserted residual and path bounds, as well as $\int \rho_C \leq 2z$ and $\|w_C\| \leq 2\alpha$. With $T_C = V_C Q_C^{-1}$ and $P_C = T_C V_C^*$, the readout is the orthogonal sum $(I - P_C)w_C + T_C(y - c_C)/\sqrt{m}$. Its squared norm is at most $4\alpha^2 + 16\alpha^2$.

Set $R_c = 5\alpha$, $U_1^c = d_1^c$, $U_j^c = 2H_c d_j^c$ for $j \geq 2$, and $F_1^c = 2sU_1^c$, $F_j^c = 2s(2H_c U_j^c + 9F_{j-1}^c)$. The latter recurrence has $F_L^c = F_c$. Backward propagation bounds $\|\delta_C^{(j)}\| \leq d_j^c R_c$, and integration bounds each hidden parameter displacement by $20U_j^c Y^2/\lambda^{3/2}$ and each feature displacement by $20F_j^c Y^2/\lambda^{3/2}$. Under the stated cap, $H_c \leq \beta^{3L}$, $F_c \leq \beta^{15L}$ and $z \leq (16H_c\sqrt{F_c})^{-1}$; hence the last bound is at most $5\sqrt{\lambda}/(64H_c^2) < \min\{1/(8H_c), \sqrt{\lambda}/8\}$. The operator and feature stops improve strictly. The normalized training feature matrix starts with smallest singular value at least $\sqrt{\lambda/2}$ and moves by less than $\sqrt{\lambda}/8$, so the Gram stop improves too. Smoothness on this strict Gram-positive domain and bounded parameters give global existence. Finite path length and (112) give convergence; $c_C \to 0$ gives fitting.

For quantitative tails, differentiating the inverse gives

$$\dot{T}_C = (I - P_C)\dot{V}_C Q_C^{-1} - T_C \dot{V}_C^* T_C, \qquad \dot{P}_C = (I - P_C)\dot{V}_C T_C^* + T_C \dot{V}_C^* (I - P_C).$$

Thus $\|\dot{T}_C\| \leq 8\|\dot{V}_C\|/\lambda$, $\|\dot{P}_C\| \leq 4\|\dot{V}_C\|/\sqrt{\lambda}$ and $\|\dot{V}_C\| \leq 2R_c F_c \rho_C$. Put $G_c = 4H_c^2 + R_c^2 F_c$ and

$$B_w = 4H_c + 48R_c F_c z + 4G_c/\sqrt{\lambda}, \qquad B_f = 2H_c B_w + 2R_c^2 F_c.$$

Differentiating the orthogonal readout formula proves $\|\dot{\widehat{w}}_C\| \leq B_w \rho_C$ and $|\dot{f}_C(v)| \leq B_f \rho_C$ uniformly over unit inputs. Therefore $|f_C(\infty, v) - f_C(t, v)| \leq 2B_f z e^{-\lambda t/2}$. For the dense tail, the RMS

bounds hold at every real time, not merely on the finite complex source domain. Indeed Lemma 2.1 bounds all normalized operators by nine and gives $\|w\|/\sqrt{n} \le 2Y/\sqrt{\lambda} \le Su_0$. The all-time RMS bounds of Proposition 3.1, or the same real forward/backward induction, give $\|h^{(j)}\|/\sqrt{n} \le u_0$ and $\|\delta^{(j)}\|/\sqrt{n} \le S\tau$ for all real times. The flow equations and sample Cauchy–Schwarz therefore imply $|\dot f_n(v)| \le 2\mathcal{K}\rho_n$, where

$$\mathcal{K} = u_0^2 + S^2\tau^2[1 + (L-1)u_0^2].$$

Comparing each flow with its value at $T$ proves the claimed extension with $\mathcal{D} = z(16\mathcal{K} + 4B_f)$. The small cap gives $\mathcal{K} \le C$, $B_f \le C(1+\lambda^{-1/2})$, which establishes the stated parameter dependence. □

**Comparison coefficients.** Here is an explicit coefficient ledger for the comparison. Its size is independent of the number of sources. Use the local envelopes $u_0, \tau, K_{\mathrm{src}}$ fixed at this proof's start, and put

$$H_r = u_0 + 3, \quad H_C = 2H_c, \quad P_h = 6u_0 + 9, \quad P_\delta = 6\tau + 9,$$
$$A_f = 18 + S^2(\tau+3)P_h, \qquad A_b = 18 + S^2 H_r P_\delta.$$

Define the forward-error recursion and the reference-response bounds by

$$\begin{aligned}
&F_1^z = 1, \quad F_1^h = 2s,\\
&F_j^z = 9F_{j-1}^h + H_r + A_f, \quad F_j^h = 2sF_j^z \quad (j \ge 2), \quad F = \max_j F_j^h,\\
&d_j^E = 2s(9s)^{L-j} + 3H_c, \quad J_C = 5\sqrt{F_c},\\
&J_R = \left[(d_1^E)^2 + H_r^2 \sum_{j=2}^{L} (d_j^E)^2\right]^{1/2},\\
&D_s = P_h + 16zP_\delta\{1 + (L-1)H_r^2\} + (L-1)(16z)^2\tau^2 P_h.
\end{aligned}$$

The backward-error recursion is

$$\begin{aligned}
B_L &= 2s(1 + 6zF + 32zP_h) + 2t_2F_L^z,\\
B_j &= 18sB_{j+1} + 2s(d_{j+1}^E zH_C + A_b) + 2t_2F_j^z \quad (j < L),\\
B_h &= \sqrt{L}\{H_C \max_j B_j + (\max_j d_j^E) zH_C F\},\\
K_1 &= 12F + 2B_h + 20z(J_C + J_R)B_h + 20D_s.
\end{aligned}$$

Increasing any of these positive coefficients is harmless. Finally set

$$\begin{aligned}
\mathcal{B}_n &= 400z^2F_c + 2zK_1 + 40zF + 64z^2K_1K_{\mathrm{src}}\sqrt{\ell},\\
\mathcal{A}_n &= [H_C(1+6zF) + 3\alpha F](1+\lambda^{-1/2})(e^{\mathcal{B}_n} - 1)\\
&\quad + 6H_CzF + 32H_CzP_h/\sqrt{\lambda} + 3\alpha F + 16zP_h.
\end{aligned} \tag{117}$$

**Claim 11** (Finite-horizon comparison)**.** *With the local coefficients* (117)*,*

$$\sup_{0\le t\le T}\sup_{x\in\mathcal{X}} |f_C(t, x/\sqrt{d}) - f_n(t,x)| \le \mathcal{A}_n\eta.$$

*Proof of the comparison claim.* We record both the source-energy estimate and the cancellation that make the parameter dependence possible. For one layer abbreviate $E = E_j$, $I = I_j$ and $M = M_j$. If $p \in E$ approximates $u$ to coordinate accuracy $e_u$, exact isometry and the mass bound give

$$\|u - p\|_n \le e_u, \quad \|u_I - p_I\|_M \le 2e_u, \quad \|u_I\|_M \le \|u\|_n + 3e_u.$$

Expanding two inner products about $p, q \in E$ therefore gives

$$|\langle u_I, v_I\rangle_M - \langle u, v\rangle_n| \le 3e_u\|v\|_n + 3e_v\|u\|_n + 9e_u e_v. \tag{118}$$

Thus feature-pair defects are at most $P_h\eta$, and response-pair defects at most $SP_\delta\eta$; here $\eta \le 1, S$ is used.

Let the proof reference have first layer and readout equal to the coordinate restrictions of the dense ones, and hidden operators

$$B_R^{(j)}(t) = B_C^{(j)}(0) + \frac{2}{m}\int_0^t \sum_a c_{n,a}(u)(\delta_{n,a}^{(j)}(u))_{I_j}(h_{n,a}^{(j-1)}(u))_{I_{j-1}}^\top M_{j-1}\,du.$$

Its preactivation, feature and response symbols below mean the restricted dense values, not a fresh forward pass; its carrier symbols mean $k_{R,a}^{(j)} = (k_{n,a}^{(j)})_{I_j}$ as well. The exact initialized pairing and operator bound eight make an initialized action defect at most $18\eta$. In each learned action, subtract the dense outer-product integral and use (118) together with $\int \rho_n \le 2z$, $\|\delta_n^{(j)}\|_n \le S\tau$ and $\|h_n^{(j)}\|_n \le u_0$. The resulting forward and reverse defects are at most $A_f\eta$ and $A_b\eta$. The identical calculation for the integrated raw readout gives

$$|\langle w_R, h_R^{(L)}(v)\rangle - f_n(v)| \le 16zP_h\eta. \tag{119}$$

Write $V_n, V_R$ for the dense and restricted top feature maps, with columns divided by $\sqrt{m}$. Their common approximants and exact isometry give, for every $a \in \mathbb{R}^m$, $\|V_R a\| \le \|V_n a\| + 3\eta\|a\|$. For $\nu = \|V_R c_n/\sqrt{m}\|$, the dense energy identity gives

$$\int_0^T \nu \le \alpha + 6\eta z \le 2\alpha. \tag{120}$$

Here let $\theta_n = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$ with its Euclidean block norm. Indeed $2\|V_n c_n/\sqrt{m}\| \le \|\dot\theta_n\|$, whose integral is at most $2\alpha$. Integrating source approximants for $\dot w_n = 2V_n c_n/\sqrt{m}$ gives a vector in $E_L$ at coordinate error at most $4\eta z$. Consequently $\|w_R\| \le 2\alpha + 12\eta z \le 3\alpha$. Likewise the restricted responses have norm at most $d_j^E\alpha$. These arguments never differentiate an approximation error.

Use the hidden-parameter Hilbert norms of the preceding local fitting claim, and let $\theta_{h,C}, \theta_{h,R}$ denote the selected and reference tuples of first-layer and hidden-layer matrices. Let $\mathcal{J}_C, \mathcal{J}_R$ have sample columns, divided by $\sqrt{m}$, equal respectively to the tuples

$$(\delta^{(1)} v_a^\top, (\delta^{(j)} h_a^{(j-1)\top} M_{j-1})_{j=2}^L).$$

Thus $K_C/m = V_C^* V_C + \mathcal{J}_C^*\mathcal{J}_C$ and $\dot\theta_{h,C} = 2\mathcal{J}_C c_C/\sqrt{m}$; the reference satisfies the latter identity with $R, n$. Moreover $\|\mathcal{J}_C\| \le J_C\alpha$, $\|\mathcal{J}_R\| \le J_R\alpha$. Put

$$a_h = \|\theta_{h,C} - \theta_{h,R}\|, \quad e = (c_C - c_n)/\sqrt{m}, \quad u = \|e\|, \quad z_w = w_C - w_R,$$

$$T_C = V_C Q_C^{-1}, \quad P_C = T_C V_C^*, \quad p = T_C e, \quad \zeta = z_w + p, \quad b_e = \|p\| + \|\zeta\|, \quad E = a_h + b_e.$$

All these error variables vanish at zero. Forward induction, first subtracting the reference action and then the nonlinear gate, gives

$$\|h_C^{(j)} - h_R^{(j)}\| \le F_j^h(a_h + \eta), \qquad \|V_C - V_R\| \le F(a_h + \eta). \tag{121}$$

The first-layer preactivation error is $a_h$; at layer $j$ it is at most $9F_{j-1}^h(a_h+\eta)+H_r a_h + A_f\eta$, which proves the displayed recursion rather than presuming an ambient Lipschitz bound.

Set $d_R = V_R^* w_R - (y-c_n)/\sqrt{m}$ and $\Delta V = V_C - V_R$. Subtracting the exact readout formula yields

$$\widehat{w}_C - w_R = (I-P_C)z_w - p - T_C\Delta V^* w_R - T_C d_R. \tag{122}$$

Since $p \in \operatorname{ran} V_C$, $(I-P_C)z_w = (I-P_C)\zeta$. Equations (119)–(121) and $\|T_C\| \le 2/\sqrt{\lambda}$ imply

$$\|\widehat{w}_C - w_R\| \le b_e + 6zF(a_h+\eta) + 32zP_h\eta/\sqrt{\lambda}, \tag{123}$$

$$|f_C(v) - f_n(v)| \le H_C\{b_e + 6zF(a_h+\eta) + 32zP_h\eta/\sqrt{\lambda}\} + 3\alpha F(a_h+\eta) + 16zP_h\eta. \tag{124}$$

Let $K_n$ be the dense training tangent Gram: it is the formula (113) with dense fields and the inner products $n^{-1}u^\top v$ in each neuron layer. The dense deficit therefore obeys $\dot{c}_n = -2K_n c_n/m$. The source-only Gram defect $D = V_R^* V_R + \mathcal{J}_R^*\mathcal{J}_R - K_n/m$ satisfies $\|D\| \le D_s\eta$. To see this without an extra factor $m$, split each product of two pairings, use (118), and recall that the sample matrix entries carry the factor $1/m$; a matrix all of whose entries have magnitude at most $a/m$ has operator norm at most $a$. With $\Delta\mathcal{J} = \mathcal{J}_C - \mathcal{J}_R$, write

$$\Delta\mathcal{K} = V_C^*\Delta V + \Delta V^* V_R + \mathcal{J}_C^*\Delta\mathcal{J} + \Delta\mathcal{J}^*\mathcal{J}_R + D.$$

The error equations are exactly

$$\begin{aligned}
\dot{e} &= -2(Q_C + \mathcal{J}_C^*\mathcal{J}_C)e - 2\Delta\mathcal{K}c_n/\sqrt{m},\\
\dot{z}_w &= 2V_C e + 2\Delta V c_n/\sqrt{m},\\
\dot{\zeta} &= 2\Delta V c_n/\sqrt{m} + \dot{T}_C e - 2T_C\mathcal{J}_C^*\mathcal{J}_C e - 2T_C\Delta\mathcal{K}c_n/\sqrt{m}.
\end{aligned}$$

The term $2V_C e$ cancels because $T_C Q_C = V_C$. This is why estimating $c_C - c_n$ by an unconstrained Gronwall inequality would lose the result.

Here are the needed norm estimates, including the dissipative one. The derivative formula for $T_C$, and $\|\dot{V}_C\| \le 10\alpha F_c\rho_C$, give $\|\dot{T}_C e\| \le 40zF_c\rho_C\|p\|$. In the derivative of $\|p\|^2/2$, the $Q_C$ term is $-2u^2$. The possible absolute contribution from the hidden Gram is at most $8\|\mathcal{J}_C\|^2u^2/\lambda \le 200z^2F_c u^2$; the label cap bounds this by $25u^2/(32H_c^2)$, leaving at least $39u^2/32$. Define

$$\mathfrak{f} = 2\|T_C\Delta\mathcal{K}c_n/\sqrt{m}\|, \qquad I(t) = \int_0^t \{40zF_c\rho_C b_e + \mathfrak{f}\}.$$

Dividing the dissipative inequality by $\|p\|$ gives

$$\|p(t)\| + \int_0^t u^2/\|p\| \le I(t), \qquad \int_0^t u \le 2I(t)/\sqrt{\lambda}.$$

At zeros use $(\|p\|^2+\epsilon^2)^{1/2}$ and let $\epsilon \downarrow 0$; $p=0$ implies $e=0$ because $Q_C$ is invertible. The second inequality also follows directly from $\|p\| \le 2u/\sqrt{\lambda}$. Integrating the $\zeta$ equation bounds its hidden-Gram term by $200z^2F_cI \le 25I/32$. Therefore

$$b_e \le 2F\int_0^t \rho_n(a_h+\eta) + 3I(t). \tag{125}$$

For completeness, the remaining response estimate can be checked layer by layer. The dense training preactivation carriers are bounded coordinatewise by $M = 1+32zK_{\mathrm{src}}\sqrt{\ell}$ on $[0,T]$. At each gate split $\phi'(z_C)k_C - \phi'(z_R)k_R$ as $\phi'(z_C)(k_C-k_R) + (\phi'(z_C)-\phi'(z_R))k_R$. The first multiplication

costs $2s$, and the second at most $2t_2MF_j^z(a_h+\eta)$. Subtracting the initialized and learned reverse actions costs $A_b\eta$, and the hidden parameter difference costs $d_{j+1}^E\alpha a_h$. Starting with (123), the displayed $B_j$ recursion, followed by the outer-product split, gives

$$\|\Delta\mathcal{J}\| \le B_h\{b_e + M(a_h+\eta) + \eta/\sqrt{\lambda}\}.$$

Only the carrier factor contains $M$; it is not multiplied once per layer. Applying this to the four terms of $\Delta\mathcal{K}$ gives

$$\begin{aligned}\mathfrak{f} \le{}& 2F\rho_n(a_h+\eta) + 4F\nu(a_h+\eta)/\sqrt{\lambda} + 4D_s\eta\rho_n/\sqrt{\lambda}\\ &+ 4z(J_C+J_R)B_h\rho_n\{b_e + M(a_h+\eta) + \eta/\sqrt{\lambda}\}.\end{aligned}$$

The hidden-parameter equation, $\int u \le 2I/\sqrt{\lambda}$ and $4zJ_C \le 5/4$ similarly yield

$$a_h \le \tfrac{5}{4}I + 2B_h\int_0^t \rho_n\{b_e + M(a_h+\eta) + \eta/\sqrt{\lambda}\}.$$

Combining with (125) proves the scalar inequality

$$E(t) \le \int_0^t \{200zF_c\rho_C + K_1M\rho_n + 20F\nu/\sqrt{\lambda}\}\{E + \eta(1+\lambda^{-1/2})\}.$$

The integral of its coefficient is at most $\mathcal{B}_n$, by $\int\rho_n, \int\rho_C \le 2z$ and (120). The integrating-factor formula gives $E \le \eta(1+\lambda^{-1/2})(e^{\mathcal{B}_n}-1)$. Substitution in (124) proves the finite-horizon claim with precisely (117). □

**Canonical error bound and retained inventory.** The local fitting claim supplies the tails and endpoint. Thus

$$\|f_{\mathrm{model}} - f_n\|_{\mathcal{X}} \le \mathcal{A}_n\eta + \mathcal{D}e^{-\lambda T/4}.$$

We verify that the explicit ledger has the promised dependence. Put $X = \beta^L \ge 100$. The local envelopes and fitting bounds give $u_0, H_c \le X^3$, $\tau \le X^4$ and $K_{\mathrm{src}} \le X^{21}$. For example $16z \le 16X^{-30}$ makes both additional terms in $A_f, A_b$ less than one; hence $A_f, A_b \le 19$. The forward recursion then gives $F \le X^6$. For further detail, $d_j^E \le X^4$, $H_C \le 2X^3$ and $P_h, P_\delta \le X^4, X^5$. In the backward recursion the additive term is at most $X^7$, and $18s \le \beta^3$. Geometric summation over at most $L$ layers gives $\max_j B_j \le X^{11}$ and then $B_h \le X^{15}$. Also $J_C, J_R \le X^9$ and $D_s \le X^5$. Substitution gives $K_1 \le X^{18}$; here the generous powers absorb $L \le X$, all numerical factors, and $H_C = 2H_c$. The selected fitting recursion already gives $F_c \le X^{15}$. Consequently

$$\mathcal{B}_n \le 400X^{-45} + 2X^{-12} + 40X^{-24} + 64X^{-21}\sqrt{\ell} \le 1 + \sqrt{\ell},$$

and $\mathcal{B}_n/z \le C(1+\sqrt{\ell})$ with $C$ depending only on $X$. Since $e^u - 1 \le ue^u$, (117) gives

$$\mathcal{A}_n \le Cz(1+\lambda^{-1/2})(1+\sqrt{\ell})e^{\sqrt{\ell}}, \qquad \mathcal{D} \le Cz(1+\lambda^{-1/2}),$$

after increasing the activation/depth constant to absorb $e$. These are exactly the canonical factors in (109). This is a small-cap proof, not a claim on a larger recurrence-defined label interval. Finally $T = 32\ell/\lambda$ makes the tail $\mathcal{D}e^{-8\ell}$. For fixed positive $Y$ it is at most $Y/(2n)$ eventually; the other half is (111).

**Retained inventory.** Retain both moving and initial copies of the first matrix, mixers and raw readout; the selected indices; each $M_j$, its inverse or factorization, and $D_j$; the training data and labels; current training forward and backward arrays; $c_C$, $V_C$, $Q_C$ and its inverse or factorization; the corrected readout; direction arrays; and one-query forward workspace. Every selected width is at most $9q$. Each mixer or metric therefore costs at most $81q^2$, each first matrix at most $9qd$, each layer's training arrays at most a fixed multiple of $9mq$, and each sample matrix at most $m^2$. Since $m, d \le q$, these arrays total at most $1020(L+1)q^2 + 36q + 10m(d+1)$, including simultaneous initial copies and working arrays. A nonprimitive activation evaluator is charged separately. Dense mixers, source bases, coefficient vectors and coefficient-compilation scratch are not retained by the runtime. The construction itself uses the finite source spaces only through its initial arrays. The next compiler establishes initialization-jet provenance; Proposition 1.2 supplies separate efficient numerical implementations of the same source-to-runtime interface. □

## 7.2 Finite initialized jets suffice

**Proposition 7.3** (Initialization-only coefficient compiler)**.** *On the initialized operator event of Lemma 2.1, fix $T, r_0, M > 0$. Let finitely many sources from Proposition 3.1 extend coordinatewise holomorphically near $[-r_0, T + r_0] + i[-r_0, r_0]$, bounded by $M$ uniformly over their compact query domain. For sphere queries assume also the joint holomorphic extension of Proposition 3.1. Any finite list of their integrals against specified polynomial time–sphere bases, or derivatives at real anchors in $[0, T]$, can be approximated to arbitrary positive accuracy using finitely many derivatives of the dense ODE and passive recursions at time zero, with exact initialized-image pairing. No later dense state or trajectory sample is needed; no bound on initialization time or working storage is asserted.*

*Proof.* We first prove the continuation assertion that makes initialized jets sufficient. For any one scalar source $g$, use its common bound $M$ on the closed rectangle $\{-r_0 \le \Re t \le T + r_0,\ |\Im t| \le r_0\}$, with a slightly larger open domain. The source rectangle, or a finite closed-disk cover with the real interval strictly inside its union, supplies such a positive $r_0$ by compactness. Put

$$\chi = \frac{\pi T}{8r_0}, \quad b_0 = \tanh\chi, \quad b_1 = \tanh(\chi + \pi/4), \quad \theta = b_0/b_1, \quad \psi(\xi) = \frac{\xi - \theta}{1 - \theta\xi},$$

$$\mathfrak{t}(\xi) = \frac{T}{2} + \frac{2r_0}{\pi} \log \frac{1 + b_1\psi(\xi)}{1 - b_1\psi(\xi)}, \qquad \xi_* = \frac{2\theta}{1 + \theta^2} < 1.$$

The logarithm is the branch equal to a real number on the real interval. For $|\xi| < 1$, $|\psi(\xi)| < 1$; the ratio in the logarithm has positive real part and modulus between $(1 - b_1)/(1 + b_1)$ and its reciprocal. Thus $|\Im\mathfrak{t}| < r_0$ and $-r_0 < \Re\mathfrak{t} < T + r_0$. Direct substitution gives $\mathfrak{t}(0) = 0$ and $\mathfrak{t}(\xi_*) = T$, and the derivative is nonzero on $[0, \xi_*]$. This proves continuation from the origin onto the entire physical interval, without stepping a dense trajectory.

The Taylor coefficients of $g \circ \mathfrak{t}$ are

$$a_j = \sum_{k=0}^{j} \frac{g^{(k)}(0)}{k!} [\xi^j]\mathfrak{t}(\xi)^k.$$

Cauchy's coefficient formula gives $|a_j| \le M$, so truncation at $N$ has error at most $M\xi_*^{N+1}/(1 - \xi_*)$ on the whole interval. This bound is uniform over all passive inputs in the compact domain and over all source coordinates. Derivatives at finitely many later anchors are also recovered: choose $\xi_* < \rho_* < 1$ and $0 < \epsilon_* < \rho_* - \xi_*$. On $|\xi| \le \rho_*$ the tail is $M\rho_*^{N+1}/(1 - \rho_*)$, hence its $k$th derivative

at any real anchor is at most $k!M\rho_*^{N+1}/\{\epsilon_*^k(1-\rho_*)\}$. Repeated application of $(\mathfrak{t}')^{-1}\partial_\xi$ converts these to the finitely many required time derivatives. All its coefficients are bounded on that compact interval. A finite $N$ therefore achieves every prescribed finite list of accuracies.

Let $\Theta = (W^{(1)}, \dots, W^{(L)}, w)$ be the dense parameter tuple and $\mathcal{F}$ its autonomous vector field. For the initialized ODE $\dot{\Theta} = \mathcal{F}(\Theta)$ the formal coefficients are computed successively by

$$\Theta_{j+1} = \frac{1}{j+1}[t^j]\mathcal{F}\left(\sum_{i=0}^{j}\Theta_i t^i\right).$$

The analytic local solution has precisely these coefficients, by coefficient comparison and uniqueness of the local ODE. The forward and passive backward recursions then give their source jets by finite addition, multiplication and composition at time zero. Any scalar activation derivatives required here are part of the fixed analytic activation evaluator; their description, evaluation and scratch are not asserted to have unit cost unless taken as primitives.

For time–sphere integral coefficients, insert the uniformly convergent continued polynomials into their defining integrals. Each integrand is continuous on the compact time interval times sphere. Finite partitions and weighted Riemann sums converge uniformly for the finite collection of integrands. A finite mesh can be chosen from their modulus of continuity: the source strip gives derivative bounds by Cauchy's formula, and the finitely many basis functions are known polynomials. Thus the required integral approximations use only finitely many evaluations of the continued origin-jet polynomial. For $d = 1$ the spatial integral is simply the two-point average.

Finally perform every scalar linear coefficient operation on a base source and set its image coefficient equal to the initialized matrix times that vector. Linear operations commute with this fixed matrix. Since its Euclidean operator norm is at most eight, approximating a base coefficient in Euclidean norm to $\epsilon/8$ approximates both the base and its image to $\epsilon$. There are only finitely many coefficients and a finite sum of their basis suprema; choose each coefficient tolerance smaller than the desired final error divided by this sum. This imposes exact pairing while retaining the chosen uniform source tolerance. No coefficient is retained after the selected initial arrays are formed. □

### 7.3 Whole-sphere polynomial sources

**Lemma 7.4** (Joint time–sphere approximation and its dimension)**.** *Let $T, r_t, r_q, M_n > 0$ and $0 < \eta \le \min\{1, M_n\}$, and put $\alpha_T = r_t/(4T)$. Suppose a real $\mathbb{R}^n$-valued source on $[0,T] \times S^{d-1}$ extends coordinatewise holomorphically to a neighborhood of*

$$([-r_t, T+r_t] + i[-r_t, r_t]) \times \{v \in \mathbb{C}^d : v^\top v = 1,\ \|\operatorname{Im} v\|_2 \le \sinh r_q\},$$

*with coordinate modulus at most $M_n$ on this set. Query holomorphy is relative to the quadric; for $d = 1$ use its two real points. For $d \ge 2$ put*

$$b_d = 2d-2, \quad D_d = 2^{d+1}d^{d-2}, \quad P_T = \frac{18D_d b_d!\, 2^{b_d+1}}{\alpha_T r_q^{b_d+1}}, \quad H_* = 2\log\frac{16M_nP_T}{\eta}.$$

*For $\alpha_T, r_q \le 1$, a real expansion in Chebyshev time polynomials and real spherical harmonics with $\alpha_T k + r_q j \le H_*$ approximates the source to coordinate error $\eta/8$ using at most*

$$N \le \frac{2\{H_* + \alpha_T + (d-1)r_q\}^d}{d!\,\alpha_T r_q^{d-1}} \tag{126}$$

*coefficient vectors. For the dense sources of Proposition 3.1, Proposition 7.3 gives initialized-jet coefficients with total coordinate error at most $\eta$ and exact initialized-image pairing. For $d = 1$, use the two points of $S^0$ and at most $2N_1$ vectors, where $N_1 = 1 + \lfloor H_1/\alpha_T \rfloor$ and $H_1 = \log(64M_n/(\alpha_T\eta))$.*

*Proof.* Fix one coordinate and call its scalar source $g$. We supply the spatial decay argument. For $d \ge 2$ let $\mathcal{H}_j$ be the restrictions of homogeneous harmonic polynomials of degree $j$ to $S^{d-1}$, with probability surface measure. Its dimension is

$$h_j = \binom{j+d-1}{d-1} - \binom{j+d-3}{d-1} \le 2\binom{j+d-2}{d-2},$$

where a binomial coefficient with negative upper index is zero. Indeed, applying the Laplacian to $|x|^{2k}H_l(x)$ gives $2k(2l+2k+d-2)|x|^{2k-2}H_l(x)$; induction decomposes every homogeneous polynomial into these summands and gives the dimension formula by subtraction. The spherical Laplacian eigenvalues $-j(j+d-2)$ give orthogonality of different degrees by integration by parts. Polynomials contain constants and separate points on the compact sphere, so their uniform density (the real Stone–Weierstrass theorem) proves completeness. For an orthonormal basis $Y_{j\nu}$, rotational invariance and integration give the addition identity $\sum_\nu Y_{j\nu}(v)^2 = h_j$; consequently $|Y_{j\nu}(v)| \le \sqrt{h_j}$ and the projection $\Pi_j$ has supremum operator norm at most $h_j$.

For fixed $v$, average an analytic function over the complex tangent circle $\cos\zeta\, v + \sin\zeta\, u$, $u \perp v$, $|u| = 1$. On $\mathcal{H}_j$ this averaging is scalar: rotations fixing $v$ leave a one-dimensional zonal eigenspace, and its normalized eigenfunction solves $(1-x^2)p'' - (d-1)xp' + j(j+d-2)p = 0$, $p(1) = 1$. Coefficient comparison gives $p(x) = C_j^{(d-2)/2}(x)/C_j^{(d-2)/2}(1)$ for $d > 2$. For real $\zeta$ the averaging is self-adjoint because the joint law of its two endpoints is symmetric. Apply this to each harmonic coefficient and continue analytically in $\zeta$; it follows that the averaged analytic source has $j$th projection equal to $p(\cos\zeta)\Pi_j g$. At $\zeta = ir_q$ its arguments satisfy $z^\top z = 1$, $|\Im z| \le \sinh r_q$, so its supremum is at most $M_n$. Thus $\|\Pi_j g\|_\infty \le h_j M_n / p(\cosh r_q)$.

For $\nu = (d-2)/2 > 0$, the generating function $(1-2xw+w^2)^{-\nu} = \sum_j C_j^\nu(x)w^j$ at $x = \cosh r$, for any $r > 0$, factors as $(1-e^r w)^{-\nu}(1-e^{-r}w)^{-\nu}$ with nonnegative coefficients. It gives $C_j^\nu(\cosh r) \ge e^{rj}(\nu)_j/j!$ and $C_j^\nu(1) = (2\nu)_j/j!$. For $j \ge 1$ their ratio is $2\prod_{i=1}^{j-1}(1+\nu/(\nu+i))$. Its logarithm is at most $\log 2 + \nu\int_0^{j-1}(\nu+x)^{-1}\,dx$. Since $\nu \ge 1/2$, the ratio is at most $2(2j)^\nu$, and hence at most $2^d(j+1)^d$; the $j = 0$ inequality is immediate. Together with $h_j \le 2d^{d-2}(j+1)^{d-2}$ this proves

$$\|\Pi_j g\|_\infty \le M_n D_d (j+1)^{b_d} e^{-r_q j}. \tag{127}$$

For $d = 2$, the basis is the constant and the sine–cosine pair at each positive integer, and the averaging multiplier is $\cosh(jr) \ge e^{jr}/2$; the same loose bound with $D_2 = 8$, $b_2 = 2$ holds directly.

Set $t = T(1+\cos u)/2$. For $|\Im u| \le \alpha_T \le 1$ this lies inside the time rectangle, since its displacement from $[0,T]$ is at most $T\alpha_T$. The resulting even periodic function has cosine coefficients whose $j$th harmonic block is bounded by $2M_nD_d(j+1)^{b_d}e^{-\alpha_T k - r_q j}$: shift the Fourier contour to $\Im u = \pm\alpha_T$ and use (127) there. The series is absolutely uniformly convergent. Discarding indices with $\alpha_T k + r_q j > H_*$ bounds the tail by

$$2M_nD_d e^{-H_*/2}\sum_{k\ge0} e^{-\alpha_T k/2}\sum_{j\ge0}(j+1)^{b_d}e^{-r_q j/2} \le M_n P_T e^{-H_*/2} \le \eta/16.$$

Here $\sum_k e^{-\alpha_T k/2} \le 3/\alpha_T$ and $\sum_j (j+1)^b e^{-r_q j/2} \le 3b!(2/r_q)^{b+1}$, obtained by comparison with the gamma integral after shifting each unit interval.

The number of retained coefficients is $N = \sum_{r_q j \le H_*} h_j(1 + \lfloor (H_* - r_q j)/\alpha_T \rfloor)$. Replace $h_j$ by twice the number of weak compositions of $j$ into $d-1$ parts. The corresponding nonnegative lattice points have weighted sum $\alpha_T k + r_q \sum l_i \le H_*$. Their disjoint unit cubes lie in the simplex with right-hand side $H_* + \alpha_T + (d-1)r_q$, whose volume is exactly the right-hand side of (126). This proves the count. For the finite collection of coefficients, choose the Euclidean approximation

tolerance in Proposition 7.3 so that its product with $8\sum_{k,j,\nu}\|Y_{j\nu}\|_\infty$ is at most $\eta/16$. Both a base source and its paired image then have total error below $\eta$, without increasing the coefficient count.

If $d = 1$, there is no spatial approximation: apply the preceding one-variable Fourier contour argument separately at the two points. The geometric tail after degree $\lfloor H_1/\alpha_T \rfloor$ is at most $6M_n e^{-H_1}/\alpha_T < \eta/8$; the same finite coefficient accuracy argument finishes this case. □

**Proposition 7.5** (Harmonic: whole-sphere accuracy and total state)**.** *Eventually at each fixed confidence, the initialized-jet construction gives a selected nonlinear model with* $\|f_{\mathrm{Harm}} - f_n\|_{\mathcal{X}} \le Y/n$ *on the whole sphere and total retained real-coordinate count at most*

$$C(m+d)^2 + \left(\frac{C}{d}\right)^{d+1}\left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^{3d+2}.$$

*Proof.* For this whole-sphere construction put $\lambda = \gamma/m$, $z = Y/\lambda$, $\ell = \log(en)$, and $T = 32\ell/\lambda$. Choose $\eta$ by the canonical tolerance formula (111). The public source proposition supplies the radii and coordinate bound

$$r_t = \frac{\gamma}{\beta^{30L}Y^2 m\sqrt{(d+3)\ell}}, \qquad r_q = \frac{1}{\beta^{3L}\sqrt{(d+3)\ell}}, \qquad M_n = \beta^{6L}\sqrt{n}.$$

Set $\alpha_T = r_t/(4T)$. For $d \ge 2$, let $N$ be the coefficient count in Lemma 7.4, with its explicitly defined $P_T, H_*$ evaluated at these local parameters. For $d = 1$, let $N_1$ be the two-point time count defined in that same statement. Apply the approximation lemma to the four joint time–sphere families from Proposition 3.1, including its passive backward family; hence the same coefficient vectors serve all training inputs. In layer $j$ include the approximate feature and response coefficients and the paired initialized forward and reverse image coefficients. Add $\mathbf{1}_n$, the at most $2m$ exact initialized training vectors needed for forward-image pairing, and the $d$ first-layer columns. Each source space has rank at most

$$R = 2m + d + 1 + 4N \quad (d \ge 2), \qquad R = 2m + d + 1 + 8N_1 \quad (d = 1).$$

The source contract is now satisfied. Invoke Proposition 7.2 with its source-dimension bound $q = R$; it selects at most $9R$ coordinates per layer and gives the claimed absolute error. All choices use initialized jets only.

We spell out the asymptotic count because dropping the label factor here would change the theorem. The exact source radius gives

$$\alpha_T^{-1} = \frac{4T}{r_t} = 128\beta^{30L}z^2\sqrt{d+3}\,\ell^{3/2}, \qquad r_q^{-1} = \beta^{3L}\sqrt{(d+3)\ell}. \tag{128}$$

For each fixed admissible problem, (109) and (111) imply $\log(1/\eta) \le 2\ell$ eventually. Every other fixed parameter inside the logarithm defining $H_*$ is additive, whereas $\log P_T = O_{d,\mathrm{problem}}(\log \ell)$. Thus $H_* + \alpha_T + (d-1)r_q \le 8\ell$ eventually. The dimension bound gives

$$N \le C^d\beta^{CLd}\frac{(d+3)^{d/2}}{d!}\,z^2\ell^{3d/2+1}.$$

For $d = 1$, the direct two-point count gives the same form with $N_1 \le C\beta^{CL}z^2\ell^{5/2}$.

The public inventory (110), with $q = R$, is $1020(L+1)R^2 + 36R + 10m(d+1)$, plus the separately charged activation evaluator. It already includes selected mixers, metrics, initial copies, data and working arrays. No dense mixer, source basis, initialized jet, quadrature table or coefficient vector is retained.

Finally, for $d \geq 2$, $R^2 \leq C(m+d)^2 + CN^2$ and $d! \geq (d/e)^d$ imply

$$\frac{(d+3)^d}{(d!)^2} \leq \frac{(8e^2)^d}{d^{d+1}}.$$

Indeed, for $d \geq 2$, $(1+3/d)^d \leq (5/2)^d$ and $d \leq 2^d$ give the stronger numerator $(5e^2)^d$. Absorb activation/depth powers into $C^d$ to obtain the asserted count. For $d = 1$, use $R^2 \leq C(m+1)^2 + CN_1^2$ and the already proved bound on $N_1$; this gives the stated logarithmic exponent five. If a finite small width makes $9R \geq n$, retaining all coordinates is a valid fallback; the claimed polylogarithmic bound concerns the eventual range only. □

### 7.4 Finite-panel sources on an adaptive time partition

**Proposition 7.6** (Taylor: panel accuracy and total state)**.** *On the declared $(m+p)$-input panel of Section 1, eventually at each fixed confidence there is an initialization-only selected model with $\|f_{\text{Taylor}} - f_n\|_{\mathcal{X}} \leq Y/n$ and total state at most*

$$C(m+p)^2 \left[ \left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^3 + \left(\frac{m}{\gamma}\right)^2 [\log(en)]^2 \{\log(e + \log(en))\}^2 \right] + C(m+p)d.$$

*Only training inputs enter the deficit equation and Gram inverse. Accuracy outside the panel is not asserted.*

*Proof.* For this finite-panel construction put $\lambda = \gamma/m$, $z = Y/\lambda$, $\ell = \log(en)$ and $T = 32\ell/\lambda$. Use the public disk-and-partition interface of Proposition 3.1: there are $0 = t_0 < \cdots < t_J = T$ and radii $r_b > 0$ such that each source is holomorphic near $|t - t_b| \leq r_b$ and $t_{b+1} - t_b \leq r_b/2$. Its count (29) gives

$$J \leq 1 + \beta^{30L} z^2 \sqrt{\ell} + \frac{\beta^{16L}}{\lambda} \log(1 + 4\ell/\log 2). \tag{129}$$

The centers and radii are deterministic functions of the problem parameters, as that public interface guarantees, so they require no later dense training data.

There are at most $2(2m+p)$ source curves per layer: forward features and their initialized images at all declared inputs, and backward responses and their initialized transpose images at training inputs. Each coordinate is bounded by $M_n = \beta^{6L}\sqrt{n}$ on its disks. Choose the source tolerance (111) and

$$K = \max\{0, \lceil \log_2(8M_n/\eta) \rceil\}.$$

On panel $b$, retain coefficient vectors $r_b^k u^{(k)}(t_b)/k!$ for $0 \leq k \leq K$, where $u$ denotes one of the source curves. They have coordinate magnitude at most $M_n$. Since $|t - t_b|/r_b \leq 1/2$, the discarded Taylor tail is at most $M_n 2^{-K} \leq \eta/8$. The vectors from all panels span one time-independent space of dimension at most $J(K+1)$ per curve. The local approximants may differ at a panel boundary: both approximate the same source there, and the comparison theorem uses values, never their derivatives or a boundary matching rule.

Apply Proposition 7.3 to approximate every scaled base coefficient in Euclidean norm to $\eta/128$, and form its initialized image exactly. The base reconstruction error is at most $(\eta/128)\sum_{k\geq 0} 2^{-k} = \eta/64$; initialized operator norm eight gives image error at most $\eta/8$. Adding their independently controlled Taylor tails leaves both below $\eta$. Include the constant, exact initial training vectors and paired images, and first-weight columns required by Proposition 7.2. Before input reduction the rank per layer is at most $2m + d + 1 + 2(2m+p)J(K+1)$. This verifies the identical source contract, so the same selected runtime, fitting, cancellation, and endpoint proofs apply. In particular taking a

maximum over the passive inputs does not introduce $p$ into any comparison coefficient. No passive deficit or additional sample Gram is introduced.

There is an exact reduction of the first-layer dimension. Let $U_0 \in \mathbb{R}^{d\times k}$ have orthonormal columns spanning all declared normalized inputs, where $k \le \min(d, m+p)$. Replace $v_i$ by $U_0^\top v_i$ and the first matrix by $W^{(1)}U_0$. Every first-layer derivative is a sum of outer products with $v_a$, hence the component on the orthogonal complement of this input span is constant. Induction through the layers proves equality of the original and reduced coupled trajectories on the panel. The reduced inputs remain unit vectors; there is no new normalization factor. Each initialized Gaussian row multiplied by $U_0$ is exactly standard Gaussian in $\mathbb{R}^k$, and the sample inner products and population feature Gram are unchanged. Thus the same source event applies to the reduced system, whose public panel-count coefficients depend only on activations and depth, not on $k$ or $d$. The source rank is now

$$R = 2m + k + 1 + 2(2m+p)J(K+1).$$

Invoke Proposition 7.2 with source-dimension bound $q = R$ and input dimension $k$. To its public inventory (110), add the passive inputs and output buffer and retain the input map. A safe explicit count is

$$1020(L+1)R^2 + 36R + 10m(k+1) + pk + (m+p) + 3(m+p)d,$$

plus the separately charged nonprimitive activation evaluator. This includes original and transformed data, the map, all metrics and inverse caches, initial and moving weights, direction arrays, training feature/response and sample-matrix buffers, and query workspace. Discard the dense arrays, source coefficients, origin jets, panel anchors and partition schedule. The optimizer is autonomous; a partition used to prove source approximation is not a retained forcing table.

For each fixed problem, (111) implies $K + 1 \le 8\ell$ eventually. The coefficients $\beta^{30L}$ and $\beta^{16L}$ in (129) depend only on activations and depth. Squaring that bound with $(a+b+c)^2 \le 3(a^2+b^2+c^2)$, and using $k \le m+p$, yields

$$R^2 \le C(m+p)^2\left[z^4\ell^3 + \lambda^{-2}\ell^2\{\log(e+\ell)\}^2\right].$$

The second term absorbs the initial rank and plain $\ell^2$ terms because $\lambda \le 1$. Crucially, the $z^2$ in the first term of the panel count was retained before squaring, so its storage contribution remains $z^4 = (Ym/\gamma)^4$; it was not replaced by a label-independent constant. This proves the stated inventory and finishes the construction. □

# 8 Efficient initialization of the selected models

The selected models also admit efficient initialization by a disposable numerical computation on the physical interval $[0, T]$, where $T = 32(m/\gamma)\log(en)$. This computation constructs the source spaces; the resulting autonomous model starts at the original time zero. Its final arrays use the original activations, exact initialized training features, and the corrected readout (112).

There are two distinct input contracts. A **supplied dense initialization** specifies the actual matrices of the reference run. A **fresh implicit Gaussian initialization** generates actions of new Gaussian matrices as needed, consistently in both directions. The second contract comes with a coupled ordinary dense reference; it does not convert a previously supplied matrix or an entry-generator seed in subquadratic work.

The proof of Proposition 1.2 first gives a finite numerical source routine with explicit orders. Real-node interpolation supplies Taylor's source space; positive time–sphere quadrature supplies Harmonic's. Exact initialized-image pairing then permits Proposition 7.2 to be applied. Finally a

two-direction Gaussian sampler replaces every initialized hidden-matrix action while preserving the entire finite computation.

## 8.1 Numerical source values from initialization

All constants until the source-space construction are local to the numerical proof. Put $\lambda = \gamma/m$, $S = 16Y/\lambda$, $\ell = \log(en)$, $T = 32\ell/\lambda$, and $W_0^{(j)} = W^{(j)}(0)$. The normalized parameter vector and its norm are

$$\theta = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n}), \qquad \|\theta\|^2 = \|W^{(1)}\|_F^2/n + \sum_{j=2}^{L} \|W^{(j)}\|_F^2 + \|w\|_2^2/n.$$

For the predictions and residuals in (1), set $\xi_a = \nabla_\theta f_n(x_a)$ and $\rho = \|r\|_2/\sqrt{m}$. The mobilities in (3) give precisely

$$\dot\theta = F(\theta), \qquad F(\theta) = -\frac{2}{m}\sum_a r_a \xi_a, \qquad \xi_a = \left( \frac{\delta_a^{(1)} v_a^\top}{\sqrt{n}}, \left( \frac{\delta_a^{(j)} h_a^{(j-1)\top}}{n} \right)_{j=2}^{L}, \frac{h_a^{(L)}}{\sqrt{n}} \right). \tag{130}$$

In the complex extension all products use algebraic transposes. The norms remain the ordinary complex Euclidean and Frobenius norms. Positivity of a complex Gram is not used.

For Harmonic condition on the fitting and whole-sphere source event of Lemma 2.1 and proposition 3.1. For Taylor use the fitting and finite-panel event of Lemma 6.2. It gives real operator caps below nine, $\|w\|_2/\sqrt{n} \le 2Y/\sqrt{\lambda}$, $\rho(t) \le Y e^{-\lambda t/2}$ and $\int_0^\infty \rho \le 2Y/\lambda$. For centers in $[0, T]$, the respective source proofs supply complex disks of radius $r_t/2$ with preactivation imaginary parts at most $3a/8$, operator caps ten, residual RMS at most $2Y$, feature RMS at most $\beta^{3L}$, and training carrier coordinates bounded by

$$r_t = \frac{\lambda}{\beta^{30L} Y^2 \sqrt{(d+3)\ell}}, \qquad M_c = 2S\beta^{21L}\sqrt{\ell}. \tag{131}$$

The displayed $r_t$ is the Harmonic choice. For Taylor set instead $r_t = h(0)/2$, where $h$ is the capped flaring function in (144). Its disk of radius $r_t/2 = h(0)/4$ lies inside the finite-panel domain at every real center. Taylor requires the complex estimates only on its declared queries; Harmonic requires them on every real unit query. Here a carrier is $k^{(L)} = w$ or $k^{(j)} = W^{(j+1)\top}\delta^{(j+1)}$. These complex estimates are only used inside the corresponding certified finite source domain. For Taylor use the input reduction below and replace $d$ by its reduced dimension in the later preactivation envelope.

### 8.1.1 Real signed stability and its constants

Use the conservative derivative envelopes $b = \beta - 1$, $s = t_2 = \beta$. Define the following real endpoint bounds in the displayed order:

$$\begin{gathered} H_1^{\mathrm{r}} = \max(1, b + 10s), \quad H_j^{\mathrm{r}} = \max(1, b + 10sH_{j-1}^{\mathrm{r}}), \quad H^{\mathrm{r}} = H_L^{\mathrm{r}}, \quad R^{\mathrm{r}} = 1 + 2Y/\sqrt{\lambda}, \\ B^{\mathrm{r}} = R^{\mathrm{r}} s(10s)^{L-1}, \quad F_z^{\mathrm{r}} = H^{\mathrm{r}}(10s)^{L-1}, \quad P_{\mathrm{layer}}^{\mathrm{r}} = 1 + (L-1)H^{\mathrm{r}}, \\ D^{\mathrm{r}}(M) = (10s)^{L-1}[s(1 + B^{\mathrm{r}}) + Lt_2 F_z^{\mathrm{r}} M], \\ J^{\mathrm{r}}(M) = \sqrt{L+1}[P_{\mathrm{layer}}^{\mathrm{r}} D^{\mathrm{r}}(M) + (1 + (L-1)B^{\mathrm{r}}) s F_z^{\mathrm{r}}], \\ C^{\mathrm{r}}(M) = sF_z^{\mathrm{r}}\sqrt{L} + LB^{\mathrm{r}} sF_z^{\mathrm{r}} + \tfrac{1}{2}L^2 t_2 (F_z^{\mathrm{r}})^2 M. \end{gathered} \tag{132}$$

Evaluate $J^{\mathrm{r}}, C^{\mathrm{r}}$ at $M = M_c$, and put

$$\begin{aligned} A_n &= 4J^{\mathrm{r}}Y/\lambda, \quad K^{\mathrm{r}}_{\mathrm{qry}} = R^{\mathrm{r}}(10s)^{L-1}, \\ C_{\mathrm{src}} &= 8\sqrt{L+1}\max\left\{sF^{\mathrm{r}}_z, (10s)^{L-1}[s(1+B^{\mathrm{r}}) + Lt_2F^{\mathrm{r}}_zK^{\mathrm{r}}_{\mathrm{qry}}]\right\}. \end{aligned} \tag{133}$$

**Lemma 8.1** (Stability of an approximate dense path)**.** *Let $u$ be an absolutely continuous path on $[0,T]$, let $d_u = u' - F(u)$, and set $E_0 = \|u(0) - \theta(0)\| + \int_0^T \|d_u\|$. If*

$$E_0 \le e^{-A_n}/4, \qquad 8(C^{\mathrm{r}} + J^{\mathrm{r}})^2 T e^{2A_n} E_0^2 \le 1,$$

*then $\sup_{[0,T]} \|u - \theta\| \le 2e^{A_n}E_0$. Every real-query source in (25) has coordinate error at most $C_{\mathrm{src}} n \|u(t) - \theta(t)\|$ when evaluated at $u(t)$ with the original activations.*

*Proof.* Stop at $E = \|u - \theta\| = 1$. Then both real states have operator caps ten and readout RMS at most $R^{\mathrm{r}}$. If $D_h$ is the sum of their hidden-parameter block discrepancies, and $D_w$ their readout RMS discrepancy, then $D_h \le \sqrt{L}E$ and $D_h + D_w \le \sqrt{L+1}E$. Subtracting forward equations gives preactivation RMS error at most $F^{\mathrm{r}}_z D_h$, and feature RMS error at most $sF^{\mathrm{r}}_z D_h$. For the backward difference write

$$\delta^{(j)}(u) - \delta^{(j)}(\theta) = \phi'(z^{(j)}(u)) \odot [k^{(j)}(u) - k^{(j)}(\theta)] + [\phi'(z^{(j)}(u)) - \phi'(z^{(j)}(\theta))] \odot k^{(j)}(\theta).$$

Its second term uses the true training carrier bound $M_c$. Propagation costs $10s$, a changed mixer costs $sB^{\mathrm{r}}\|\Delta W\|_F$, and the top readout costs $sD_w$. Summing these finite recursions gives RMS response error $D^{\mathrm{r}}(M_c)(D_h + D_w)$ and, from the displayed gradient blocks, $\|\xi_a(u) - \xi_a(\theta)\| \le J^{\mathrm{r}}E$.

There is also the endpoint Taylor estimate

$$|f_n(u, x_a) - f_n(\theta, x_a) - \langle \xi_a(\theta), u - \theta\rangle| \le C^{\mathrm{r}}E^2.$$

To see every contribution, put $a^{(j)} = \Delta h^{(j)} - \phi'(z^{(j)}(\theta)) \odot \Delta z^{(j)}$. Scalar Taylor's formula gives $|a_i^{(j)}| \le t_2 |\Delta z_i^{(j)}|^2/2$. Forward differentiation followed by backward substitution expresses the remainder as

$$\frac{1}{n}\sum_j k_a^{(j)}(\theta)^\top a_a^{(j)} + \frac{1}{n}\sum_{j=2}^{L} \delta_a^{(j)}(\theta)^\top \Delta W^{(j)} \Delta h_a^{(j-1)} + \frac{1}{n}\Delta w^\top \Delta h_a^{(L)}.$$

The respective bounds are $L^2 t_2 M_c (F^{\mathrm{r}}_z)^2 E^2/2$, $LB^{\mathrm{r}} sF^{\mathrm{r}}_z E^2$, and $sF^{\mathrm{r}}_z \sqrt{L} E^2$. If $e_{f,a} = f_n(u, x_a) - f_n(\theta, x_a)$, subtracting (130) and retaining its negative square now gives

$$\tfrac{1}{2}(E^2)' \le -2\|e_f\|_2^2/m + 2(C^{\mathrm{r}} + J^{\mathrm{r}})\|e_f\|_2 E^2/\sqrt{m} + 2J^{\mathrm{r}}\rho E^2 + E\|d_u\|.$$

Maximizing the first two terms over $\|e_f\|_2/\sqrt{m}$ gives

$$E' \le 2J^{\mathrm{r}}\rho E + \tfrac{1}{2}(C^{\mathrm{r}} + J^{\mathrm{r}})^2 E^3 + \|d_u\|. \tag{134}$$

At zero use $\sqrt{E^2 + \varepsilon^2}$ and let $\varepsilon \downarrow 0$. The integrating factor has exponent $2J^{\mathrm{r}}\int_0^t \rho \le A_n$. As long as its product with $E$ is at most $2E_0$, integration bounds that product by $E_0 + 4(C^{\mathrm{r}} + J^{\mathrm{r}})^2 T e^{2A_n} E_0^3 \le 3E_0/2$. A first crossing of $2E_0$ is impossible; also $E \le 1/2$, removing the original stop. The zero-defect case follows by the same inequality.

For any passive real query, a carrier has RMS at most $K^{\mathrm{r}}_{\mathrm{qry}}$ and hence coordinate maximum at most $\sqrt{n}K^{\mathrm{r}}_{\mathrm{qry}}$. Repeat the backward subtraction with this maximum, then convert the resulting

RMS bound to coordinate norm. The two factors $\sqrt{n}$ give the displayed $n$; initialized images cost at most eight in operator norm. This proves the stated source bound and also the same bound between any two real states in this operator and readout neighborhood. At fixed parameters $A_n = O(\sqrt{\ell})$. □

### 8.1.2 Complex restarts and scalar activation approximation

The following recurrences give usable neighborhoods of computed anchors. They are different from the real coefficients just defined. Set

$$\begin{gathered}
R_{\mathrm{loc}} = 1 + S\beta^{3L}, \quad H_1 = \max(1, b + 11s), \quad H_j = \max(1, b + 11sH_{j-1}), \\
B_j = R_{\mathrm{loc}} s(11s)^{L-j}, \quad P_1 = 1, \quad P_j = H_{j-1} + 11sP_{j-1}, \quad P_* = \max_j P_j, \\
\kappa_L(M) = 1, \quad \kappa_j(M) = 11s\kappa_{j+1}(M) + 11t_2 M P_{j+1} + B_{j+1}, \\
D_j(M) = s\kappa_j(M) + t_2 M P_j, \quad \kappa_*(M) = \max_j \kappa_j(M), \\
G = H_L + B_1 + \sum_{j=2}^{L} B_j H_{j-1}, \\
J_{\mathrm{loc}}(M) = sP_L + D_1(M) + \sum_{j=2}^{L} [H_{j-1} D_j(M) + B_j s P_{j-1}].
\end{gathered} \tag{135}$$

For complex states of operator cap eleven and readout RMS cap $R_{\mathrm{loc}}$, forward subtraction gives preactivation RMS error $P_j E$. Backward subtraction anchored at an endpoint with carrier maximum $M$ gives carrier and response RMS errors $\kappa_j(M)E, D_j(M)E$. Applying these bounds to (130) gives $\|\xi_a\| \le G$, $\|\Delta\xi_a\| \le J_{\mathrm{loc}}(M)E$, and

$$\|F(u) - F(v)\| \le [2G^2 + 2\rho(v) J_{\mathrm{loc}}(M)] \|u - v\|.$$

Indeed, the residual change costs $G\|u - v\|$ by integration along the parameter segment; split the residual-gradient product into its two endpoint differences. All scalar preactivation segments remain in the half-strip in the neighborhoods used next.

Define

$$\begin{gathered}
b_n = \min\{1/4, a/(16\sqrt{n}P_*), 1/(\sqrt{n}\kappa_*(M_c))\}, \\
L_n = 2G^2 + 2(2Y + G) J_{\mathrm{loc}}(M_c + 1), \quad V = 2G(2Y + G), \quad \widehat{R}_n = \min\{r_t/2, (8L_n)^{-1}\}.
\end{gathered} \tag{136}$$

Every ball of radius $b_n$ around the true complex trajectory inside a restart disk stays in the half-strip: before a hypothetical first exit, each preactivation changes by at most $\sqrt{n}P_* b_n \le a/16$, so its imaginary part stays below $7a/16$. Operators remain below eleven. The same endpoint subtraction gives carrier maximum $M_c + 1$ and residual RMS at most $2Y + G$. Thus $L_n$ is a pairwise Lipschitz bound throughout each ball, not an assumption about a numerical trajectory. The current source proof also supplies the following sufficient bound for the real part at its centers:

$$Z_* = [\beta^{21L} + 16\sqrt{d+3}\,\beta^{3L} + 1 + \beta^{26L}\sqrt{d+3}\,S^2]\sqrt{\ell} + a/8.$$

This follows by integrating its coordinate velocity bound $2\rho S\beta^{26L}\sqrt{d+3}\sqrt{\ell}$, and using its initial Gaussian maximum and short complex-time bound. For the finite-panel event the same envelope

follows from its initialized finite-list maximum and response bound $SU_{\mathrm{fin}}\sqrt{\ell}$, since $U_{\mathrm{fin}} \le \beta^{26L}$. The moving balls therefore have preactivations in $|\Re z| \le Z_* + a/16$, $|\Im z| \le 7a/16$.

Here is a complete finite activation-value backend. For a requested source-vector tolerance $\varepsilon_v > 0$, put $\tau = \varepsilon_v/(32\sqrt{n})$ and define

$$\begin{aligned} e_d &= e^{-A_n}\min\left\{\frac{1}{4}, \frac{1}{\sqrt{8}(C^{\mathrm{r}}+J^{\mathrm{r}})\sqrt{T}}, \frac{\tau}{4C_{\mathrm{src}}n}, \frac{b_n}{16}\right\}, \quad \zeta_F = \min\{e_d/(2T), b_nL_n/4, V/2\}, \\ U_0 &= 0, \quad U_j = 1 + 11sU_{j-1}, \quad U_* = \max_j U_j, \quad K_j^{\mathrm{act}} = R_{\mathrm{loc}}(11s)^{L-j}, \\ Q_{L+1}(M) &= 0, \quad Q_j(M) = 11(s+1)Q_{j+1}(M) + K_j^{\mathrm{act}} + 11t_2MU_{j-1}, \\ Q_\xi &= U_L + Q_1(M_c+1) + \sum_{j=2}^{L}[Q_j(M_c+1)(H_{j-1}+1) + B_jU_{j-1}], \\ Q_f &= R_{\mathrm{loc}}U_L, \quad C_F = 2Q_f(G+1) + 2(2Y+G)Q_\xi, \\ M_q &= \sqrt{n}\max_j K_j^{\mathrm{act}}, \quad C_p = 8\sqrt{n}\max\{U_*, \max_j Q_j(M_q)\}, \\ \epsilon &= \min\{1, U_*^{-1}, a/(5632\sqrt{n}U_*), Q_\xi^{-1}, \zeta_F/C_F, \tau/(2C_p)\}. \end{aligned} \tag{137}$$

Every denominator is positive. Set

$$\begin{aligned} B &= 8(Z_*+a+1), \quad \tau_o = \operatorname{arsinh}(31a/(64B)), \quad \tau_i = \operatorname{arsinh}(29a/(64B)), \quad \Delta = \tau_o - \tau_i, \\ M_\phi &= b + s(B+a), \quad C_a = \max\{1, 512/a, 2(512/a)^2\}, \quad \epsilon_p = \epsilon/C_a, \\ D &= \max\left\{1, \left\lceil \Delta^{-1}\log\max\{e, 24M_\phi/(\epsilon_p\Delta)\}\right\rceil\right\}, \qquad N_\phi = 4(D+1). \end{aligned} \tag{138}$$

With $x_i = B\cos(2\pi i/N_\phi)$, compute for each layer

$$\begin{aligned} c_{j,0} &= N_\phi^{-1}\sum_i \phi_j(x_i), \qquad c_{j,k} = 2N_\phi^{-1}\sum_i \phi_j(x_i)\cos(2\pi ki/N_\phi)(k>0), \\ \psi_j(z) &= \sum_{k=0}^{D} c_{j,k}T_k(z/B). \end{aligned} \tag{139}$$

Finite value-call errors of magnitude at most $\epsilon_p/[4(D+1)e^{D\tau_i}]$ are allowed in this formula only. Initialized training features below still use exact original values.

**Lemma 8.2** (Certified numerical source polynomials)**.** *Let $C_g = \max_j\{H_j+1, B_j+1\}$ and choose*

$$\begin{aligned} K = \max\Big\{&8, \lceil 2\log_2\max\{1, 16TV/e_d\}\rceil, \left\lceil\log_2\max\{1, 8V\widehat{R}_n/b_n\}\right\rceil, \\ &\left\lceil\log_2\max\{1, 128V\widehat{R}_nC_{\mathrm{src}}n^{3/2}/\varepsilon_v\}\right\rceil, \lceil\log_2\max\{1, 64\sqrt{n}C_g/\varepsilon_v\}\rceil\Big\}, \\ J = \lceil 2T/\widehat{R}_n\rceil &= \lceil\max\{4T/r_t, 16TL_n\}\rceil. \end{aligned} \tag{140}$$

*Use $J$ equal physical-time panels. Starting at the original initialization, compute Taylor coefficients through degree $K$ for the polynomial-activation field $\widetilde{F}$ by*

$$[u]_{k+1} = \frac{1}{k+1}[h^k]\widetilde{F}\left(\sum_{i=0}^{k}[u]_ih^i\right), \qquad 0 \le k < K, \tag{141}$$

*and advance to each polynomial endpoint. The forward and backward source jets through degree $K$, computed with the same coefficients and $\psi_j$, approximate every base source $h^{(j)}$ or $\delta^{(j)}$ in Euclidean norm by at most $\varepsilon_v/16$ at every real time in its numerical panel. Queries are all real unit inputs for Harmonic and the declared input list for Taylor. If $\log(1/\varepsilon_v) = O(\ell)$ at fixed parameters, then*

$$J = O(\ell^{3/2}), \qquad K = O(\ell), \qquad D = O(\ell^{3/2}). \tag{142}$$

*Proof.* First verify the activation approximation. The ellipse with boundary $z = B(\zeta + \zeta^{-1})/2$, $|\zeta| = e^{\tau_o}$ is inside the half-strip, and $|\phi(z)| \le M_\phi$. The inner ellipse contains disks of radius $a/512$ around all points at distance at most $a/512$ from the moving-ball preactivation rectangle: the squared normalized real and imaginary coordinates are at most $(1/8)^2 + (226/232)^2 < 1$. Also $0 < \Delta < 1$ and $\Delta \ge a/(64B)$.

Shifting the Fourier contour of $\phi(B\cos u)$ bounds its degree-$k$ Chebyshev coefficient by $2M_\phi e^{-k\tau_o}$. On the inner ellipse $|T_k(z/B)| \le e^{k\tau_i}$, so the tail after $D$ is at most $4M_\phi\Delta^{-1}e^{-(D+1)\Delta}$. The discrete coefficient at frequency $k$ sums the Fourier coefficients at $k + qN_\phi$, $q \in \mathbb{Z}$. Summing these aliases for $k \le D$ bounds their polynomial error by $8M_\phi(D+1)e^{-2(D+1)\tau_o} \le 8M_\phi\Delta^{-1}e^{-(D+1)\Delta}$. The degree prescription makes their sum at most $\epsilon_p/2$. The allowed value-call error adds at most $\epsilon_p/2$ by summing $D+1$ coefficient errors of size at most twice that error. Cauchy's formula on the verified disks proves simultaneous errors at most $\epsilon$ in values and the first two derivatives.

At a fixed state in a moving ball, subtract the original and polynomial forward passes. Feature RMS error is at most $U_j\epsilon$; preactivation RMS error is at most $11U_{j-1}\epsilon$. The constraint $11\sqrt{n}U_*\epsilon \le a/512$ keeps every polynomial argument in the verified approximation domain. For carriers of coordinate maximum $M$, backward subtraction gives response RMS error $Q_j(M)\epsilon$: its propagated term costs $11(s+1)Q_{j+1}$, direct gate approximation costs $K_j^{\mathrm{act}}$, and the changed original gate costs $11t_2MU_{j-1}$. The gradient-block and prediction differences are thus at most $Q_\xi\epsilon$ and $Q_f\epsilon$. Splitting their product in (130) gives $\|\widetilde{F} - F\| \le C_F\epsilon \le \zeta_F$ throughout the moving ball. The passive-source substitution error, including initialized images, is at most $C_p\epsilon \le \tau/2$ in each coordinate.

On its concentric ball of radius $b_n/2$, vector-valued Cauchy in each unit parameter direction gives $\|D(\widetilde{F} - F)\|_{\mathrm{op}} \le 2\zeta_F/b_n \le L_n/2$. In particular $\widetilde{F}$ is $2L_n$-Lipschitz there. For a real anchor $u_0$ with $\|u_0 - \theta(t)\| \le b_n/8$, consider holomorphic errors on $|h| \le \widehat{R}_n$ and the map

$$e(h) \longmapsto u_0 - \theta(t) + \int_0^h [\widetilde{F}(\theta(t+z) + e(z)) - F(\theta(t+z))]\,dz.$$

On the supremum ball of radius $b_n/2$ its contraction factor is at most $1/4$. Its image norm is at most $b_n/8 + \widehat{R}_n\zeta_F + b_n/8 \le 9b_n/32$. Successive iterates converge uniformly; Cauchy's formula shows the limit is holomorphic. The fixed point is the unique local solution $v$ of the polynomial field, and has discrepancy at most $5b_n/24 < b_n/4$ from the true complex reference and velocity at most $2V$. Conjugation and uniqueness make it real on the real diameter.

Its degree-$K$ Taylor polynomial $u$ on a half-radius real panel has parameter tail at most $2V\widehat{R}_n 2^{-K}$ and derivative tail at most $2V(2K+4)2^{-K}$. It stays in the smaller moving ball, and its defect against the original field is at most $2V(2K+5)2^{-K} + \zeta_F$. Since $2K + 5 \le 4\,2^{K/2}$ for $K \ge 8$, the numerical order gives total integrated defect at most $e_d$. Lemma 8.1 on each already computed prefix gives

$$\|u - \theta\| \le 2e^{A_n}e_d \le \min\{b_n/8, \tau/(2C_{\mathrm{src}}n)\}.$$

This verifies the hypothesis for the next restart, starting with exact initialization. Finite induction constructs the entire path, without a test of an unknown trajectory norm or rejected panels.

For a base source $g$, let $g^0, g^p$ denote original and polynomial activation evaluation, and $g_K$ its source Taylor polynomial along $v$. On the complex disk $\|g^p(v)\|_2 \le \sqrt{n}C_g$: the original RMS bounds are $H_j, B_j$ and the proved substitution error is less than one. Thus $\|g_K - g^p(v)\|_2 \le \sqrt{n}C_g 2^{-K} \le \varepsilon_v/64$. Split the remaining error exactly as

$$g_K - g^0(\theta) = [g_K - g^p(v)] + [g^p(v) - g^0(v)] + [g^0(v) - g^0(u)] + [g^0(u) - g^0(\theta)].$$

The second and fourth terms have Euclidean norm at most $\sqrt{n}\tau/2 = \varepsilon_v/64$. For the third use the real source bound and the parameter tail, giving $2V\widehat{R}_n C_{\mathrm{src}} n^{3/2} 2^{-K} \le \varepsilon_v/64$. This proves the four-term error bound in the statement.

Finally $M_c, L_n, A_n = O(\sqrt{\ell})$, $b_n^{-1} = O(\sqrt{n\ell})$, $\widehat{R}_n^{-1} = O(\sqrt{\ell})$, $C_p = O(n)$ and $Z_* = O(\sqrt{\ell})$ at fixed parameters. Hence the logarithms of all inverse tolerances are $O(\ell)$, $\Delta^{-1} = O(\sqrt{\ell})$, and the explicit formulas give (142). Recurrence (141) is causal and differentiates only the finite polynomials $\psi_j$. □

## 8.2 Two source spaces with the original retained sizes

Let $\mathcal{A}_n$ be the comparison coefficient in (117), and let $\mathcal{D}$ be the explicit endpoint coefficient in the proof of Claim 10. Use

$$\eta = \min\{1, Y, S, Y/(2n\mathcal{A}_n)\}, \qquad \mathcal{D}e^{-8\ell} \le Y/(2n). \tag{143}$$

The second inequality is an eventual scalar gate. The comparison proof gives $\log(1/\eta) \le 2\ell$ eventually for each fixed positive admissible problem. Once each source space meets the hypotheses of Proposition 7.2 with this $\eta$, its all-time error is at most $\mathcal{A}_n\eta + \mathcal{D}e^{-8\ell} \le Y/n$. For Taylor set $C_{\mathrm{sel}} = \beta^{60L}$ as in the polynomial-width assembly. Its coefficient bounds show that this $\eta$ is no smaller than the conservative tolerance used there, and (104) implies the displayed tail inequality. The precise degree gate used below is (105) itself, without enlarging its exponent. There is no additional qualitative width threshold for Taylor.

### 8.2.1 Taylor from real source nodes

Write $M = m + p$. Let $U_0 \in \mathbb{R}^{d\times k}$ have orthonormal columns spanning the declared normalized inputs, $k \le \min(d, M)$. Replace them by $U_0^\top v_i$ and the first matrix by $W_0^{(1)}U_0$. Their norms and Gram are unchanged. First-layer updates lie in the same span, so the original and reduced dense paths agree on every declared query. A Gaussian first row projected by $U_0$ is standard Gaussian in $\mathbb{R}^k$. This is the exact reduction used in Proposition 7.6; the retained map costs at most $Md$.

Use the deterministic capped finite-panel source domain from Lemma 6.2. Its constants $\eta_*, U_{\mathrm{fin}}, \mathcal{K}$ are the explicit source recurrences in the source coefficient definitions, (34) and (35). In particular they do not involve observed trajectory norms. Put

$$\begin{aligned} \nu = \lambda/4, \quad a_0 = \min(a,1), \quad b_{\mathrm{cap}} = \log(e+\ell), \quad B_{\mathrm{cap}} = 1 + (S^2/\eta_*)b_{\mathrm{cap}}, \\ r_{\mathrm{cap}} = \min\left\{ \frac{a_0}{64YSU_{\mathrm{fin}}\sqrt{\ell}}, \frac{1}{\beta^{300L}(1+\lambda B_{\mathrm{cap}})\sqrt{b_{\mathrm{cap}}}} \right\}, \\ h(t) = \frac{\log(1 + 2\mathcal{K}r_{\mathrm{cap}}e^{\nu t})}{2\mathcal{K}}, \qquad d_h = \lambda/(8\mathcal{K}). \end{aligned} \tag{144}$$

Starting at $t_0 = 0$, set $r_b = h(t_b)/[2(1+d_h)]$ and $t_{b+1} = \min\{T, t_b + r_b/2\}$. These are the disks and panels certified by Lemma 6.2. Since $h$ is nondecreasing and $d_h \le 1/8$, the certified disk radius

at every real center is at least $4h(0)/9 > h(0)/4$. This verifies the uniform numerical disk used earlier. If $J_s$ denotes the number of source panels, then

$$J_s \le 1 + \beta^{400L}\left[(Y/\lambda)^2\sqrt{\ell} + \lambda^{-1}\log(e+\ell)\right]. \tag{145}$$

For the count, $h' \le 1/8$ makes each full panel contribute at least $1/5$ to $\int dt/h(t)$. Split this integral where $2\mathcal{K}r_{\mathrm{cap}}e^{\nu t} = 1$, using $\log(1+x) \ge x/2$ before the split and logarithmic growth of slope at least $\nu/2$ afterward. This gives

$$J_s \le 1 + \frac{40}{\lambda r_{\mathrm{cap}}} + \frac{80\mathcal{K}}{\lambda}\log(1 + 4\ell/\log 2).$$

The reciprocal minimum is bounded by

$$\frac{1}{\lambda r_{\mathrm{cap}}} \le \frac{1024U_{\mathrm{fin}}}{a_0}(Y/\lambda)^2\sqrt{\ell} + \beta^{300L}\lambda^{-1}\sqrt{b_{\mathrm{cap}}} + \beta^{300L}B_{\mathrm{cap}}\sqrt{b_{\mathrm{cap}}}.$$

Writing $x = \log\ell \ge 0$, one has $b_{\mathrm{cap}} \le 2 + x$ and $(2+x)^{3/2}e^{-x/2} \le 4$. Thus $b_{\mathrm{cap}}^{3/2} \le 4\sqrt{\ell}$. Use $S^2 = 256(Y/\lambda)^2$, $\eta_*^{-1} \le \beta^{25L+4}$, $U_{\mathrm{fin}} \le \beta^{26L}$ and $a_0^{-1} \le \beta$ to obtain (145), with no confidence factor or new logarithmic storage power. Also $2\mathcal{K}r_{\mathrm{cap}} \le 1$, so $h(0) \ge r_{\mathrm{cap}}/2$ and $h(0)^{-1} = O(\sqrt{\ell})$ at fixed parameters. Therefore the numerical orders remain (142). The cap need never be inactive.

Put $M_n = \beta^{6L}\sqrt{n}$ and

$$K_s = \max\{8, \lceil 2\log_2(64M_n/\eta)\rceil\}, \quad N_s = J_s(K_s+1), \quad \varepsilon_v = \eta/[128(K_s+1)]. \tag{146}$$

On each panel $[t_b, t_{b+1}]$, use its $K_s+1$ transported Chebyshev roots $t_b + (t_{b+1}-t_b)[1+\cos((i+1/2)\pi/(K_s+1))]/2$. Obtain base-source vectors from Lemma 8.2. Use forward vectors on all $M$ inputs and backward vectors on the $m$ training inputs. Insert their exact initialized images. The nodal vectors themselves generate the space; interpolation coefficients need not be retained or computed.

To verify accuracy, interpolation $I_{K_s}$ at those roots obeys $\|I_{K_s}g\|_\infty \le (2K_s+1)\max_i|g_i|$ by discrete cosine orthogonality and $|T_j| \le 1$. The same inequality holds in Euclidean norm. The degree-$K_s$ Taylor polynomial on a half-radius panel has coordinate error at most $M_n 2^{-K_s}$. As interpolation reproduces this polynomial,

$$\|g - I_{K_s}g\|_\infty \le (2K_s+3)M_n 2^{-K_s} \le 4M_n 2^{-K_s/2} \le \eta/16.$$

This applies separately to each exact source and its initialized image. The approximate base node has Euclidean error at most $\varepsilon_v$; an initialized image has at most $8\varepsilon_v$. Its interpolated perturbation is therefore at most $16(K_s+1)\varepsilon_v = \eta/8$. Both families have coordinate error below $\eta$, and $I_{K_s}(W_0\widetilde{g}) = W_0 I_{K_s}\widetilde{g}$ exactly. The transpose identity has the same proof.

After adding the constant, exact initialized training features and images, and first-layer columns, the rank $r$ and actual selected width $q$ obey

$$r \le R = 2m + k + 1 + 2(M+m)N_s, \qquad q \le 9r. \tag{147}$$

Apply Proposition 7.2 with its source-dimension argument equal to $R$. We verify the numerical source degree at the stated polynomial gate rather than identifying it with the initialized-jet degree. Put $B_0 = 32\beta^{6L}(1+C_{\mathrm{sel}})\lambda^{-3/2}(1+Y^{-1})$. At $n \ge B_0^4$ and the source gate, one has $\ell \ge 16$ and $\log B_0 \le (\log n)/4$. For the active comparison-tolerance branch, using $1+\lambda^{-1/2} \le 2\lambda^{-1/2}$ gives

$$\log(64M_n/\eta) \le \tfrac{3}{2}\log n + \log B_0 + \log 8 + \sqrt{\ell} + \log(1+\sqrt{\ell}) \le \tfrac{9}{4}\ell + \log 8.$$

For the branches $\eta = Y$ and $\eta = 1$ the bound is smaller; $S \geq 16Y$ cannot give a smaller branch. Consequently $K_s + 1 \leq \max\{9, (9/(2\log 2))\ell + 8\} \leq 8\ell$ for $\ell \geq 16$. Squaring (145) and retaining its $(Y/\lambda)^2$ factor gives precisely the parameter order in (10), including $(Y/\lambda)^4\ell^3$. The map and declared data add $O(Md)$. Passive labels are never used.

### 8.2.2 Harmonic from positive coefficient quadrature

Use the current source radii $r_t$ in (131) and $r_q = 1/[\beta^{3L}\sqrt{(d+3)\ell}]$, with $M_n = \beta^{6L}\sqrt{n}$ and $\alpha = r_t/(4T)$. Require the eventual scalar gates $\alpha, r_q \leq 1$. For $d \geq 2$, take exactly the mode set of Lemma 7.4:

$$\begin{gathered} b_d = 2d-2, \quad D_d = 2^{d+1}d^{d-2}, \quad P_T = 18D_d b_d! 2^{b_d+1}/(\alpha r_q^{b_d+1}), \quad H_* = 2\log(16M_nP_T/\eta), \\ \Lambda = \{(k,j,b) : \alpha k + r_q j \leq H_*,\ 1 \leq b \leq h_j\}, \quad h_j = \binom{j+d-1}{d-1} - \binom{j+d-3}{d-1}, \\ N = |\Lambda|, \quad p_t = \lfloor H_*/\alpha \rfloor, \quad j_* = \lfloor H_*/r_q \rfloor, \quad H_{\mathrm{sph}} = \sum_{j \leq j_*} h_j, \quad \mathcal{Y} = \max_{j \leq j_*} \sqrt{h_j}. \end{gathered} \tag{148}$$

Impossible binomials are zero. For a scalar source $g$, its coefficient is

$$c_{kjb} = \gamma_k \int_0^{2\pi} \int_{S^{d-1}} g(T(1+\cos u)/2, v)\cos(ku) Y_{jb}(v)\, d\sigma(v) \frac{du}{2\pi}, \quad \gamma_0 = 1, \quad \gamma_k = 2\ (k>0), \tag{149}$$

where $Y_{jb}$ are real orthonormal harmonics for probability surface measure. The proof of Lemma 7.4 bounds the omitted-mode tail by $\eta/16$.

Here are finite quadrature orders and their construction. For $d \geq 3$ use polar angles $\theta_1, \ldots, \theta_{d-2}$ and azimuth $\varphi$ in the usual recursive coordinates $v = (\cos\theta_1, \sin\theta_1 u)$; for $d = 2$ use only $v = (\cos\varphi, \sin\varphi)$. Define

$$\begin{gathered} I_0 = \pi, \quad I_1 = 2, \quad I_b = (b-1)I_{b-2}/b, \quad A_d = \prod_{b=1}^{d-2} \pi/I_b, \\ a_t = \alpha/2, \quad h_{\mathrm{ang}} = r_q/[2(d-1)], \quad \sigma = \operatorname{arsinh}(2h_{\mathrm{ang}}/\pi), \\ \epsilon_c = \eta/(16N\mathcal{Y}), \quad \delta_{\mathrm{node}} = \eta/(64NA_d\mathcal{Y}^2), \\ B_{\mathrm{int}} = 2M_n\mathcal{Y}A_d e^{a_t p_t + j_*(d-1)h_{\mathrm{ang}}} (\cosh h_{\mathrm{ang}})^{(d-2)(d-1)/2}, \\ N_t = \max\{1, \lceil a_t^{-1}\log(1 + 4dB_{\mathrm{int}}/\epsilon_c)\rceil\}, \\ N_\varphi = \max\{1, \lceil h_{\mathrm{ang}}^{-1}\log(1 + 4dB_{\mathrm{int}}/\epsilon_c)\rceil\}, \\ N_\theta = \max\{1, \lceil \sigma^{-1}\log[8dB_{\mathrm{int}}/(\epsilon_c(1-e^{-\sigma}))]\rceil\} \quad (d \geq 3), \\ N_x = N_\varphi N_\theta^{d-2} \quad (d \geq 3), \qquad N_x = N_\varphi \quad (d = 2). \end{gathered} \tag{150}$$

Empty products equal one. Apply equal-weight periodic trapezoids with $N_t, N_\varphi$ nodes to $u, \varphi$. For each polar coordinate, transport to $\theta = \pi(1+x)/2$ the nodes and weights

$$\omega_i = (2i+1)\pi/(2N_\theta), \quad x_i = \cos\omega_i, \quad w_i = \frac{1}{N_\theta}\left[1 - 2\sum_{k=1}^{\lfloor (N_\theta-1)/2 \rfloor} \frac{\cos(2k\omega_i)}{4k^2-1}\right]. \tag{151}$$

Multiply the spatial weights by $\prod_{i=1}^{d-2}(\pi/I_{d-1-i})\sin^{d-1-i}\theta_i$. The identity $I_b = \int_0^\pi \sin^b\theta\, d\theta$ follows by integration by parts; the product is the normalized spherical Jacobian.

For completeness, the quadrature error follows directly from convergent series. A periodic function bounded by $B$ on an analytic strip of half-width $a$ has Fourier coefficients bounded by $Be^{-a|k|}$. The trapezoid retains exactly the frequencies divisible by its order, so its error is at most $2B/(e^{aN}-1)$. Discrete cosine orthogonality at the polar roots gives the interpolant $N_\theta^{-1}\sum_i g_i[1+2\sum_{k=1}^{N_\theta-1}\cos(k\omega_i)T_k(x)]$. Its normalized integral is (151) because the integral of an odd Chebyshev mode is zero and that of mode $2k>0$ is $-1/(4k^2-1)$. Thus this rule is exact through degree $N_\theta-1$. Its weights are positive since $2\sum_{k=1}^{q}(4k^2-1)^{-1}=1-(2q+1)^{-1}<1$, and sum to one. A Chebyshev tail on the ellipse of parameter $e^\sigma$ is at most $2Be^{-N_\theta\sigma}/(1-e^{-\sigma})$. The exact integral and positive rule each have norm one, so their difference is at most twice this tail.

The domains needed for these bounds hold jointly. A complex plane rotation of imaginary angle $y$ sends a point of the quadric whose imaginary norm is $\sinh r$ to one with imaginary norm at most $\sinh(r+|y|)$. This follows by writing its real and imaginary parts, of norms $\cosh r, \sinh r$, and bounding the rotation's two parts by $\cosh|y|, \sinh|y|$. Successive rotations with all angle imaginary parts at most $h_{\mathrm{ang}}$ therefore stay in the quadric tube of radius $r_q/2$. Each harmonic is separately a trigonometric polynomial of degree at most $j_*$ in each angle. The maximum-modulus argument for its Laurent polynomial gives bound $\mathcal{Y}e^{j_*(d-1)h_{\mathrm{ang}}}$. The angular Jacobian has bound $A_d(\cosh h_{\mathrm{ang}})^{(d-2)(d-1)/2}$. The coefficient integrand is consequently bounded by $B_{\mathrm{int}}$. The polar ellipse has imaginary angular height $(\pi/2)\sinh\sigma = h_{\mathrm{ang}}$.

Telescoping the product of $d$ one-dimensional integral/rule operators now bounds exact-value coefficient error by

$$B_{\mathrm{int}}\left[\frac{2}{e^{a_tN_t}-1}+\frac{2}{e^{h_{\mathrm{ang}}N_\varphi}-1}+\frac{4(d-2)e^{-\sigma N_\theta}}{1-e^{-\sigma}}\right]\le\epsilon_c/2.$$

Run Lemma 8.2 with $\varepsilon_v=\delta_{\mathrm{node}}$. Its base error is smaller than $\delta_{\mathrm{node}}/8$ in Euclidean norm; an initialized image therefore has coordinate error at most $\delta_{\mathrm{node}}$. The spatial weights have total mass at most $A_d$, so the nodal perturbation in any coefficient is at most $2A_d\mathcal{Y}\delta_{\mathrm{node}}=\epsilon_c/2$. Summing coefficient errors and the independent analytic tail gives $\eta/16+N\mathcal{Y}\epsilon_c=\eta/8$.

Accumulate global coefficients only for the base sources and form their initialized images afterward. Every coefficient is the same finite scalar sum for each member of a pair, so $\widetilde{c}^{W_0g}=W_0\widetilde{c}^{g}$ exactly. Apply the preceding error proof separately to the analytic source and its analytic image. This avoids converting coordinate errors through an uncontrolled matrix norm. The source rank remains $R=2m+d+1+4N$, and Proposition 7.2 applies.

For $d=1$, use the two points, $N_x=H_{\mathrm{sph}}=2$, $H_1=\log(64M_n/(\alpha\eta))$, $N_1=1+\lfloor H_1/\alpha\rfloor$, $p_t=N_1-1$, $N=2N_1$, and $R=2m+d+1+8N_1$. Take $\epsilon_c=\eta/(16N_1)$, $\delta_{\mathrm{node}}=\eta/(64N_1)$, $B_{\mathrm{int}}=2M_ne^{a_tp_t}$, and $N_t=\max\{1,\lceil a_t^{-1}\log(1+4B_{\mathrm{int}}/\epsilon_c)\rceil\}$. Only the time trapezoid is used. Its error allocation is identical; the one-dimensional analytic tail is at most $\eta/8$ by Lemma 7.4, still leaving total error below $\eta$.

The count proof of Proposition 7.5 applies to this identical mode set and the same logarithmic tolerance. In particular its factor $\alpha^{-1}=128\beta^{30L}(Y/\lambda)^2\sqrt{d+3}\ell^{3/2}$ is preserved before squaring the rank. This proves the complete retained bound (9), with its label and dimension factors.

### 8.3 Exact implicit Gaussian execution

The next finite-dimensional sampler is useful because the numerical construction accesses an initialized hidden matrix only through its actions. All its zero tests and normalizations are exact.

**Lemma 8.3** (Adaptive queries to both directions of a Gaussian matrix)**.** *For each of finitely many layers let the reference matrix have independent $N(0,1/n)$ entries, independently across layers and*

*of the client's independent randomness. The client may choose the next layer, direction and query vector measurably from that randomness and its complete past transcript, and may stop within deterministic finite query budgets. There is an online sampler whose entire transcript has exactly the dense reference law. On an extension of its probability space, independent Gaussian reference matrices can be completed so that every returned reply is simultaneously the requested product with that one matrix.*

*For $Q$ requests to one layer, the sampler uses at most $4nQ^2+6nQ+2$ scalar arithmetic operations, at most $n\min\{Q,2n-1\}$ standard normal samples, and $2n\min\{Q,2n\}+8n+5\min\{Q,2n\}+64$ real or address words. These counts exclude the client's own work and retained arrays. The completion matrices are proof objects, not arrays constructed by the algorithm.*

*Proof.* For one layer keep matrices $V,Y\in\mathbb{R}^{n\times p}$ and $U,R\in\mathbb{R}^{n\times q}$, initially empty, with

$$V^\top V=I_p,\quad U^\top U=I_q,\quad U^\top Y=R^\top V.$$

Their intended meanings are $Y=WV$ and $R=W^\top U$. For a forward query $v$, compute $a=V^\top v$, $v_\perp=v-Va$, $\alpha=\|v_\perp\|_2$. If $\alpha=0$, return $Ya$. Otherwise put $e=v_\perp/\alpha$, draw a fresh standard Gaussian vector $g$ if $q<n$, and set

$$h=UR^\top e+n^{-1/2}[g-U(U^\top g)]. \tag{152}$$

If $q=n$, omit the Gaussian term. Return $Ya+\alpha h$ and append $(e,h)$ to $(V,Y)$. For a transpose query $u$, compute $b=U^\top u$, $u_\perp=u-Ub$, $\alpha=\|u_\perp\|_2$. If this is zero return $Rb$. Otherwise set $f=u_\perp/\alpha$ and

$$h=VY^\top f+n^{-1/2}[g-V(V^\top g)], \tag{153}$$

using a fresh standard Gaussian vector only when $p<n$. Return $Rb+\alpha h$ and append $(f,h)$ to $(U,R)$. Each layer has its own quadruple; all Gaussian vectors are independent. No dense projector is formed in these operations.

Here is the conditional-law proof, including adaptation. For proof only write $P=VV^\top$, $Q_0=UU^\top$ and $A=YV^\top+UR^\top(I-P)$. Compatibility gives $AV=Y$, $A^\top U=R$. The homogeneous solution space to these observations is exactly $\{(I-Q_0)B(I-P):B\in\mathbb{R}^{n\times n}\}$, and $A$ is Frobenius-orthogonal to it. The induction invariant is the conditional kernel

$$W=A+(I-Q_0)G(I-P),\qquad G_{ij}\ \text{independent}N(0,1/n). \tag{154}$$

At the initial empty state this is the dense Gaussian law. Given the entire past, the next query is fixed. For a new forward direction $e$, the action is $We=UR^\top e+(I-Q_0)Ge$, which has exactly (152)'s law. Put $P'=P+ee^\top$. The remaining matrix splits as

$$(I-Q_0)G(I-P)=((I-Q_0)Ge)e^\top+(I-Q_0)G(I-P').$$

The vector and remaining matrix are independent: their coordinate cross-covariances are $n^{-1}(I-Q_0)_{ia}[e^\top(I-P')]_b=0$. Jointly Gaussian linear functions with zero cross-covariance are independent even in singular cases, since their joint characteristic function $\exp(-\|A^\top s+B^\top t\|^2/(2n))$ factors when $AB^\top=0$. Thus observing the returned action changes only the conditional mean by $(h-UR^\top e)e^\top$ and replaces $P$ by $P'$. Expanding the new stored-state formula for $A$ gives exactly this new mean. Also $U^\top h=R^\top e$, preserving compatibility.

For a new transpose direction $f$, put $Q_0'=Q_0+ff^\top$. The split is $f((I-P)G^\top f)^\top+(I-Q_0')G(I-P)$, with zero cross-covariance for the same reason. The new mean is $A+f(h-VY^\top f)^\top$,

exactly its new stored-state formula. Dependent queries are already determined by the past and leave the kernel unchanged; full opposite rank makes the innovation identically zero.

Conditional on the global transcript, kernels of different layers remain a product. A query to one layer conditions only its factor; its choice reveals nothing further because it is already measurable from the past. Induction over the bounded number of requests proves equality of complete transcript laws. A bounded stopping time is handled by partitioning into its finitely many stopping events. At the stop, define completed matrices by (154) with fresh independent $G$'s. Their conditional kernels agree with the dense model; integration over the identical transcript law proves the entire joint dense law, including independence from client randomness. Their actions on the final $V, U$ are the stored $Y, R$, so every earlier reply is pathwise the correct action of the single completed matrix.

A new forward query at current ranks $p, q$ costs at most $6np + 8nq + 9n + 1 \le 8n(p + q) + 10n$ operations by direct dot-product counts. A dependent query is cheaper, and the transpose count exchanges the two ranks. Before request $j$, $p + q \le j - 1$, giving $4nQ^2 + 6nQ$ operations plus two for $1/\sqrt{n}$. There are at most $2n - 1$ drawing rank increments before one rank reaches $n$; after that all new opposite directions are determined. The four column lists use $2n(p + q)$ words. Eight length-$n$ scratch vectors, five words per column for coefficients and addresses, and 64 scalar words give the displayed memory bound. Exact zero tests and rank comparisons add at most two comparisons per request. No condition-number or nonsingular query-Gram hypothesis is needed in this real-arithmetic statement. □

### 8.3.1 Factored numerical anchors and output coupling

Fix the numerical orders, scalar routines, generator ordering, Gram–Schmidt signs, selector tie-breaking and other deterministic choices. On numerical panel $b$, write $\Delta_b$ for its length and use unscaled coefficients in $t - \tau_b$. For the polynomial-activation training network let $U_i^{(j)} = ([r_a \delta_a^{(j)}]_i)_{a=1}^m$ and $H_k^{(j)} = ([h_a^{(j)}]_k)_{a=1}^m$, both $n$ by $m$ matrices. Coefficient comparison in the actual gradient equations gives, for $s \ge 1$,

$$\begin{aligned} [W^{(1)}]_s &= -\frac{2}{ms} U_{s-1}^{(1)} (v_a)_{a=1}^{m\ \top}, \\ [W^{(j)}]_s &= -\frac{2}{mns} \sum_{i+k=s-1} U_i^{(j)} H_k^{(j-1)\top}, \\ [w]_s &= -\frac{2}{ms} \sum_a [r_a h_a^{(L)}]_{s-1}. \end{aligned} \tag{155}$$

Only lower orders are needed for an order-$s$ parameter coefficient. The increment's action on a query series $X$ at order $s$ is

$$-\frac{2}{mn} \sum_{i+k+c=s-1} \frac{U_i^{(j)} (H_k^{(j-1)\top} X_c)}{i + k + 1}. \tag{156}$$

Cache all inner products $H_k^\top X_c$ before taking the outer linear combinations. For a batch of $b'$ queries the work is $O(mb'K^2(n + K))$ per layer: inner products and vector combinations cost $O(nmb'K^2)$, scalar weighted sums $O(mb'K^3)$. The transpose calculation exchanges the two factors. This calculation is causal for training jets and applies to all passive jets after the training coefficients are available.

The endpoint update groups into $mK$ outer products per hidden layer:

$$W_{b+1}^{(j)} - W_b^{(j)} = -\frac{2}{mn}\sum_{i=0}^{K-1} U_i^{(j)} V_i^{(j-1)\top}, \qquad V_i^{(j-1)} = \sum_{k=0}^{K-1-i} \frac{\Delta_b^{i+k+1}}{i+k+1} H_k^{(j-1)}. \tag{157}$$

For the explicit execution, form each $V_i$ then add its outer products to the dense anchor. For implicit execution append these pairs to factors $A_b^{(j)}, B_b^{(j)} \in \mathbb{R}^{n\times bmK}$, including the scalar factor in $A_b$, so that

$$W_b^{(j)} = W_0^{(j)} + A_b^{(j)} B_b^{(j)\top}, \qquad W_b^{(j)} X_c = \mathrm{Action}_j(X_c) + A_b^{(j)}(B_b^{(j)\top} X_c). \tag{158}$$

Transpose actions use the same factors exchanged. This is an exact representation of the finite numerical anchor, not a rank assertion for the continuous dense trajectory. The first matrix remains explicit; its higher coefficient acts through the training input inner products in (155). It needs no array of $K$ dense first-matrix coefficients. The readout is an ordinary vector.

The complete list of hidden initialized-matrix requests is:

1. $m$ forward actions per mixer for exact original initialized training features and their images. These use $\phi_j$, not $\psi_j$. Initial backward fields are zero.
2. Constant-anchor actions on each source/training coefficient through $K$ on each numerical panel. The current-panel higher coefficients and previous-panel increments use the finite factors just displayed.
3. Initialized images of completed Harmonic coefficients or Taylor nodes.
4. $W_0^{(j)} U_{j-1}$ for the final source basis columns, after orthogonalization. Their number is at most the actual source rank $r$. Previously paired sources need not span all these image requests.

For Harmonic, all $N_x$ quadrature inputs require both directions; training uses $m$ in both directions. For Taylor the $M$ forward inputs already include training, and only its $m$ training inputs need backward jets. Thus sufficient per-mixer query counts are

$$Q_{\mathrm{H}} = m + 2J(m+N_x)(K+1) + 2N + r, \qquad Q_{\mathrm{T}} = m + J(M+m)(K+1) + (M+m)N_s + r. \tag{159}$$

To invoke Lemma 8.3, replace random $r$ in the budget by the deterministic $\min(n, R)$. The cost may still be bounded afterward using its actual value. No entry, norm, singular value or other information about a hidden initialized matrix is accessed.

Under the sampler's completion coupling, run the explicit numerical initializer with the completed matrices. Its exact initialized features agree by layer induction. Assuming all earlier coefficients agree, (158) and (156) give the same next matrix action. All remaining operations use identical real operands, so coefficient induction and then panel induction give the same sources and initialized images. Their ordered generator lists agree, hence so do exact ranks, bases, selector comparisons and selected weights. For source bases $U_j^\top U_j/n = I$, selected row matrices $P_j = (U_j)_{I_j}$, and source metrics $M_j$, item 4 gives the same core matrix $U_j^\top W_0^{(j)} U_{j-1}/n$ and thus the same initialized mixer

$$B_C^{(j)}(0) = P_j \frac{U_j^\top W_0^{(j)} U_{j-1}}{n} P_{j-1}^\top M_{j-1}.$$

The first-weight restriction, zero raw readout, initial deficit $c_C = y$, exact initial training Gram, and corrected-readout caches likewise agree. Thus the final arrays and their autonomous future agree pathwise, not only in marginal law. The full source/fitting event retains its probability with no extra failure budget. All sampler transcripts and trained factors are discarded once the retained arrays have been formed.

## 8.4 Complete operation and storage ledgers

All big-$O$ constants in the finite-order formulas below are numerical. They hide no problem parameter, numerical order, scalar-evaluator cost or confidence dependence. Assume $n \geq \max(m, d)$; the theorem is eventual, and full-width exact retention is available otherwise when the source budget is not smaller than $n$. Throughout, $R$ is the generator count bound, $r$ the maximum actual rank, and $q$ the maximum actual selected support size, so $r \leq \min(n, R)$ and $r \leq q \leq 9r$. They are not the source-dimension symbol used in Proposition 7.2.

### 8.4.1 Scalar geometry, global coefficients, and selection

For Harmonic, assign each time node $t_a = T(1 + \cos u_a)/2$ to one numerical panel, retaining multiplicity if two time angles have the same physical time. Let $I_b$ be those node indices. Form

$$\mathcal{B}_{kc}^{(b)} = \frac{\gamma_k}{N_t} \sum_{a \in I_b} \cos(k u_a)((t_a - \tau_b)/\Delta_b)^c.$$

If $[g]_c(v_s)$ is an unscaled local source coefficient, first multiply it by $\Delta_b^c$ to obtain $G_{b,c}(v_s)$. Its global coefficient is exactly

$$\widetilde{c}_{kjb'} = \sum_b \sum_s \omega_s Y_{jb'}(v_s) \sum_{c=0}^{K} \mathcal{B}_{kc}^{(b)} G_{b,c}(v_s). \tag{160}$$

Process one panel's shared training jets, then its spatial queries, then advance the anchor. Accumulate spatial coefficients first and transform only the retained mode pairs. The scalar tables cost $O((N_t + J)(p_t + 1)(K + 1))$ work, and projection costs $O(LnJ(K + 1)(N_x H_{\mathrm{sph}} + N))$ work with $O(Ln(K + 1)H_{\mathrm{sph}})$ extra vector words. Only $O(LnR)$ global coefficient words persist across panels. Local panels do not create separate retained Harmonic source blocks.

We give the finite basis evaluation needed by these costs. On a circle use $1, \sqrt{2}\cos(j\varphi), \sqrt{2}\sin(j\varphi)$ and angle-addition recurrences. In dimension $d' \geq 3$, writing $v = (\cos\theta, \sin\theta\, u)$, recursively use

$$(\sin\theta)^l C_{j-l}^{l+(d'-2)/2}(\cos\theta) Y_{lb}(u), \qquad 0 \leq l \leq j,$$

with scalar normalization. To verify this basis, substitution into the sphere Laplacian gives eigenvalue $-j(j + d' - 2)$. Lower harmonics are orthogonal, and for a fixed lower harmonic the remaining inner product has weight $(1 - z^2)^{l+(d'-3)/2}$. The Gegenbauer equation is $(1 - z^2)y'' - (2\mu+1)zy' + s(s+2\mu)y = 0$; multiply it by $(1-z^2)^{\mu-1/2}$ and integrate by parts. The boundary term vanishes because $\mu \geq 1/2$, and unequal eigenvalues give orthogonality. The harmonic dimension decomposition in Lemma 7.4 makes this orthogonal family complete. Its recurrence is

$$(s + 1)C_{s+1}^{\mu}(z) = 2(s + \mu)zC_s^{\mu}(z) - (s + 2\mu - 1)C_{s-1}^{\mu}(z), \quad C_0^{\mu} = 1, \quad C_1^{\mu} = 2\mu z.$$

It follows by differentiating the generating function $(1 - 2zt + t^2)^{-\mu}$. If $N_{l,s}$ is the squared angular norm for probability-sphere measure and $\mu = l + (d' - 2)/2$, then

$$N_{0,0} = 1, \quad \frac{N_{l+1,0}}{N_{l,0}} = \frac{2l + d' - 1}{2l + d'}, \qquad \frac{N_{l,s+1}}{N_{l,s}} = \frac{(s + 2\mu)(s + \mu)}{(s + 1)(s + \mu + 1)}.$$

The first ratio is the integration-by-parts ratio for powers of sine. For the second, express $zC_s^{\mu}$ using the three-term recurrence and equate its inner products with $C_{s+1}^{\mu}$ by symmetry of multiplication by $z$. Thus norms need only these recurrences and inverse square roots, not a

gamma-function oracle. There are $O((j_*+1)^2)$ degree pairs per intermediate dimension; combining with lower bases uses at most $O(H_{\mathrm{sph}})$ products per dimension.

For definiteness recompute spatial basis values on each numerical panel. The following matched time and memory geometry bounds include quadrature weight construction, node generation and the preceding basis recurrence:

$$\mathcal{G}_{\mathrm{time}} = \begin{cases} N_t + J, & d = 1, \\ N_t + N_x + JN_x(j_*+1), & d = 2, \\ N_\theta^2 + dN_\theta + N_t + N_\varphi + d + d(j_*+1)^2 + JdN_x[1 + (j_*+1)^2 + H_{\mathrm{sph}}], & d \geq 3, \end{cases}$$

$$\mathcal{G}_{\mathrm{memory}} = \begin{cases} N_t + 1, & d = 1, \\ N_t + N_x + j_* + 1, & d = 2, \\ dN_\theta + N_t + N_\varphi + d[1 + (j_*+1)^2 + H_{\mathrm{sph}}], & d \geq 3. \end{cases} \tag{161}$$

In (151), cosine recurrences cost $O(N_\theta^2)$ total; precomputed sine powers cost $O(dN_\theta)$, so coordinates and Jacobian weights cost $O(d)$ per spatial node. These operations account for every term just displayed.

For either model, process at most $R$ source generators against at most $r$ basis columns, costing $O(LnRr)$. The constructive selector in Lemma 7.1 takes at most $9r$ steps per layer. Each step forms shifted $r$ by $r$ inverses in $O(r^3)$ and tests all $n$ rows using $O(nr^2)$ quadratic-form work. Since $r \leq n$, total selector work is $O(Lnr^3)$. This term is retained in every ledger.

For the final metric, suppress layer indices and let $P$ be the selected basis rows, $\mathsf{D}$ the positive diagonal selector weights and $G = P^\top \mathsf{D} P$. Formula (108) and direct multiplication give

$$M = \mathsf{D} + \mathsf{D}P(G^{-2} - G^{-1})P^\top \mathsf{D}, \quad M^{-1} = \mathsf{D}^{-1} + P(I - G^{-1})P^\top, \quad P^\top M = G^{-1}P^\top \mathsf{D}.$$

All inverses here are of the positive $r$ by $r$ Gram, whose eigenvalues lie in $[1, 4]$. Forming metric caches and composing the initialized core mixers costs $O(L(qr^2 + r^3 + q^2 r)) = O(Lq^2 r)$. An explicit initialized basis image costs $O(n^2 r)$ per mixer; implicit execution instead pays its requests in $Q$ and then $O(nr^2)$ to contract with the other basis. The initial training Gram and its factorization cost $O(nm^2 + m^3)$ and $O(m^2)$ words. Exact source isometry makes it the same initial Gram in the selected model.

One scalar degree-$D$ activation polynomial composed with a degree-$K$ series costs $O(DK^2)$ and $O(DK)$ scratch: at time coefficient $k$ each of the $O(D)$ Chebyshev recurrence products costs $O(k+1)$, summed over $k \leq K$. Preparing coefficients and the derivative polynomial costs $O(LD^2)$ once. Both online training compositions and offline source compositions use this algorithm.

### 8.4.2 Supplied dense initialization

For Harmonic put $P_d = (L-1)n^2 + n(d+1)$. A sufficient complete arithmetic envelope is

$$\begin{aligned} O\big(&P_d + LD^2 + mP_d + nm^2 + m^3 + L + d + m + 1 \\ &+ JP_d(m + N_x)K + JLm(m + N_x)K^2(n + K) + JLn(m + N_x)DK^2 \\ &+ (N_t + J)(p_t + 1)K + LnJK(N_x H_{\mathrm{sph}} + N) + Ln^2 N \\ &+ LnRr + Lnr^3 + Ln^2 r + Lq^2 r + \mathcal{G}_{\mathrm{time}} + N_t \log(2 + N_t)\big). \end{aligned} \tag{162}$$

Its sufficient peak real-coordinate count is

$$\begin{aligned} O\big(&P_d + LmnDK + Lm^2K^2 + LnR + Lq^2 + LnKH_{\mathrm{sph}} + LD \\ &+ (p_t+1)K + N_t + J + m(d+1) + m^2 + L + d + \mathcal{G}_{\mathrm{memory}}\big). \end{aligned} \tag{163}$$

The constant-anchor actions cost $P_d(m+N_x)K$ per panel. Cached higher-coefficient actions give the next term. Forming each grouped endpoint factor in (157) costs $O(nmK^2)$, and adding its $mK$ outer products costs $O(n^2mK)$ per layer and panel, absorbed in the same ledger. The $Ln^2N$ and $Ln^2r$ terms separately pay initialized source images and final basis images. The moment table, projection and selection terms were proved above. A single numerical anchor, original initialized arrays, current-panel jets and global coefficients are live; no $JP_d$ trajectory or $KP_d$ dense jet array is stored.

For Taylor let $k$ be its reduced input dimension and $P_k = (L-1)n^2 + n(k+1)$. Its sufficient complete work is

$$\begin{aligned} O\big(&P_d + dM^2 + ndk + LD^2 + mP_k + nm^2 + m^3 + L + k + m + 1 \\ &+ J\{P_k(M+m)K + Lm(M+m)K^2(n+K) + Ln(M+m)DK^2\} \\ &+ LnMKN_s + Ln^2R + LnRr + Lnr^3 + Lq^2r + N_s\log(2+N_s) + J_s + N_s\big), \end{aligned} \tag{164}$$

with sufficient peak memory

$$\begin{aligned} O\big(&P_d + LnmDK + LnMK + LmMK^2 + LnR + Lq^2 + LD \\ &+ N_s + J + J_s + Md + dk + m^2 + L + k + m + 1\big). \end{aligned} \tag{165}$$

The input reduction costs $O(dM^2)$ and the supplied first-matrix projection $O(ndk)$. Evaluating source jets at the $N_s$ nodes costs $O(Ln(M+m)KN_s)$, absorbed by the displayed $M$ term since $m \le M$. Images and final basis actions are bounded by $O(Ln^2R)$ together. Sort and merge the source nodes with the numerical panels before the single continuation pass. Source nodes and numerical panels are distinct partitions.

### 8.4.3 Fresh implicit Gaussian initialization

For Harmonic use $Q = Q_{\mathrm{H}}$ from (159). A complete arithmetic envelope is

$$\begin{aligned} O\big(&nd(1+m) + m^2d + nm^2 + m^3 + L + d + m + 1 + L(D+1)^2 \\ &+ (L-1)nQ^2 + (L-1)nm(m+N_x)J(J-1)K(K+1) \\ &+ J[nd(m+N_x) + dmN_x] + JLm(m+N_x)(K+1)^2(n+K+1) \\ &+ JLn(m+N_x)(D+1)(K+1)^2 \\ &+ (N_t+J)(p_t+1)(K+1) + LnJ(K+1)(N_xH_{\mathrm{sph}} + N) \\ &+ LnRr + Lnr^3 + (L-1)nr^2 + L(qr^2 + r^3 + q^2r) + qd \\ &+ \mathcal{G}_{\mathrm{time}} + N_t\log(2+N_t)\big). \end{aligned} \tag{166}$$

Its matched peak-memory envelope is

$$
\begin{aligned}
O\big(&nd + (L-1)n[Q + JmK] + Lmn(D+1)(K+1) + Lm^2(K+1)^2 \\
&+ LnR + Ln(K+1)H_{\mathrm{sph}} + Lq^2 + q(d + Lm + 1) + m^2 + m(d+1) \\
&+ L(D+1) + (p_t+1)(K+1) + N_t + J + L + d + m + 1 + \mathcal{G}_{\mathrm{memory}}\big).
\end{aligned}
\tag{167}
$$

The sampler pays $O((L-1)nQ^2)$ work and $O((L-1)nQ)$ memory. At panel $b$ an accumulated trained factor has $bmK$ columns; its action costs $O(nbmK)$ per coefficient vector. Summing the $2(m+N_x)(K+1)$ possible vectors per panel gives the second term, using $\sum_{b<J} b = J(J-1)/2$. Its factors occupy $O((L-1)nJmK)$ words. Their grouped formation costs $O(JLnmK^2)$, covered by the current-panel contraction term. The explicit first matrix, input pairings, coefficient projection and selector account for the remaining terms already proved above.

For Taylor, sample only the $n$ by $k$ reduced first matrix, and use $Q = Q_{\mathrm{T}}$. It has the same law as $W_0^{(1)}U_0$; the independent Gaussian component on the orthogonal input complement may be completed in the proof. No request concerns that component. A sufficient complete arithmetic bound is

$$
\begin{aligned}
O\big(&dM^2 + nk(1+m) + km^2 + nm^2 + m^3 + L + k + m + 1 + L(D+1)^2 \\
&+ (L-1)nQ^2 + (L-1)nm(M+m)J(J-1)K(K+1) \\
&+ J[nk(M+m) + kmM] + JLm(M+m)(K+1)^2(n+K+1) \\
&+ JLn(M+m)(D+1)(K+1)^2 + Ln(M+m)(K+1)N_s \\
&+ LnRr + Lnr^3 + (L-1)nr^2 + L(qr^2 + r^3 + q^2 r) + qk \\
&+ N_s \log(2+N_s) + N_s + J + J_s\big).
\end{aligned}
\tag{168}
$$

Its sufficient peak real-coordinate count is

$$
\begin{aligned}
O\big(&Md + dk + nk + (L-1)n[Q + JmK] + Lmn(D+1)(K+1) \\
&+ LnM(K+1) + LmM(K+1)^2 + LnR + Lq^2 + q(k + Lm + 1) \\
&+ m^2 + Mk + L(D+1) + N_s + J + J_s + L + k + m + 1\big).
\end{aligned}
\tag{169}
$$

The factor-history term is charged exactly as for Harmonic, now with $M$ forward and $m$ backward coefficient vectors per order. All Taylor initialized node images and final basis images are included in $Q$. The ledger includes node evaluation, source orthogonalization, the $nr^3$ selector and the final Gram and metric solves. Thus a small source rank has not been used to omit either reference matrix queries or trained-factor multiplication.

### 8.4.4 Primitive costs and fixed-parameter bounds

In each model there are $4L(D+1) + Lnm$ original real activation-value calls, plus at most $L$ if the values at zero used in an envelope are not supplied. Only the first $4L(D+1)$ calls may use the finite tolerance in (139). There are no original derivative calls during setup. The retained runtime uses original activation and first-derivative evaluations as before.

Supplied dense initialization needs no Gaussian draws. Fresh explicit initialization, if desired, adds $nd + (L-1)n^2$ draws. The implicit Harmonic draw count is at most $nd + (L-1)nQ_{\mathrm{H}}$; the implicit Taylor count is at most $nk + (L-1)nQ_{\mathrm{T}}$. No readout draw is needed. The unobserved

completion draws are not algorithmic operations. For Harmonic the additional elementary-function call count is at most

$$O(L+d+m+1+L(D+1)+N_t+A_{\mathrm{ang}}+(L-1)Q+Lr),\quad A_{\mathrm{ang}}=\begin{cases}0, & d=1,\\ N_x, & d=2,\\ N_\theta+N_\varphi+d(j_*+1)^2, & d\ge 3.\end{cases}$$

For Taylor replace the time/angular terms by $N_s+J_s$ and add $O(M)$ input-basis square roots. For explicit execution omit the sampler's $Q$ term. These counts include scalar certificates, ellipse orders, geometry tables, query normalizations and orthogonalization. Rounding and mode enumeration require at most the displayed finite table sizes. If scalar calls have nonconstant cost, add the sum of their actual work and the largest simultaneously used evaluator scratch. The formulas do not assume that analyticity computes its own envelopes, population gap or evaluator.

For separately fixed admissible parameters, (142) and the source formulas give

| | Harmonic | Taylor |
|---|---|---|
| $J, D$ | $O(\ell^{3/2})$ | $O(\ell^{3/2})$ |
| $K$ | $O(\ell)$ | $O(\ell)$ |
| source sizes | $N_t, p_t+1=O(\ell^{5/2}),\quad N_x, H_{\mathrm{sph}}=O(\ell^{3(d-1)/2})$ | $J_s=O(\ell^{1/2}),\quad K_s+1=O(\ell)$ |
| $R, r, q$ | $O(\ell^{3d/2+1})$ | $O(\ell^{3/2})$ |
| $Q$ | $O(\ell^{3d/2+1})$ | $O(\ell^{5/2})$. |

(170)

The two-point convention gives the same Harmonic exponents at $d=1$. For the supplied dense variants the full ledgers respectively imply

$$O(n^2\ell^{3d/2+1}+n\ell^{9d/2+3}+\mathrm{polylog}(n)),\qquad O(n^2\ell^{5/2}+n\ell^5+\mathrm{polylog}(n)).$$

At fixed parameters their second and third terms are eventually smaller than their first, and their memory ledgers are $O(n^2)$. For implicit Harmonic the sampler term is $O(n\ell^{3d+2})$; trained-factor and activation work are $O(n\ell^{3d/2+7/2})$, and projection exponents are at most $3d-1/2$ and $3d/2+7/2$. The selector is $O(nr^3)=O(n\ell^{9d/2+3})$, dominating every displayed width-proportional term for $d\ge 1$. The remaining terms are fixed powers of $\ell$. Its transcript, source coefficients and other buffers fit $O(n\ell^{3d/2+1})$ memory.

For implicit Taylor the sampler term, accumulated trained-factor work, and activation composition are each $O(n\ell^5)$:

$$nQ_{\mathrm{T}}^2=O(n\ell^5),\qquad nm(M+m)J^2K^2=O(n\ell^5),\qquad Jn(M+m)DK^2=O(n\ell^5).$$

The selector costs $O(n\ell^{9/2})$, and the other terms fit the same work bound. Transcript, trained-factor and online activation storage are $O(n\ell^{5/2})$; all other buffers fit this bound. These substitutions prove all four rows of Proposition 1.2, including source construction, original initialized additions, selection and final assembly.

At completion the dense anchors, source generators, bases, panel schedules, activation polynomials, Gaussian transcripts and trained factors are discarded. The retained inventories and nonlinear optimizer are exactly those of the source-to-runtime proposition. Approximate generator construction does not require preservation of an accidental rank deficiency: the selector uses the span actually computed and the guaranteed generator bound $R$. Exact initialized additions preserve

the initial Gram and pairing hypotheses. Their replacement by finite-precision values would require an additional perturbation theorem, as would finite-precision rank decisions, barrier comparisons and infinite-time runtime evolution. The bounds here count real coordinates and operations, not bits.

The setup covers $T = 32\ell/\lambda$ in physical time. Its small number of numerical panels is not a claim about a short physical training prefix. Ordinary dense Euler at physical step $h$ has work $\Theta(mP_dT/h)$ before source assembly. At fixed parameters the explicit Taylor and Harmonic setup/Euler ratios are bounded by $O(h\ell^{3/2})$ and $O(h\ell^{3d/2})$, respectively. This comparison specifies an Euler step; it neither proves that such a step is necessary nor compares favorably against every high-order dense solver.

# 9 Nonlinearity and feature learning

We prove Theorem 1.3 using the real bounds already established in Lemma 2.1. All auxiliary notation below is local; constants and $O(\cdot)$ bounds may depend on the entire fixed problem, but not on width. The key simplification is to test feature increments against their initial adjoints, rather than follow every feature acceleration.

**Lemma 9.1** (Nonlinear initial velocity)**.** *Every $Q^{(\ell)}$, $\ell \geq 1$, is positive definite. The limiting initial prediction velocity is nonaffine on the sphere, witnessed by an affine-annihilating combination of four deterministic inputs chosen before initialization.*

*Proof.* Write $v = x/\sqrt{d}$. Gaussian covariance recursion gives $Q^{(\ell)}_{ab} = F_\ell(v_a^\top v_b)$, $F_0(s) = s$, with

$$F_\ell(s) = \sum_{k\geq 0} c_{\ell,k} s^k, \qquad c_{\ell,k} \geq 0, \quad \sum_k c_{\ell,k} < \infty, \tag{171}$$

and unbounded positive coefficient support. Here is a direct justification. The orthonormal Gaussian Hermite expansion of $\phi_\ell(\sqrt{q}G)$ has coefficients $\alpha_k$ and covariance $\sum_k \alpha_k^2 r^k$ for Gaussian correlation $r$. The covariance identity follows by comparing coefficients in $\mathbb{E}[e^{zG-z^2/2}e^{wG'-w^2/2}] = e^{rzw}$. Completeness follows because a function orthogonal to all polynomials has entire transform $\mathbb{E}[h(G)e^{zG}]$ identically zero, hence zero Fourier transform. Linear growth gives Gaussian $L^2$ integrability. Finite Hermite support would make the activation a polynomial, hence affine by its bounded derivative. Composing the covariance series with $F_{\ell-1}(s)/F_{\ell-1}(1)$ proves (171): Tonelli at $s = 1$ gives summability, and a positive inner coefficient of degree $j \geq 1$ and outer degree $k$ produces degree $jk$. These arguments allow nonzero means and justify uniform absolute convergence on $[-1, 1]$. The tensor Grams $[(v_a^\top v_b)^k]$ are positive semidefinite and converge to $I_m$ by the nonparallel-input hypothesis. Thus every $Q^{(\ell)} \succ 0$.

Let $P$ be the orthogonal projection onto affine functions in spherical $L^2$, with normalized surface measure $\sigma$: $Ph(v) = \int h\, d\sigma + d\, v^\top \int uh(u)\, d\sigma(u)$. Projecting the tensor feature in both variables gives the positive kernel

$$R_k(v, u) = (v^\top u)^k - a_k - b_k v^\top u, \quad a_k = \int (z^\top u)^k\, d\sigma(z), \quad b_k = d \int (z^\top u)^{k+1}\, d\sigma(z).$$

Here $a_k, b_k \to 0$ by dominated convergence, since $d \geq 2$, so $[R_k(v_a, v_b)] \to I_m$. The limiting velocity $g(v) = (2/m)\sum_a y_a F_L(v^\top v_a)$ is not affine: otherwise

$$0 = \sum_a y_a[(I - P)g](v_a) = \frac{2}{m} \sum_k c_{L,k}\, y^\top [R_k(v_a, v_b)]y > 0.$$

Every term is nonnegative and some large-degree term is positive. Uniform convergence justifies the projection and sums.

Some great circle witnesses this nonaffinity. Otherwise the antipodal average of $g$ is constant, and the odd remainder extends homogeneously to a function linear on every plane, hence globally linear. On such a circle, interpolate three distinct points by an affine function and take a fourth where it fails. The resulting dependence can be signed and normalized so that, for unit vectors $u_i$,

$$\sum_i |c_i| = 1, \qquad \sum_i c_i = 0, \qquad \sum_i c_i u_i = 0, \qquad \sum_i c_i g(u_i) > 0. \tag{172}$$

These inputs depend on the fixed problem only. The finite-query Gram convergence proved in Lemma 2.1, applied to this augmented list, and $\dot f_n(0,x) = (2/mn)\sum_a y_a h_a^{(L)}(0)^\top h^{(L)}(0,x)$ give convergence of their initial velocities. □

**Lemma 9.2** (Initial backward fields and hidden forces)**.** *The initialized forward fields and first backward derivatives, including the vectors before derivative gates, have deterministic joint empirical row-law limits with second moments. This includes the full first-layer weight and acceleration rows. Every hidden block's squared initial acceleration in inverse-mobility norm converges to a positive constant.*

*Proof.* All unmarked fields in this proof are at zero. Put $k = 2/m$ and define the pre-gated vectors $P$ and their gated versions $B$ by

$$P_a^{(L)} = \sum_b y_b h_b^{(L)}, \qquad B_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}) \odot P_a^{(\ell)}, \qquad P_a^{(\ell-1)} = W^{(\ell)\top} B_a^{(\ell)}.$$

Thus $\dot\delta_a^{(\ell)}(0) = kB_a^{(\ell)}$. Differentiating (3), since all hidden velocities initially vanish, gives

$$\ddot W^{(1)}(0) = k^2 \sum_a y_a B_a^{(1)} v_a^\top, \qquad \ddot W^{(\ell)}(0) = \frac{k^2}{n} \sum_a y_a B_a^{(\ell)} h_a^{(\ell-1)\top}. \tag{173}$$

Only a forward pass and a backward pass are needed for their joint limits. For one hidden Gaussian matrix $W$, after revealing $WH = Z$, its conditional law is $Z(H^\top H)^{-1}H^\top + \widetilde W(I - \Pi_H)$, where $\Pi_H = H(H^\top H)^{-1}H^\top$ and $\widetilde W$ is fresh with entries $N(0, 1/n)$. This is orthogonal Gaussian conditioning on the linear constraint. For a reverse query $U$, known from the previous answers,

$$W^\top U \overset{d}{=} HQ_n^{-1}C_n + (I - \Pi_H)\Xi D_n^{1/2}, \quad Q_n = H^\top H/n,\ C_n = Z^\top U/n,\ D_n = U^\top U/n, \tag{174}$$

with independent standard Gaussian rows in $\Xi$. The removed projection has conditional squared RMS $\operatorname{rank}(H)\operatorname{tr}(D_n)/n$, hence vanishes in probability. Adaptive queries cause no extra dependence: conditional on the global transcript, the next query is fixed; revealing one Gaussian action leaves the other matrices' residuals independent. In forward-then-backward order no matrix is queried again after its reverse answer.

For precision, the induction uses empirical quadratic Wasserstein convergence. Appending independent Gaussian rows preserves a deterministic such limit: conditional variances for bounded tests are $O(1/n)$, and the appended second moment obeys the law of large numbers. Continuous coordinate maps of at most linear growth preserve this convergence, by an $L^2$ coupling and uniform integrability. These include $\phi(z)$ and $\phi'(z)u$. Same-layer products consequently converge; inverse Grams and positive covariance square roots converge when their limiting Grams are, respectively, positive definite or positive semidefinite. The vanishing projected term in (174) does not change

the row-law limit. Starting with the full iid first-layer rows, these rules prove the forward limits and then the backward limits, including all pre-gated fields. The forward Grams being inverted are positive by Lemma 9.1; $Q^{(0)}$ is never inverted. The first-row acceleration is a linear combination of the $B$ fields, so its joint limit is included.

Let $D^{(\ell)}$ be the limiting Gram of $B^{(\ell)}$. At the top it is positive definite: a null vector $c$ would give

$$\Big(\sum_b y_b \phi_L(z_b)\Big)\Big(\sum_a c_a \phi_L'(z_a)\Big) = 0 \qquad (z \in \mathbb{R}^m).$$

Indeed $L \geq 2$ gives full Gaussian support. The first factor is not identically zero since $y^\top Q^{(L)} y > 0$; analyticity forces the second to vanish everywhere. Varying each coordinate, the nonconstant derivative gives $c = 0$. Downward, the fresh reverse innovation in (174) is independent of the lower-layer fields, with covariance $D^{(\ell+1)}$, so

$$c^\top D^{(\ell)} c \geq \lambda_{\min}(D^{(\ell+1)}) \sum_a c_a^2 \mathbb{E}[\phi_\ell'(Z_a^{(\ell)})^2] > 0.$$

Each Gaussian marginal is nondegenerate and a nonzero analytic derivative has a zero set of measure zero. Finally, expanding (173), the layer-$\ell$ squared acceleration limit is $k^4 y^\top (Q^{(\ell-1)} \circ D^{(\ell)}) y > 0$: for $Q \succeq 0$, $D \succ 0$, $Q \circ D \succeq \lambda_{\min}(D)\,\mathrm{diag}(Q)$ by decomposing $Q$ into rank-one terms, and every relevant $Q$ has positive diagonal. This also covers singular input Grams. □

**Lemma 9.3** (Adjoint test of the actual motion)**.** *Every hidden layer satisfies* (15)*. On the training set,*

$$y^\top (f_n(t) - f_{\mathrm{NTK},n}(t)) = \frac{m}{3} t^3 \left[ \frac{\|\ddot{W}^{(1)}(0)\|_F^2}{n} + \sum_{\ell=2}^{L} \|\ddot{W}^{(\ell)}(0)\|_F^2 \right] + o_*(t^3). \tag{175}$$

*Here $o_*(t^j)$ means: for every $\varepsilon > 0$, some deterministic $T_\varepsilon > 0$ makes the probability that the normalized remainder exceeds $\varepsilon$ anywhere on $0 < t \leq T_\varepsilon$ tend to zero.*

*Proof.* Use $\|u\|_n = \|u\|_2/\sqrt{n}$ and the inverse-mobility norm $\|U\|_{\mathrm{mob}}^2 = \|U^{(1)}\|_F^2/n + \sum_{\ell \geq 2} \|U^{(\ell)}\|_F^2$ on hidden matrices. On the event of Lemma 2.1, all relevant operators, feature RMS and residuals have deterministic bounds. The readout equation therefore gives $\|w(t)\|_n = O(t)$; backpropagation gives $\|\delta_a^{(\ell)}(t)\|_n = O(t)$; integration of the hidden updates and forward subtraction give, uniformly on the sphere for $0 \leq t \leq 1$,

$$\|W(t) - W(0)\|_{\mathrm{mob}} = O(t^2), \qquad \|\Delta z^{(\ell)}\|_n + \|\Delta h^{(\ell)}\|_n = O(t^2), \qquad f_n(t,x) = t\dot{f}_n(0,x) + O(t^2). \tag{176}$$

Here $\Delta$ means the increment from zero. For the last assertion, the readout equation gives $\|\dot{w}(t) - \dot{w}(0)\|_n = O(t)$; integrate and use $f = w^\top h/n$. No new bootstrap is needed.

The remainder control uses one elementary estimate. For bounded Lipschitz $g$ and an initial vector $u$,

$$\|[g(z) - g(z_0)] \odot u\|_n^2 \leq \mathrm{Lip}(g)^2 D^2 \|z - z_0\|_n^2 + 4\|g\|_\infty^2 \|u \mathbf{1}_{|u|>D}\|_n^2. \tag{177}$$

Split at $|u_i| = D$. The deterministic row-law limits with second moments in Lemma 9.2 give, for every $\eta > 0$, a fixed cutoff $D$ for which the probability that any needed initial squared tail exceeds $\eta$ tends to zero. Choose $D$ first, then a small time in (176). Thus gate differences times initial fields are $o_*(1)$, uniformly also along the segment from $z_0$ to $z$.

For this proof define $k = 2/m$, $P_a^{(L)} = \sum_b y_b h_b^{(L)}(0)$, $B_a^{(\ell)} = \phi_\ell'(z_a^{(\ell)}(0)) \odot P_a^{(\ell)}$, and $P_a^{(\ell-1)} = W^{(\ell)}(0)^\top B_a^{(\ell)}$. Since $w(t)/t = \dot{w}(0) + O_{\|\cdot\|_n}(t)$, downward subtraction of the backward recursion yields

$$\delta_a^{(\ell)}(t) = kt B_a^{(\ell)} + o_{*,\|\cdot\|_n}(t), \qquad \Delta W = \tfrac{1}{2} t^2 \ddot{W}(0) + o_{*,\|\cdot\|_{\mathrm{mob}}}(t^2). \tag{178}$$

Indeed each backward difference is the next-layer error through bounded operators, an $O(t^2)$ operator increment acting on $B$, and a gate difference acting on $P$, controlled by (177). Insert this into the hidden updates with $r = -y + O(t)$ and $h = h(0) + O_{\|\cdot\|_n}(t^2)$, then integrate. The normalized outer-product identities $\|uv^\top\|_F/\sqrt{n} = \|u\|_n\|v\|_2$ and $\|uh^\top\|_F/n = \|u\|_n\|h\|_n$ justify the matrix-norm remainder.

Instead of expanding each forward feature, test its increment against $P$. Put $E_{1,n} = \|\ddot{W}^{(1)}(0)\|_F^2/n$ and $E_{\ell,n} = \|\ddot{W}^{(\ell)}(0)\|_F^2$ for $\ell \geq 2$. The integral formula for an activation increment and (177) give

$$\langle P_a^{(\ell)}, \Delta h_a^{(\ell)}\rangle_n = \langle B_a^{(\ell)}, \Delta z_a^{(\ell)}\rangle_n + o_*(t^2), \qquad \langle u, v\rangle_n = u^\top v/n.$$

The error is bounded by the RMS gate difference times $P$, multiplied by $\|\Delta z\|_n = O(t^2)$. In $\Delta z = \Delta W h(0) + W(0)\Delta h + \Delta W \Delta h$, the last term is $O(t^4)$ in RMS. Summing the first term's pairings with $y_a B_a$ and using (173) gives $t^2 E_{\ell,n}/(2k^2) + o_*(t^2)$; the second pairs the lower feature increment with $P_a^{(\ell-1)}$. Telescoping to the first layer yields

$$\sum_a y_a \langle P_a^{(\ell)}, \Delta h_a^{(\ell)}\rangle_n = \frac{t^2}{2k^2} \sum_{j\leq \ell} E_{j,n} + o_*(t^2). \tag{179}$$

All energies have positive limits. Cauchy–Schwarz, with bounded $\sum_a y_a^2 \|P_a^{(\ell)}\|_n^2$, therefore proves the feature-motion lower bound in every layer on one deterministic small interval.

The tangent kernel is the readout Gram plus the hidden terms

$$K_{ab}(t) = \langle h_a^{(L)}, h_b^{(L)}\rangle_n + (v_a^\top v_b)\langle \delta_a^{(1)}, \delta_b^{(1)}\rangle_n + \sum_{\ell=2}^{L} \langle \delta_a^{(\ell)}, \delta_b^{(\ell)}\rangle_n \langle h_a^{(\ell-1)}, h_b^{(\ell-1)}\rangle_n.$$

Equation (176) gives $K(t) - K(0) = O(t^2)$. Its readout part, contracted with $y$, is twice the top pairing in (179), plus $O(t^4)$. By (178) and the squared outer products in (173), the hidden contribution has the same coefficient. Consequently

$$y^\top [K(t) - K(0)] y = \frac{2t^2}{k^2} \sum_\ell E_{\ell,n} + o_*(t^2).$$

Finally $\dot{f}_n = kK(t)(y - f_n)$, whereas the frozen flow uses $K(0)$. Variation of constants gives their difference as $k\int_0^t e^{-kK(0)(t-s)}[K(s) - K(0)](y - f_n(s))\, ds$. Here $f_n(s) = O(s)$ and the exponential is $I + O(t-s)$ because $K(0)$ is bounded. Pair with $y$ and integrate the last display. The additional terms are $O(t^4)$, giving $2/(3k) = m/3$ in (175). □

**Lemma 9.4** (The nonlinear feature component moves)**.** *Let $V = \operatorname{span}\{x_1, \ldots, x_m\}$, with normalized surface measure $\sigma_V$ on its unit sphere. For sufficiently small fixed $t > 0$, with probability tending to one,*

$$\frac{1}{n}\sum_i \inf_{a\in V,\, b\in\mathbb{R}} \int_{S(V)} |h_i^{(1)}(t, \sqrt{d}\, v) - h_i^{(1)}(0, \sqrt{d}\, v) - a^\top v - b|^2 \, d\sigma_V(v) \geq c_* t^4. \tag{180}$$

*Proof.* Since $\dim V \geq 2$, for nonzero $g, r \in V$ the function $\phi_1'(g^\top v)(r^\top v)$ is not affine on $S(V)$. Otherwise its values on $r^\perp$ and antipodes force the affine function to be $\lambda r^\top v$. Off the equator, division and then continuity make $\phi_1'$ constant on $[-\|g\|, \|g\|]$, contradicting analyticity and nonaffineness. Project the initial first-layer row onto $V$ and call it $g_i$; call its acceleration row $r_i \in V$. By Lemma 9.2, their joint empirical law converges with second moments; $g$ is nondegenerate Gaussian

and $\mathbb{E}\|r\|^2 > 0$. The squared distance of $\phi_1'(g^\top v)(r^\top v)$ from affine functions in $L^2(\sigma_V)$ is continuous in $(g, r)$, bounded by $\|\phi_1'\|_\infty^2\|r\|^2$, and positive when both are nonzero. Its empirical mean therefore tends to a positive constant.

The first-layer expansion in (178) and bounded $\phi_1'$ reduce the actual increment to the fixed row direction $t^2 r_i/2$. On $\|r_i\| \le D$, its Taylor remainder divided by $t^2$ is at most $\|\phi_1''\|_\infty t^2 D^2/8$; on the complement it is at most $\|\phi_1'\|_\infty\|r_i\|$. The deterministic second-moment limit, choosing $D$ and then time, gives $o_*(t^2)$ in neuron-index-times-sphere $L^2$. Orthogonal projection off affine functions is contractive, so the positive limiting projected acceleration proves the claim. □

*Proof of Theorem 1.3.* The four-point combination (172), initial velocity convergence and (176) give a positive multiple of $t$ for the dense predictor on a width-independent small interval. It annihilates every affine predictor and has coefficient absolute sum one, so it bounds their uniform distance below. The positive energy limit in (175), divided by $\|y\|_1$, similarly gives a training-input gap of order $t^3$ from the frozen kernel. The two preceding lemmas supply all internal feature conclusions.

For each fixed positive time, the vanishing absolute compression errors in Propositions 4.2, 7.5 and 7.6 transfer both prediction gaps by the triangle inequality; eventual estimates at every confidence imply convergence in probability. Choose common smaller $c_*, t_* > 0$. Legendre and Harmonic already cover the witnesses. Taylor appends them before initialization, without labels. Its bound replaces $p$ by at most $p + 4$, multiplying the quadratic panel factor by at most 9 and the linear data factor by at most 3, since $m + p \ge 2$. No endpoint gap or identification of compressed coordinates with dense neurons is used or asserted. □

## 10 Assembly of the headline theorem

*Proof of Theorem 1.1.* The preceding constructions provide the stated retained counts and the initialization-jet provenance alternative; the efficient alternatives have their own execution proofs. Here we first verify the original asymptotic comparison with actual dense variability, not an upper certificate. The quantitative width refinement uses its separately proved finite dense lower bound rather than claiming polynomial onset for the sharper coefficient used in this first argument. Fix a confidence budget $0 < \delta_0 < 1$. By Proposition 5.2, for sufficiently large width,

$$\Pr\left\{\left\|f_n - \widetilde{f}_n\right\|_{\mathcal{X}} \ge c_{\phi,L,\delta_0}\frac{Y\sqrt{\gamma}}{\sqrt{n}[\log(en)]^{5/2}}\right\} \ge 1 - \delta_0.$$

Its witness is a training input, so the same lower bound applies on either promised domain. The source and fitting events give all required limits. The Legendre order of Proposition 4.2 gives error at most $CY/(\sqrt{n}[\log(en)]^3)$; its ratio to the displayed lower bound is at most a fixed-problem, fixed-confidence multiple of $[\log(en)]^{-1/2}$. For the two selected models, Propositions 7.5 and 7.6 give error at most $Y/n$, and the corresponding ratio is at most

$$\frac{[\log(en)]^{5/2}}{c_{\phi,L,\delta_0}\sqrt{\gamma n}} \longrightarrow 0.$$

Intersecting each approximation event with the lower event costs at most $2\delta_0$ by the union bound, without any independence requirement. For every fixed positive ratio tolerance the limiting superior of its failure probability is therefore at most $2\delta_0$. Letting $\delta_0 \downarrow 0$ proves (7), including the stated failure convention. For (11), use the selected-model event at the requested confidence. The label assumption stays fixed throughout; no label inequality is used to reduce a displayed sample/gap power. For the polynomial sufficient width (12), apply Theorem 6.1 with the explicit maxima in

equations (106) and (107). That argument supplies its own quantitative actual-dense lower event and retains exactly equations (8) and (10); it does not attach polynomial onset to the sharper coefficient in the display above. □

# 11 Literal storage of a frozen-top neural tangent hierarchy

Literal neural tangent hierarchy (NTH) arrays provide a separate benchmark for how a prescribed representation affects storage. We study the ordered hierarchy associated with Huang and Yau [1], with its highest retained rank frozen, in the same feature-learning metric and zero-readout initialization as Section 1. On a correlated growing dataset, its necessary and sufficient retained size is $\exp[\Theta(\log m \log n)]$ on a fixed short interval.

This section has its own assumptions and probability convention. The fixed-data, small-label, Gram-gap and complete-trajectory hypotheses of Theorem 1.1 are not imposed here; the result is an independent benchmark, and gives no same-family storage separation from the preceding compression constructions. No theorem for another parameterization or readout initialization is transferred.

## 11.1 Model, closure, and storage

There are $m \geq 1$ deterministic samples, independent of initialization, $v_a = x_a/\sqrt{d} \in \mathbb{R}^d$, with $\|v_a\|_2 \leq 1$ and $|y_a| \leq 1$. Predictions indexed by $v$ mean the physical network input $x = \sqrt{d}\, v$. We specialize (1) to two identity hidden activations and use its readout $w$. Thus $W^{(1)} \in \mathbb{R}^{n\times d}$, $W^{(2)} \in \mathbb{R}^{n\times n}$, $w \in \mathbb{R}^n$, and

$$\begin{aligned} h^{(1)} = W^{(1)} v, \quad h^{(2)} = W^{(2)} h^{(1)}, \\ f_n(v) = \frac{w^\top h^{(2)}}{n}, \qquad r_a = f_n(v_a) - y_a, \\ \mathcal{L}_{1/2} = \frac{1}{2m} \sum_a r_a^2. \end{aligned} \tag{181}$$

Initially the first two matrices have independent entries with laws $\mathcal{N}(0,1)$ and $\mathcal{N}(0,1/n)$, respectively, and $w(0) = 0$. All blocks train with mobilities $(n, 1, n)$. The half-MSE is deliberate: the full-MSE flow (3) at physical time $t$ equals this flow at time $2t$. Throughout this section, $t$ and dots refer to half-MSE physical time. Thus $T = 2^{-15}$ here corresponds to $[0, T/2] = [0, 2^{-16}]$ in the paper's full-MSE time.

Put

$$\begin{gathered} B = W^{(1)}/\sqrt{n}, \qquad W = W^{(2)}, \qquad c = w/\sqrt{n}, \\ \theta = (B, W, c). \end{gathered} \tag{182}$$

Then $f_n(v) = c^\top W B v$ and training is ordinary Euclidean/Frobenius gradient flow in $\theta$. In particular $B_0, W_0$ have independent $\mathcal{N}(0, 1/n)$ entries. Define the fixed input second moment $\Sigma \in \mathbb{R}^{d\times d}$, label-input mean $b \in \mathbb{R}^d$, and prediction coefficient $u \in \mathbb{R}^d$ by

$$\begin{aligned} \Sigma = \frac{1}{m} \sum_a v_a v_a^\top, \qquad b = \frac{1}{m} \sum_a y_a v_a, \\ u = B^\top W^\top c, \qquad f_n(v) = v^\top u. \end{aligned} \tag{183}$$

Here $\Sigma$ is an input-space second moment, distinct from the sample-space population Grams $Q^{(\ell)}$ in (4). We have $\|\Sigma\|_{\mathrm{op}} \le 1$, $\|b\|_2 \le 1$, and the exact dense equations are

$$\begin{aligned} \dot{B} = W^\top c(b - \Sigma u)^\top, \qquad \dot{W} = c[B(b - \Sigma u)]^\top, \\ \dot{c} = WB(b - \Sigma u). \end{aligned} \tag{184}$$

For arbitrary input vectors define the source field and ordered tensors by

$$\begin{aligned} V(v;\theta) = \nabla_\theta f_n(v) = (W^\top c v^\top,\ c(Bv)^\top,\ WBv), \\ K_1(v;\theta) = f_n(v), \\ K_{r+1}(v_0,\dots,v_r;\theta) = D_\theta K_r(v_0,\dots,v_{r-1};\theta)[V(v_r;\theta)]. \end{aligned} \tag{185}$$

Every input slot is linear; permutation symmetry is not assumed. $K_r(\mathbf{v};0)$ means evaluation at $\theta_0$. For an integer $q \ge 2$, copy these initialized tensors through rank $q$, freeze rank $q$, and evolve all lower ranks by

$$\begin{aligned} \dot{K}_r^{(q)}(\mathbf{v};t) = -\frac{1}{m}\sum_a r_a^{(q)}(t) K_{r+1}^{(q)}(\mathbf{v}, v_a; t), \\ r_a^{(q)} = f_{n,q}(v_a) - y_a, \\ f_{n,q}(v) = K_1^{(q)}(v). \end{aligned} \tag{186}$$

The closure uses its own residual. In each array the first slot is a training input or a passive coordinate query $e_i$, $1 \le i \le d$; all later slots are training indices. Passive queries have no labels. Linearity gives the exact decoder $f_{n,q}(t,v) = v^\top u_q(t)$ for any later query, where $u_q(t) = (f_{n,q}(t,e_i))_{i=1}^d$. The moving and frozen array counts are, respectively,

$$\begin{aligned} N_{\mathrm{mov}}(q) = (m+d)\sum_{r=1}^{q-1} m^{r-1}, \\ N_{\mathrm{top}}(q) = (m+d)m^{q-1}, \qquad N_{\mathrm{arr}} = N_{\mathrm{mov}} + N_{\mathrm{top}}. \end{aligned} \tag{187}$$

The dataset adds at most $md + m$ fixed scalars. These are retained real coordinates, excluding coefficient-generation work, initialization peak memory, numerical integration and bit precision. They do not charge a factorized or implicit encoding. Once the initial tensors have been formed from $B_0, W_0$ and the data, those dense matrices need not be retained.

Fix throughout

$$\begin{aligned} T = 2^{-15}, \\ E_n(q) = \sup_{0\le t\le T}\sup_{\|v\|_2=1} |f_n(t,v) - f_{n,q}(t,v)|, \\ D_n = \sup_{0\le t\le T}\sup_{\|v\|_2=1} |f_n(t,v) - \widetilde{f}_n(t,v)|, \end{aligned} \tag{188}$$

where $\widetilde{f}_n$ is an independent dense initialization on the same dataset. Both errors use the same physical window and sphere. In probability statements, nonexistence on the window counts as failure; $E_n(q) \le AD_n$ includes $E_n(q) = D_n = 0$ without taking a ratio.

**Theorem 11.1** (Matched literal storage)**.** *Fix $0 < Y \le 1$, a success probability $0 < \pi < 1$, and an accuracy factor $A > 0$. For $4 \mid m$, $4 \le m \le \sqrt{n}$, take $d = m+1$, orthonormal coordinates $e_0,\dots,e_m$, and*

$$\begin{aligned} v_a = \frac{e_0 + e_a}{\sqrt{2}}, \qquad y_a = Y s_a, \\ s_a = \begin{cases} 1, & a \le 3m/4, \\ -1, & a > 3m/4. \end{cases} \end{aligned} \tag{189}$$

*Let $S_n(\pi, A)$ be the minimum of $N_{\mathrm{arr}}(q) + md + m$ over deterministic orders $q \geq 2$ for which $\Pr\{E_n(q) \leq AD_n\} \geq \pi$. For sufficiently large $n$, uniformly over the displayed sample range,*

$$\exp[c_Y \log m \log n] \ \leq S_n(\pi, A) \leq \exp[C_Y \log m \log n] \tag{190}$$

*for positive constants depending only on $Y$; the width threshold may also depend on $\pi, A$. The same order holds for the minimum moving-array count, and survives omission of a zero odd frozen top. The lower bound excludes all insufficient orders simultaneously, including initialization-dependent order selection under a deterministic budget. For $m = 4\lfloor n^a/4 \rfloor$, fixed $0 < a \leq 1/2$, the size is $\exp[\Theta_{Y,a}((\log n)^2)]$.*

**Corollary 11.2** (A uniform local upper bound for deterministic datasets)**.** *For every dataset in the setup, $n \geq 4$ and $1 \leq d \leq n$, put*

$$q_n = \left\lceil \frac{\log(2^{37} n^3)}{\log 1024} \right\rceil, \qquad c_* = 4 - \tfrac{3}{2} \log 9 > 0. \tag{191}$$

*With probability at least $1 - 4e/\sqrt{n} - 4e^{-c_* n}$, the dense and order-$q_n$ solutions exist and $E_n(q_n) \leq D_n/n$. For $m \geq 2$ their arrays satisfy $N_{\mathrm{mov}} \leq 2(m+d)m^{q_n-2}$ and $N_{\mathrm{arr}} \leq 2(m+d)m^{q_n-1}$; for $m = 1$ the exact counts are $(1+d)(q_n - 1)$ moving and $1 + d$ frozen. No Gram gap or positive label mean is required. If $b = 0$, both dense predictors and the closure vanish identically. The probability is for each deterministic dataset, not one simultaneous anti-concentration event for all datasets.*

The proof separates three issues. Ordered source integration controls the unchanged closure with its own residual. A nonzero omitted derivative and analytic continuation force a lower error at every finite order. Upper and lower probability bounds on the *actual* dense-pair discrepancy then give the two storage inequalities.

**Lemma 11.3** (Ordered source convergence and a common analytic disk)**.** *Suppose deterministically that $\|B_0\|_{\mathrm{op}}, \|W_0\|_{\mathrm{op}} \leq 4$. The dense solution and every finite closure exist on $[0, T]$. Their prediction coefficients extend holomorphically to $|t| < R_0 = 2^{-12}$, with norms at most one there; the dense coefficient satisfies $\|u(t)\|_2 \leq \|b\|_2(e^{1875|t|} - 1) < \|b\|_2$ when $b \neq 0$. Simultaneously for every $q \geq 2$, with $j = 2\lfloor q/2 \rfloor + 1$,*

$$E_n(q) \leq 64(j+1)(j+2)(\|b\|_2/4096)^j \leq 64\|b\|_2\, 1024^{-q}. \tag{192}$$

*Proof.* In the maximum block norm, a source derivative replaces one of the three parameter leaves of $c^\top W B v_0$ by a quadratic expression. After $k$ derivatives there are at most $(k+2)!/2$ terms, each with $k+3$ parameter leaves and one occurrence of each input. Products, transposes, outer products and scalar contractions are bounded by their operator/vector norm products; this construction introduces no free trace. Consequently, on a block-norm ball of radius $s$,

$$\begin{aligned} |K_{k+1}(v_0, \dots, v_k; \theta)| &\leq \tfrac{1}{2}(k+2)!\, s^{k+3} \prod_i \|v_i\|_2, \\ |K_{k+1}(\mathbf{v}; 0)| &\leq 32\, 4^k (k+2)! \prod_i \|v_i\|_2. \end{aligned} \tag{193}$$

Readout reflection $J(B, W, c) = (B, W, -c)$ obeys $f_n \circ J = -f_n$ and $V(v; J\theta) = -JV(v; \theta)$. Induction gives $K_r(\mathbf{v}; J\theta) = (-1)^r K_r(\mathbf{v}; \theta)$. Thus all odd initialized ranks vanish.

Extend (184) over complex coordinates using ordinary transposes. Within distance one of $\theta_0$, all block norms are at most five, $\|u\|_2 \leq 125$, and $\|b - \Sigma u\|_2 \leq 126$. Each block velocity is at most

$25 \cdot 126 = 3150$. A unit variation changes $u$ by at most 75, so each velocity derivative is at most $5 \cdot 126 + 5 \cdot 126 + 25 \cdot 75 = 3135$. Picard integration on $|t| \le R_0$ maps this path ball into itself and contracts, since $3150R_0, 3135R_0 < 1$. Its uniform iterates give a holomorphic solution, unique also as a real polynomial ODE solution. Differentiating $u = B^\top W^\top c$ gives $\|u'\|_2 \le 1875(\|b\|_2 + \|u\|_2)$. Integration along a radial segment and scalar Gronwall give the stated bound, using $1875R_0 < \log 2$.

For a vector path $z$, put $a_z(t) = b - \Sigma z(t)$. Define a vector-valued map by its pairing with every $v$:

$$v^\top \mathcal{F}_q[z](t) = \sum_{k=1}^{q-1} \int_{0<s_k<\cdots<s_1<t} K_{k+1}(v, a_z(s_1), \dots, a_z(s_k); 0)\, ds_k \cdots ds_1. \tag{194}$$

The complex integrals follow the straight segment to $t$; $\mathcal{F}_\infty$ uses all $k \ge 1$. The first appended slot is always at the latest integration time. For $\zeta_a(t) = (y_a - v_a^\top z(t))/m$, define $I_{a_1 \dots a_k}[\zeta]$ as the ordered integral of $\prod_i \zeta_{a_i}(s_i)$ over that simplex, and $I_\varnothing = 1$. For each retained rank the exact reconstruction is

$$K_r^{(q)}(\mathbf{v}; t) = \sum_{k=0}^{q-r} \sum_{a_1, \dots, a_k} K_{r+k}(\mathbf{v}, v_{a_1}, \dots, v_{a_k}; 0) I_{a_1 \dots a_k}[\zeta](t). \tag{195}$$

Differentiating the outer integration limit gives (186) whenever $z = \mathcal{F}_q[z]$; multilinearity identifies its rank-one equation with (194). Thus a fixed point supplies all actual arrays, their initialized values, and their frozen top. Polynomial ODE uniqueness identifies this solution with the prescribed closure.

On the unit ball of holomorphic coefficient paths on $|t| \le R_0$, $\|a_z\| \le 2$. The simplex volume and (193) bound every finite map by $32 \sum_{k \ge 1} (k+1)(k+2)(8R_0)^k < 0.38$. Telescoping its $k$ source slots bounds its Lipschitz constant by $16 \sum_{k \ge 1} k(k+1)(k+2)(8R_0)^k < 0.2$. Contraction gives all the asserted analytic closure paths.

On real $[0, T]$, if $\|b\|_2 > 0$, use instead the path ball of radius $\|b\|_2$ and put $x = 8T = 1/4096$. Uniformly over finite and infinite maps,

$$\begin{aligned} \|\mathcal{F}_q[z]\|_\infty &\le 64\|b\|_2[(1-x)^{-3} - 1] < \|b\|_2/16, \\ L &\le \frac{96x}{(1-x)^4} < \frac{1}{32}. \end{aligned} \tag{196}$$

These follow from $\|a_z\| \le 2\|b\|_2$ and $\|a_z - a_{\widetilde{z}}\| \le \|z - \widetilde{z}\|$; the series converge uniformly. The dense path is the infinite fixed point: repeated integration of its exact hierarchy leaves, after $N$ steps, an $N$-fold integral of $K_{N+1}(v, a_u(s_1), \dots, a_u(s_N); \theta(s_N))$. Previously selected source vectors are held fixed during each integration. By the dense bound and (193), this remainder is at most $\frac{125}{2}(N+1)(N+2)(10\|b\|_2 T)^N$, which tends to zero.

Parity makes $j = 2\lfloor q/2 \rfloor + 1$ the first possible omitted index. For $z_0 = x\|b\|_2$, the omitted tail is at most

$$32 \sum_{k \ge j} (k+1)(k+2) z_0^k \le 32(j+1)(j+2) z_0^j \frac{1 + z_0}{(1 - z_0)^3}.$$

Here $(j + \ell + 1)(j + \ell + 2) \le (j+1)(j+2)(\ell+1)^2$ and $\sum_{\ell \ge 0} (\ell+1)^2 z^\ell = (1+z)/(1-z)^3$. The difference of the two own-residual fixed points is bounded by this tail divided by $1 - L$. Since $(1+x)/[(1-x)^3(1-L)] < 2$ and $(j+1)(j+2) \le 4^j$ for $j \ge 2$, (192) follows. If $b = 0$, the dense initial state is stationary and the zero path is a fixed point of every finite map; uniqueness proves the omitted case. □

**Lemma 11.4** (Gaussian events and dense concentration)**.** *For $d \le n$ the matrix-norm event in Lemma 11.3 has probability at least $1 - 2e^{-c_* n}$. For the hard family, with probability at least $1 - n^{-1} - Ce^{-cn}$,*

$$D_n \le C\sqrt{\frac{m + \log(en)}{n}}. \tag{197}$$

*Here and below unspecified $c, C > 0$ are absolute.*

*Proof.* A maximal 1/4-separated sphere set has at most $9^d$ elements by comparing disjoint radius-1/8 balls with the radius-9/8 ball, and is a 1/4 net. It gives $\|M\|_{\mathrm{op}} \le (4/3)\max_v \|Mv\|_2$. For a fixed unit $v$, $n\|B_0 v\|_2^2$ is $\chi^2_n$. The Gaussian integral gives $\mathbb{E}e^{t\chi^2_n} = (1-2t)^{-n/2}$. Exponential Markov at $t = 4/9$ yields $\Pr(\|B_0 v\|_2 > 3) \le e^{-4n} 9^{n/2}$. The net union bound proves the asserted $e^{-c_* n}$ failure for $B_0$; the same calculation proves it for $W_0$.

We supply the Gaussian Lipschitz concentration used here. For standard Gaussian measure $\gamma$ in any finite dimension define $P_s h(x) = \mathbb{E}h(e^{-s}x + \sqrt{1-e^{-2s}}\, Z)$, $Z \sim \gamma$. Write $\mathscr{A} = \sum_i \partial^2_{x_i} - x \cdot \nabla$. For positive bounded smooth $h$ with bounded derivatives and bounded away from zero, Gaussian integration by parts gives $\partial_s P_s h = \mathscr{A} P_s h$ and

$$-\frac{d}{ds}\operatorname{Ent}_\gamma(P_s h) = \mathbb{E}_\gamma \frac{|\nabla P_s h|^2}{P_s h},$$
$$\operatorname{Ent}_\gamma(h) = \mathbb{E}_\gamma(h \log h) - \mathbb{E}_\gamma h \log \mathbb{E}_\gamma h.$$

Indeed $\mathbb{E}_\gamma[\zeta \mathscr{A} v] = -\mathbb{E}_\gamma \nabla\zeta \cdot \nabla v$, and $P_s$ preserves $\gamma$. Since $\nabla P_s h = e^{-s} P_s \nabla h$, weighted Cauchy–Schwarz gives $|\nabla P_s h|^2 / P_s h \le e^{-2s} P_s(|\nabla h|^2/h)$. As $s \to \infty$, bounded convergence gives $P_s h \to \mathbb{E}_\gamma h$ and entropy tending to zero. Integrating proves $\operatorname{Ent}_\gamma(h) \le \frac{1}{2}\mathbb{E}_\gamma(|\nabla h|^2/h)$. For bounded smooth $L$-Lipschitz $F$, with $L > 0$, substitute $h = e^{\lambda F}$. With $\psi(\lambda) = \log \mathbb{E}e^{\lambda F}$ this gives $\lambda\psi'(\lambda) - \psi(\lambda) \le \lambda^2 L^2/2$. Integrating $(\psi/\lambda)'$ from zero yields $\log \mathbb{E}e^{\lambda(F - \mathbb{E}F)} \le \lambda^2 L^2/2$. Markov, optimized at $\lambda = s/L^2$, and the same argument for $-F$ give

$$\Pr\{|F - \mathbb{E}F| \ge s\} \le 2e^{-s^2/(2L^2)}. \tag{198}$$

For an arbitrary Lipschitz $F$, truncate its values and smooth by convolution, preserving the Lipschitz bound. Pass first through smoothing and then truncation by dominated convergence: $|F(x)| \le |F(0)| + L|x|$, whose exponential is Gaussian-integrable. The case $L = 0$ is constant. This also justifies the preceding smooth calculations by bounded approximations.

On the matrix-norm event, all real states have block norms at most five. For two such trajectories, measure parameter differences by the sum of Frobenius/vector norms. Their $u$ values differ by at most 25 times this sum. Expanding (184) bounds each velocity difference by 1255 times the sum: the two direct-product differences each cost 630 on distinct blocks, while the residual difference costs 625 on each block. Gronwall gives factor $e^{3765t}$ in the sum norm. Therefore a fixed unit-query output at any $t \le T$ is $L_T = 50e^{4000T}$-Lipschitz in the Euclidean list of initial entries. Each such scalar map extends from the good set by $F(x) = \inf_{z\ \mathrm{good}}\{F(z) + L_T\|x - z\|_2\}$, with unchanged values and Lipschitz constant. Apply (198) to the variance-$1/n$ Gaussian entries. Two independent copies differ by at most $2s$ except with probability $4e^{-ns^2/(2L_T^2)}$, since their extension means agree.

The coefficient path has time Lipschitz constant at most $3 \cdot 25 \cdot 3150 < 250000$. Take a 1/2 sphere net with at most $5^d$ points and a time grid with $M = \lceil nT \rceil + 1$ points and mesh at most $1/n$. A linear functional's sphere norm is at most twice its net maximum. For $s = L_T\sqrt{2\log(4\,5^d M/\delta)/n}$ a union bound and the time Lipschitz estimate give $D_n \le 4s + 500000/n$, except with probability $\delta + 4e^{-c_* n}$. Set $\delta = 1/n$, $d = m + 1$ to obtain (197). □

**Lemma 11.5** (A nonzero omitted derivative at every effective order)**.** *For the hard family put* $\bar v = m^{-1}\sum_a s_a v_a$, $\lambda = \|\bar v\|_2$, *and* $v_* = \bar v/\lambda$. *With probability at least* $1 - Ce^{-cn}$, *simultaneously for every* $q \ge 2$, *the signed prediction difference* $g_q(t) = m^{-1}\sum_a s_a[f_n(t, v_a) - f_{n,q}(t, v_a)]$ *satisfies*

$$\begin{aligned} g_q^{(k)}(0) &= 0 \ (k < j), \\ g_q^{(j)}(0) &\ge j!(Y/64)^j, \qquad j = 2\lfloor q/2\rfloor + 1. \end{aligned} \tag{199}$$

*Proof.* Directly, $G = (v_a^\top v_b)_{a,b} = \frac{1}{2}(I_m + \mathbf{1}\mathbf{1}^\top)$, $\lambda^2 = 1/8 + 1/(2m)$, label RMS is $Y$, and label variance is $3Y^2/4$. Identity activations preserve this population feature Gram. For $\xi_0 = B_0 v_*$, the Gaussian norm ratios $\|\xi_0\|_2^2$ and $\|W_0\xi_0\|_2^2/\|\xi_0\|_2^2$ are independent $\chi_n^2/n$. Each is at least $3/4$ except with exponentially small probability: Markov applied to $e^{-\chi_n^2/6}$ gives failure at most $\exp[n/8 - (n/2)\log(4/3)]$. Intersecting with the matrix-norm event gives $\|\xi_0\|_2^2, \|W_0\xi_0\|_2^2 \ge 1/2$.

An odd frozen top is zero, so its predictor equals that of the preceding even top: for $h = 2\lfloor q/2\rfloor$, parity gives $\mathcal{F}_q = \mathcal{F}_h$. Put $e = u - u_h$ and $j = h + 1$. The first omitted index $h$ vanishes, so the ordered series gives

$$\begin{aligned} v^\top(\mathcal{F}_\infty[u] - \mathcal{F}_h[u])(t) = \frac{t^j}{j!}K_{j+1}(v, b, \dots, b; 0) \\ + O(t^{j+1}). \end{aligned}$$

If $e(t) = O(t^\ell)$, telescoping source slots in the finite map gives $\mathcal{F}_h[u] - \mathcal{F}_h[u_h] = O(t^{\ell+1})$: one slot contributes $-\Sigma e$, and there is at least one time integration. Since $e(0) = 0$, repeated substitution in the difference of the two fixed-point equations proves $e = O(t^j)$; the finite-map difference is then $O(t^{j+1})$. This proves both the vanishing derivatives and the displayed leading coefficient, with the closure's own feedback retained. Contracting $v = \bar v$ and using $b = Y\bar v$ gives exactly

$$g_q^{(j)}(0) = Y^j K_{j+1}(\bar v, \dots, \bar v; 0). \tag{200}$$

For source ascent of $F = c^\top W B\bar v$, write $\xi = Bv_*$ and use the source clock $\tau = \lambda s$. The exact equations are $\xi' = W^\top c$, $W' = c\xi^\top$, $c' = W\xi$. Their invariants $WW^\top - cc^\top = W_0W_0^\top$ and $\|\xi\|_2^2 - \|c\|_2^2 = \|\xi_0\|_2^2$ imply

$$\begin{aligned} c'' = (\|\xi_0\|_2^2 I + W_0W_0^\top)c \\ + 2\|c\|_2^2 c, \qquad c(0) = 0, \quad c'(0) = W_0\xi_0. \end{aligned} \tag{201}$$

Diagonalize the positive semidefinite matrix, choosing eigenvector signs so that $a = W_0\xi_0$ is coordinatewise nonnegative. The Taylor recurrence then has nonnegative coefficients. With $A_0 = \|\xi_0\|_2^2$, $H = \|a\|_2^2$, coefficientwise induction compares $c$ from below with $a\zeta$, where $\zeta'' = A_0\zeta + 2H\zeta^3$, $\zeta(0) = 0$, $\zeta'(0) = 1$. Indeed the linear term dominates $A_0 a_i\zeta$ in coordinate $i$, and $\|c\|_2^2 c_i$ dominates $Ha_i\zeta^3$ at every successive coefficient. Since $A_0, H \ge 1/2$, it also compares $\zeta$ from below with $2\tan(\tau/2)$, which solves $\zeta'' = \frac{1}{2}\zeta + \frac{1}{8}\zeta^3$ with the same initial values. Writing $\tan z = \sum_{k\ge0} t_k z^{2k+1}$, $(2k+1)t_k = \sum_{i=0}^{k-1} t_i t_{k-1-i}$ for $k \ge 1$, and $t_0 = 1$, give $t_k \ge 3^{-k}$ by induction. Hence $\zeta \ge \tau/(1 - \tau^2/12)$ coefficientwise. The degree-$j = 2k+1$ coefficient of $c^\top W\xi = c^\top c'$ is at least $H(k+1)^2 12^{-k} \ge 8^{-j}$. Returning to $F$ and clock $s$ multiplies its $j$th derivative by $\lambda^{j+1}$. Since $\lambda \ge 2^{-3/2}$, $K_{j+1}(\bar v, \dots, \bar v; 0) \ge j!\lambda^{j+1}8^{-j} \ge j!64^{-j}$, proving the claim on one event for every order. □

**Lemma 11.6** (From an omitted derivative to error on a real interval)**.** *If $g$ is holomorphic on a neighborhood of $|t| \le R$, bounded there by two, and $|g^{(j)}(0)| \ge j!\rho^j$, where $j \ge 1$ and $0 < \vartheta = \rho R/8 < 1$, then, for $\kappa = 16[1 + \log(1/\vartheta) + \log(8/3)]$,*

$$\|g\|_{[0,R/4]} \ge \frac{1}{8\kappa}\left(\frac{\vartheta}{8e^2\kappa^2}\right)^j. \tag{202}$$

*Proof.* For a degree-$N$ polynomial $P$, map $[0, R/4]$ to $[-1, 1]$. Its nonconstant Chebyshev coefficients have modulus at most $2\|P\|$, by their cosine integral formula, and

$$|T_k^{(j)}(-1)| = \prod_{r=0}^{j-1} \frac{k^2 - r^2}{2r+1} \le k^{2j}/j!,$$
$$|P^{(j)}(0)| \le \frac{2N}{j!}(8N^2/R)^j \|P\|_{[0,R/4]}.$$

For $k < j$ the derivative is zero; for $k \ge j$ the product formula follows by successively differentiating $(1-x^2)T_k'' - xT_k' + k^2T_k = 0$ at $x = -1$. Take the degree-$N = \lceil \kappa j \rceil$ Taylor polynomial of $g$. Cauchy's coefficient bound gives remainder at most $(2/3)4^{-N}$. Since its $j$th derivative agrees with that of $g$,

$$\|g\|_{[0,R/4]} \ge \frac{\vartheta^j (j!)^2}{2N^{2j+1}} - \frac{2}{3}\,4^{-N}.$$

To verify that the tail is at most half the first term, put $\zeta = \log(1/\vartheta)$, $v = \log(8/3)$, $D = 1 + \zeta + v$. Using $N \le 2\kappa j$, $\log(j!) \ge j\log j - j$, and $\log j \le j$,

$$j\zeta + (2j+1)\log N - 2\log(j!) + v \le j[\zeta + v + 3\log(2\kappa) + 3] \le 15jD < \kappa j \log 4 \le N \log 4.$$

The second inequality uses $\kappa = 16D$, $\log 32 < 4$ and $\log D \le D - 1$. Exponentiation proves the tail comparison. The resulting lower bound $\vartheta^j (j!)^2/(4N^{2j+1})$, with the same factorial bound, $N \le 2\kappa j$ and $j \le 2^j$, yields (202). □

**Lemma 11.7** (An actual dense-pair lower bound)**.** *For every fixed deterministic dataset in the setup with $n \ge 4$, $1 \le d \le n$, and $b \ne 0$, with probability at least $1 - 4e/\sqrt{n} - 4e^{-c_* n}$,*

$$D_n \ge \frac{\|b\|_2}{2^{31} n^2}. \tag{203}$$

*Proof.* Use $v_* = b/\|b\|_2$ and $g(t) = f_n(t, v_*) - \widetilde{f}_n(t, v_*)$. On the pair's matrix-norm event, Lemma 11.3 bounds $|g| \le 2\|b\|_2$ on $|t| < R_0$. Set $R = 2^{-13}$, so $T = R/4$. For $0 \le t \le T$ the radius-$R/2$ circle centered at $t$ lies in this disk; Cauchy's formula gives $|g''(t)| \le 2^{30}\|b\|_2$. At initialization,

$$f_n'(0, v_*) = \|b\|_2 H,$$
$$H = \|W_0 B_0 v_*\|_2^2 \overset{\text{law}}{=} UV,$$

where $U, V$ are independent $\chi_n^2/n$ variables. Indeed $U = \|B_0 v_*\|_2^2$, and conditional on $B_0 v_*$ the ratio $H/U$ has that same chi-square law independently of the conditioning vector.

For $n \ge 4$, the density $p(\zeta) = C_n \zeta^{n/2-1} e^{-n\zeta/2}$ is maximal at $\zeta_0 = 1 - 2/n \ge 1/2$. On $[\zeta_0, \zeta_0 + n^{-1/2}]$ its logarithm has zero initial derivative and second derivative at least $-2n$. Thus $p(\zeta) \ge e^{-1} p(\zeta_0)$ there and $\|p\|_\infty \le e\sqrt{n}$. Integration by parts in the same density gives $\mathbb{E}V^{-1} =$

$n/(n-2) \le 2$. The product density satisfies $\|p_H\|_\infty \le 2e\sqrt{n}$, and convolution gives the same bound for $H - \widetilde{H}$. Therefore $\Pr(|H - \widetilde{H}| \le a) \le 4e\sqrt{n}\, a$. Taking $a = 1/n$ gives $|g'(0)| \ge \|b\|_2/n$ outside an event of probability $4e/\sqrt{n}$, without conditioning on the matrix norms. Intersect afterward with their good event. At $t_n = 1/(2^{30}n) \le T$, Taylor's integral remainder now gives $|g(t_n)| \ge \|b\|_2/(2^{31}n^2)$, proving the assertion. □

*Proof of Theorem 11.1 and corollary 11.2.* For the general upper, combine (192) and (203) to get $E_n(q)/D_n \le 2^{37}n^2\, 1024^{-q}$ when $b \ne 0$. The choice (191) proves Corollary 11.2, including its probability. For $b = 0$, uniqueness gives the claimed exact zero predictions instead. The counts follow by summing (187).

For the lower, apply Lemma 11.6 to $g_q$ from Lemma 11.5 with $R = 2^{-13}$ and $\rho = Y/64$; the common analytic bound is two. Since $g_q = \bar{v}^\top(u - u_q)$ and $\lambda \le 1$, $E_n(q) \ge \|g_q\|_{[0,T]}$. Writing

$$\kappa_Y = 16[1 + \log(2^{22}/Y) + \log(8/3)],$$
$$a_Y = (8\kappa_Y)^{-1}, \qquad b_Y = \log(2^{25}e^2\kappa_Y^2/Y),$$

gives, on one event of probability $1 - Ce^{-cn}$, $E_n(q) \ge a_Y e^{-b_Y j}$ simultaneously for every $q \ge 2$. Put $R_n = n/(m + \log(en))$. If $q \le (4b_Y)^{-1}\log R_n$, then $j \le q + 1$ gives $E_n(q) \ge a_Y e^{-b_Y} R_n^{-1/4}$, whereas (197) gives $D_n \le CR_n^{-1/2}$. Because $R_n \to \infty$ uniformly in $4 \le m \le \sqrt{n}$, these bounds imply $E_n(q) > AD_n$ simultaneously over these orders. Thus their success probability for any fixed factor $A$ tends to zero, and eventually is below $\pi$.

Literal training arrays alone have at least $m^{q-1}$ entries, even after omitting a zero odd top; nonfrozen arrays have at least $m^{q-2}$. The necessary order therefore costs $\exp[c_Y \log m \log R_n]$ in either convention. Uniformly in the sample range $\log R_n$ is bounded above and below by positive absolute multiples of $\log n$ for sufficiently large $n$. For the upper, $q_n = O(\log n)$ has success tending to one and error at most $D_n/n$. With $d = m + 1$, its count is at most $5m^{q_n} + md + m$. These prove Theorem 11.1 and its moving-array assertion. For an adaptive order under a deterministic smaller budget, every permitted order lies in the simultaneously excluded range, proving the stated budget interpretation. □

The matched result concerns a growing-data, deep-linear model, a specified literal tensor representation, and the window $[0, 2^{-15}]$ in half-MSE physical time. All layers are trainable, but the input-output function is linear. No longer-time, nonlinear-activation, arbitrary-encoding, or finite-precision conclusion is asserted. Fixed $m, d$ give polynomial size; the superpolynomial example grows $m$ and $d$ together with width. Here $Y$ is fixed independently of $m, n$. The fixed-data small-label hypotheses of Theorem 1.1 are therefore not assumptions of this result. The preceding fitting and dense-variability theorems are not used, and this result establishes no same-family storage separation from those compression constructions.

## References


[1] Jiaoyang Huang and Horng-Tzer Yau. Dynamics of deep neural networks and neural tangent hierarchy, 2020.